

BEYOND INDIVIDUAL CLAIMS: PROBABILISTIC LOGICAL COHERENCE FOR LONG-FORM LLM EVALUATION

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ABSTRACT

Long-form factuality assessment decomposes a response into atomic claims and verifies each against retrieved evidence. The paradigm is effective and now standard, but it embeds an assumption — that the truth of each claim can be settled in isolation — which discards the structure that makes prose an argument. A response can therefore pass per-claim verification while contradicting itself, and two responses built from *identical* claims can differ in coherence through the order in which they assert them. We formalize *logical coherence* as a property of the joint distribution over claim truth values induced by a Markov random field whose pairwise factors are the relations the response itself draws. The relation vocabulary is a two-level taxonomy: twelve interpretable discourse senses, grounded in PDTB 3.0 and RST, compile through a deterministic map onto five inferential couplings, each fixing a factor table by naming the joint assignment it penalizes. We characterize the tables coherence requires by a row-stochasticity property and show that those used for claim–evidence factuality are wrong here, collapsing the measure’s dynamic range; claim priors come from an existing factuality pipeline, making coherence and factuality two stages over one decomposition. On a corpus of response families whose coherence ordering is fixed by construction, the measure recovers 82.7% of declared orderings from gold relation graphs and 34.7–37.1% from graphs mined out of the response text — and since the two conditions differ only in the relation graph, that gap localizes the open problem in relation extraction rather than in the probabilistic model. Against five families of evidence-free baselines it is the only measure that both orders responses correctly (37/50 against the best baseline’s 30/50) and stays exactly invariant under meaning-preserving edits (20/20), the latter *by construction*: reordering sentences leaves the Markov network identical, a guarantee no surface statistic can offer and one that an LLM judge violates while swinging 42 points between judge models on identical prose. The score also decomposes into per-claim marginals, so an incoherence is located rather than merely detected.

1 INTRODUCTION

The dominant paradigm for assessing the factuality of long-form generation is decompose-then-verify: split a response into atomic claims, check each against retrieved evidence, and aggregate the verdicts (Min et al., 2023; Wei et al., 2024; Song et al., 2024). It is effective, it is now standard, and it embeds an assumption. Treating each claim as independently checkable discards precisely the structure that distinguishes an *argument* from a *list*.

Two failure modes make this concrete. A report may assert both that no passengers were harmed and that a wire service reported three deaths; per-claim verification reports one claim true and one false, but not the more basic defect — that the response contradicts *itself*. Separately, the inferential links a response draws may fail even when their endpoints are individually true: a chain of accurate statements joined by a spurious “therefore” is not a sound argument, and every atom in it verifies. Xie et al. (2026) quantify how often this matters: on logic-related samples, SAFE, VeriScore and FactCheck-GPT omit the logical dimension on 60%, 78% and 48% of cases respectively. The two axes are orthogonal — a response can be a bag of true facts that is incoherent, or internally impeccable while entirely false — and our sharpest fixture is a pair of summaries of one appellate judgment built from the *same* eighteen claims, one a permutation of the other, differing only in whether self-serving

assertions precede or follow the findings that qualify them: every per-atom check returns an identical verdict, yet our measure separates them (0.95 against 0.90).

A motivating example. Figure 1 shows the failure in the smallest instance we can enumerate exactly, and it is worth reading in full because every later section is an answer to something visible in it. Five claims about Voyager 1 are fixed once and for all: three true of the world and two false. *Both* responses assert all five, so the two are indistinguishable to any method that scores claims one at a time — every retrieval, every NLI check, every per-atom verdict is byte-identical across the pair, and a precision-style aggregate returns 3/5 for both. What differs is the argument the response builds out of them. Response A names the two falsehoods, attributes them, and refuses each on a stated ground; Response B asserts them as fact, offers the false a_4 (“the plasma wave instrument had failed”) as the *reason* the true a_3 (“therefore in interstellar space”) holds, and in the same breath denies a_3 outright — then denies a_4 and a_5 in its closing sentence. Read as prose, B is fluent; read as an argument, it is incoherent, and its incoherence is not located in any one claim but in the wiring between them.

Two changes in the relation graph carry the entire difference (callouts 1 and 2 in the figure): the support edge $a_2 \rightarrow a_3$ becomes a CONTRADICTION, and the edge between a_3 and a_4 *reverses* — what A uses to reject a falsehood, B uses to lean on one. Our measure scores A at $LCS_{mm} = 0.590$ against B’s 0.494 (exact, by 2^5 enumeration), and because the priors enter only through unary factors that are identical in the two panels, the whole drop is attributable to arrangement alone. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. The score also localizes the damage, which an aggregate cannot: the *true* claim a_3 finishes at 0.513 under B against 0.992 under A, dragged below its own 0.9 prior because its only support now arrives from a claim the response itself denies — and a premise the text refuses cannot transmit support. A claim true of the world is demoted to a coin flip by the argument built around it. §3.7 works this pair formally once the model is in place, and App. G gives both responses verbatim alongside three further examples.

We therefore ask: how much of what a response asserts does its argument, taken as a whole, uphold? That is answered by a *marginal* in a graphical model whose variables are claim truth values and whose factors are the asserted relations — not by counting violated rules, which cannot propagate a conflict from its endpoints to what they support, and which yields a measure a response maximizes by asserting no relations at all (Prop. 3).

Our contributions are as follows. (i) A **coherence MRF and a formal two-level relation taxonomy** (§3.1–§3.2): twelve discourse senses compile through a deterministic map onto five inferential couplings, each defined by the joint assignment it penalizes, which fixes its factor table — separating the interpretable layer from the inferential one lets the annotation vocabulary be rich while the model stays a closed three-parameter pairwise family. (ii) A **characterization of why factuality’s factor tables are wrong for coherence** (§3.3): they charge a premise for being a premise, and the alternative collapses the measure’s dynamic range by more than half. (iii) A **selected readout and a two-stage composition with factuality** (§3.4, §3.6): of four candidates evaluated by exact inference against five criteria, only two are monotone in conflict resolution. (iv) **An evaluation separating relation-extraction error from scoring error** (§5), by holding the readouts byte-identical and swapping only the relation graph: the measure satisfies 82.7% of 202 declared constraints from gold graphs against 34.7–37.1% from mined ones, localizing the open problem in extraction; and against five families of evidence-free baselines it is the only one that both orders responses correctly (37/50 against the best baseline’s 30/50) and is exactly invariant under meaning-preserving edits (20/20) — the latter *by construction*, a guarantee no surface statistic can offer.

2 RELATED WORK

Atomic factuality. FActScore (Min et al., 2023) introduced per-atom precision against a knowledge source; SAFE (Wei et al., 2024) added search-augmented verification and an $F_1@K$ metric, VeriScore (Song et al., 2024) restricted attention to *verifiable* claims, and FLEEK (Bayat et al., 2023) retrieves evidence to detect and correct errors. All aggregate independent per-claim verdicts and are therefore blind to inter-claim structure by construction. FactReasoner (Marinescu et al., 2025) replaced verdict counting with probabilistic inference over a factor graph of claim and evidence variables with pairwise factors from NLI relations; it is the closest antecedent — we reuse its decomposition, relation extractor and inference stack, and take its marginals as our priors (§3.6) — but its factors

Query Where is Voyager 1 now, and has it left the Solar System?

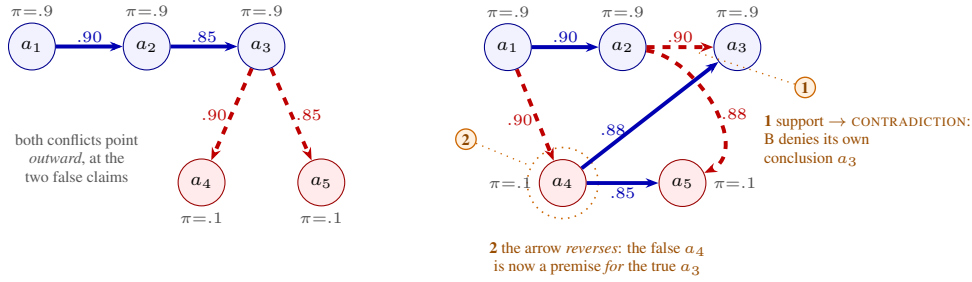
Response A *coherent*

Voyager 1 crossed the heliopause in August 2012, [a₁] and in the days that followed its instruments measured a sharp rise in plasma density [a₂] — the measurement that established the boundary had been passed. Voyager 1 is therefore in interstellar space. [a₃] Two persistent misconceptions are worth clearing up. The first is that the plasma wave instrument had failed permanently before the crossing; [a₄] it cannot have, since it is precisely that instrument which returned the density profile the conclusion rests on. The second is that Voyager 1 has left the Solar System altogether. [a₅] Being in interstellar space is not the same thing: the Sun's gravitational reach extends through the Oort cloud.

Response B *incoherent*

Voyager 1 crossed the heliopause in August 2012, [a₁] and after the crossing its instruments measured a sharp rise in plasma density. [a₂] That said, the probe cannot really be counted as being in interstellar space. [a₃] The plasma wave instrument had in fact failed permanently before the crossing, [a₄] and it is precisely that failure which tells us the probe reached interstellar space. The instrument failure also means Voyager 1 has left the Solar System entirely. [a₅] Of course, the crossing rules out any such failure, and the density measurement is inconsistent with the probe having left the Solar System.

The same five claims $a_1, \dots, a_5$ ([a_i] marks each first mention) $\Rightarrow$ identical per-claim factuality verdicts. Only the italicized relation-bearing spans differ.



A: a support chain over the true claims; each false claim attributed, then refused from a_3 . **B:** the same five claims, rearranged. Half its conflicts point inward, at what it asserts.

Claim	$\Pr(a_i=1)$ under A	$\Pr(a_i=1)$ under B
a_1 (true) heliopause crossing Aug 2012	0.9386	0.9601
a_2 (true) plasma-density rise after it	0.9891	0.9732
a_3 (true) therefore in interstellar space	0.9920	0.5126
a_4 (false) instrument failed pre-crossing	0.0129	0.0055
a_5 (false) left the Solar System entirely	0.0199	0.0177

The diagnostic. a_3 is true of the world, yet B drags it to a coin flip — 0.992 $\rightarrow$ 0.513, far below its own 0.9 prior — because its only support now arrives from a_4 , a claim the response itself denies. No aggregate score reveals this; a per-claim marginal locates it.

Coherence score LCS_{mm} : 0.590 (A) $\rightarrow$ 0.494 (B) $\Delta = -0.097$
 exact, by $2^5=32$ -world enum. LCS_{cn} : 0.963 $\rightarrow$ 0.414 • $\log Z$: $-0.95 \rightarrow -3.78$ • prior-only baseline 0.580 (A above it, B below)

The priors enter *only* through the unary factors, which are identical in the two responses. The entire 0.590 $\rightarrow$ 0.494 drop is produced by the pairwise relation factors — that is, by **arrangement alone**.

Figure 1: Two responses, the same five claims, different coherence. Five claims about Voyager 1 are fixed once and for all: a_1, a_2, a_3 true of the world (priors $\pi_i=0.9$, blue) and a_4, a_5 false ($\pi_i=0.1$, red). Both responses answer the *same* query and assert *all* five claims, so every per-claim factuality check returns the same verdict on both and no decompose-then-verify score can separate them. **A** attributes the two falsehoods and then refuses each on a stated ground; **B** asserts them and reasons *from* them, offering the false a_4 as the reason the true a_3 holds while denying a_3 outright — then denying a_4 and a_5 in its closing sentence. Middle: the relation graphs those spans induce (solid blue ENTAILMENT, dashed red CONTRADICTION, line width growing with the mined weight p), drawn at one shared scale; the two numbered orange callouts **1** and **2** mark the only two changes, which carry the whole difference. Bottom: the posterior marginals of the coherence MRF, and the score. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. §3.7 works this pair formally and App. G gives both responses in full.

run between a claim and the *evidence* retrieved for it, so its graph has no claim–claim edges and its marginals cannot express internal consistency. Xie et al. (2026) states the gap directly, classifying inter-atom dependencies as a labelling task over claim pairs; our departure is to treat them as *factors in a joint distribution*, which lets a local conflict propagate to distant claims and makes the score a functional of the whole graph rather than a per-edge average.

Discourse coherence. Entity-grid models (Barzilay & Lapata, 2008) and their neural successors (Li & Jurafsky, 2017) score coherence from entity-mention distributions, evaluated against *synthetic perturbations* — a methodology we extend in §5.1 by making the perturbation an operator on a declared relation graph rather than on raw text, so its intended effect is known per item. DiscoScore (Zhao et al., 2023) uses BERT-based discourse representations. What these share is that coherence is read off surface statistics or learned representations: none exposes an inspectable relation graph, and none yields per-claim attributions. Our inventory takes its senses from PDTB 3.0 (Prasad et al., 2019) and RST (Mann & Thompson, 1988) but treats them as an interpretable layer compiling to a smaller inferential vocabulary. Closest to our mining stage is Liu & Strube (2025), who label sentence pairs with PDTB 3.0 senses via a discourse parser (App. F).

Consistency checking, judges, and relational models. SelfCheckGPT (Manakul et al., 2023) detects hallucination by pairwise NLI over sampled responses — the natural ablation isolating what global inference adds — and ROSCOE (Golovneva et al., 2023) scores reasoning chains with a self-consistency term. G-Eval (Liu et al., 2023) is the practical incumbent, an LLM judge scoring coherence from a rubric; judges are strong on aggregate correlation and weak on per-claim attribution and on any guarantee that a meaning-preserving edit cannot change the score. Steen & Markert (2022) argue coherence measures must be checked against system-level confounders, which we adopt in our ladder-blind controls. Markov logic (Richardson & Domingos, 2006) is the natural alternative encoding; App. I shows the obvious one-clause-per-relation encoding is *not* equivalent to ours and is degenerate at its mode, while a three-clause expansion reproduces it exactly on the three-coupling fragment. We keep the pairwise MRF because its factor family is closed, and the Markov logic view because the beyond-pairwise rules addressing our main limitation live there.

3 PROBABILISTIC LOGICAL COHERENCE

This section builds the measure in six steps. We define the probabilistic object — a Markov random field over claim truth values whose factors are the relations a response draws (§3.1) — then fill in the two things that definition leaves open: *which* relations the vocabulary admits (§3.2) and *what factor table* each contributes (§3.3). With the model fixed, §3.4 reads a score off it, §3.5 estimates the relations from text, §3.6 couples coherence to factuality through the free prior slot, and §3.7 works a five-claim pair end to end. Each step closes a choice the previous one left open, and the two that matter most — the coupling vocabulary and the factor tables — are exactly where a coherence model diverges from a factuality one.

3.1 THE COHERENCE MARKOV RANDOM FIELD

Definition 1 (Coherence MRF). *Let $\mathcal{A} = \langle a_1, \dots, a_n \rangle$ be the atomic claims of a response in assertion order, each binary with $a_i = 1$ read as “the claim holds”. Let $\mathcal{R} = \{(s_r, t_r, \tau_r, p_r)\}_{r=1}^{|\mathcal{R}|}$ be a set of relations, where $s_r \neq t_r$ index claims, τ_r is an inferential coupling (Def. 2) and $p_r \in [0, 1]$ is the relation’s weight. Given priors $\pi \in [0, 1]^n$, the coherence MRF induced by $(\mathcal{A}, \mathcal{R}, \pi)$ is*

$$\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \prod_{i=1}^n \phi_i(a_i) \prod_{r \in \mathcal{R}} \psi_{\tau_r, p_r}(a_{s_r}, a_{t_r}), \quad Z = \sum_{\mathbf{a} \in \{0,1\}^n} \prod_i \phi_i(a_i) \prod_r \psi_r, \quad (1)$$

where $\phi_i(a_i) = [1 - \pi_i, \pi_i]$ over the states $(0, 1)$ and $\psi_{\tau, p}$ is given by §3.3.

Three structural properties are deliberate commitments, each load-bearing later.

Remark 1 (The factor family is closed). *Every $\psi_{\tau, p}$ is determined by p , the source prior π_s and the penalized set, so the whole inferential vocabulary is a three-parameter pairwise family — a guarantee about what the model cannot do: no relation type, however elaborately named at the discourse level, can smuggle in expressiveness beyond a pairwise potential.*

Remark 2 (Order-sensitivity enters only through the relations). *Nothing in Eq. 1 references claim position; assertion order enters exclusively through which pairs are related and in which direction. An edit preserving the relation set — reordering claims whose relations are symmetric, or exchanging a Precedence for a Succession, which compiles to no factor — therefore provably cannot change any readout, a falsifiable prediction the invariance ladders of §5.1 test.*

Remark 3 (Coherence is evaluated without evidence). *The graph contains claim variables only, and with uniform priors $\pi_i \equiv \frac{1}{2}$ the model consults no external knowledge whatever: it scores how the response hangs together, not whether it is true, which is what makes the two axes separable.*

3.2 TWO LEVELS: INFERENTIAL COUPLINGS AND DISCOURSE SENSES

Definition 2 (Inferential couplings). *A coupling τ is specified by the set $V(\tau) \subseteq \{0, 1\}^2$ of joint assignments (a_s, a_t) it penalizes — the configurations at which the relation is violated:*

Coupling	Reading	Directed	$V(\tau)$
ENTAILMENT	$a_s \Rightarrow a_t$	yes	$\{(1, 0)\}$
CONTRADICTION	$a_s \Rightarrow \neg a_t$	yes	$\{(1, 1)\}$
EQUIVALENCE	$a_s \Leftrightarrow a_t$	no	$\{(0, 1), (1, 0)\}$
EXCLUSIVE	<i>exactly one of a_s, a_t holds</i>	no	$\{(0, 0), (1, 1)\}$
CO_NECESSITY	<i>at least one of a_s, a_t holds</i>	no	$\{(0, 0)\}$
NONE	<i>no coupling</i>	—	$\emptyset$ (no factor)

Write $\mathcal{T}$ for the five edge-producing couplings and $\mathcal{T}_{\text{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$ for the conflict couplings, whose penalized set contains the both-true world.

The first three are exactly the NLI edge types of an existing factuality pipeline (App. B), which lets a mined relation reuse the existing graph machinery unchanged. The last two are additions required by argumentative prose: CONTRADICTION forbids only the both-true world, whereas *exhaustive* alternatives also forbid the both-false one — “no one was harmed” and “three people died” can neither both hold nor both fail, so treating them as mere opposition leaves a world in which *neither* is true and under-penalizes — and CO_NECESSITY is its dual (App. C).

Couplings are the right vocabulary for a factor table and the wrong one for annotation, since what is observable in the text is a discourse fact rather than a logical one. We therefore mine an interpretable layer and compile it down through a total, deterministic map $\mathcal{C}$ assigning each sense a coupling, a strength prior ($\perp$ meaning “use the estimator’s own value”) and flags:

Level-2 sense	$\mathcal{C}$ coupling	prior	note
Cause-Effect*, Effect-Cause*, Evidence*	ENTAILMENT	$\perp$	directed
Condition, Instantiation	ENTAILMENT	$\perp$	directed; inadmissible (§3.5)
Restatement*	EQUIVALENCE	0.90	definitional identity
Contrast*	CONTRADICTION	$\perp$	opposed, <i>not</i> exhaustive
Concession*	CONTRADICTION	$\perp$	a tension the text may resolve
Alternative*	EXCLUSIVE	$\perp$	exhaustive: exactly one holds
Disjunction*	CO_NECESSITY	$\perp$	at least one holds
Precedence*, Succession	NONE	—	ordering-only: no factor

Five senses collapse onto ENTAILMENT, a deliberate loss: a pairwise potential over truth values cannot distinguish causation from evidence. What survives compilation is the inferential force, the sense being retained for interpretability. The two ordering-only senses draw a relation with no truth-functional content — exactly what makes an order-only edit provably score-invariant (Rem. 2).

A Concession asserts a tension and then overrides it, and treating that as a live contradiction would penalize a response for doing what careful argument requires. When a resolving *holding* claim h is present we discount the conflict’s weight,

$$p_r^{\text{eff}} = p_r (1 - \lambda \pi_h), \quad (2)$$

with $\lambda = 0.45$; absent a resolver the tension counts at full strength. Holdings are found by adjudicative cue phrases on a non-endpoint claim maximizing lexical overlap with the endpoints — a heuristic which, with λ , is the one place hand-set constants enter.

3.3 THE PAIRWISE POTENTIALS, AND WHY FACTUALITY’S TABLES ARE WRONG HERE

A coupling names the world it penalizes; it does not say by how much. Filling that in is the second half of Def. 1, and it is where a coherence model parts company with the factuality model it otherwise reuses.

Listing $\psi_{\tau,p}(a_s, a_t)$ row-major over $(0, 0), (0, 1), (1, 0), (1, 1)$, with π_s the source prior and bold marking $V(\tau)$:

ENTAILMENT	CONTRADICTION	CO_NECESSITY	EQUIVALENCE	EXCLUSIVE
$[1 - \pi_s, \pi_s, \mathbf{1} - p, p]$	$[1 - \pi_s, \pi_s, p, \mathbf{1} - p]$	$[\mathbf{1} - p, \pi_s, \pi_s, p]$	$[p, \mathbf{1} - p, \mathbf{1} - p, p]$	$[\mathbf{1} - p, p, p, \mathbf{1} - p]$

Proposition 1 (Row-stochasticity of the asymmetric couplings). *The two asymmetric tables are row-stochastic in the source state: for $\tau \in \{\text{ENTAILMENT}, \text{CONTRADICTION}\}$ each row sums to one, so the row indexed by a_s is the conditional $\mathbb{P}(a_t | a_s)$, giving $\mathbb{P}(a_t=1 | a_s=0) = \pi_s$ and $\mathbb{P}(a_t=1 | a_s=1) = p$ for ENTAILMENT and $1 - p$ for CONTRADICTION. Consequently such a factor constrains the target only when the source is true; when the source is false it returns the target to its prior. The symmetric couplings are not of this form and are not claimed to be: CO_NECESSITY = $[1 - p, \pi_s, \pi_s, p]$ has rows summing to $(1 - p) + \pi_s$ and $\pi_s + p$, both one only in the degenerate case $p = \pi_s = \frac{1}{2}$.*

Proof. The $a_s=0$ row is $[1 - \pi_s, \pi_s]$ in both tables, and the $a_s=1$ row is $[1 - p, p]$ for ENTAILMENT and $[p, 1 - p]$ for CONTRADICTION; each sums to one, so dividing by the row sum is the identity. For CO_NECESSITY, row-stochasticity needs $\pi_s = p$ and $\pi_s = 1 - p$ at once, hence $p = 1 - p$. $\square$

Row-stochasticity is a property of the *table*, not an invariance of the marginals: it says each row is already normalized over the target, so $\psi(a_s, \cdot)$ reads as $\mathbb{P}(a_t | a_s)$ and the flat $a_s=0$ row makes a false source uninformative — the formal content of “an implication says nothing when its antecedent fails”. Once factors are multiplied and renormalized both endpoints can still move. Concretely, at $p = 0.9, \pi_s = 0.5$: ENTAILMENT = $[.5, .5, .1, .9]$ and CONTRADICTION = $[.5, .5, .9, .1]$ have rows summing to 1 and 1, reading $\mathbb{P}(a_t=1 | a_s=0) = 0.5 = \pi_s$ with $\mathbb{P}(a_t=1 | a_s=1) = 0.9 = p$ and $0.1 = 1 - p$ respectively; CO_NECESSITY = $[.1, .5, .5, .9]$ has rows summing to 0.6 and 1.4, and its *informative* row is $a_s=0$ (reading 0.833, against 0.643 for $a_s=1$) — a disjunction constrains its target when the source *fails*, which no source-conditioned normalization can express. Example 1 works the CO_NECESSITY joint through; both endpoints rise together, $0.5 \rightarrow 0.7$, the factor being symmetric.

The factuality literature instead uses a *no-priors* family whose ENTAILMENT table is $[p, p, \mathbf{1} - p, p]$: the $(0, \cdot)$ cells are both p , so the table charges the *source* being true whenever the target is uncertain. That suits claim–evidence factuality, where the source is a retrieved passage whose own reliability is in question, but it is wrong for coherence, where the source is another claim of the same response and a well-supported premise must not be penalized for being a premise. The consequence is not a matter of degree: on the sixteen-claim recall report of App. G, joined by a twelve-edge entailment backbone, the no-priors tables drive the root to $\mathbb{P}(a_1=1) = 0.19$, far below its 0.5 prior, and the mean marginal sits at 0.490, moving only to 0.498 when every conflict edge is deleted — a range of 0.008 against 0.607 to 0.620 under the with-priors tables, all exact by 2^{16} enumeration. A coherence measure built on the factuality tables spends its dynamic range penalizing premises and has little left for conflicts.

3.4 THE COHERENCE SCORE

The model is now fully specified: Def. 1 gives the joint, §3.2 the admissible relations, §3.3 their tables. What remains is to summarize the resulting distribution in a single number — and, as it turns out, not every natural way of doing so is admissible.

We report the *mean posterior marginal* $\text{LCS}_{\text{mm}}(y) = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(a_i = 1)$, read from a single MAR query: the expected fraction of the response’s claims that the argument, taken as a whole, upholds. A conflict suppresses the marginals of its endpoints, a satisfied entailment chain propagates support along itself, and a claim no relation touches stays at its prior. Three diagnostics come free with the same query: the per-claim marginals, which *localize* an incoherence; the count of claims dragged *below their own prior*; and the mean normalized binary entropy.

App. D defines three alternatives — a two-term consistency probability, a reified coherence node and a normalized log-partition — states five criteria, and evaluates all four by exact inference. Only the mean marginal and the normalized log-partition are monotone in conflict resolution, and the two failures share one cause: they measure whether a conflict is *active* rather than what should be *believed*, and softening a resolved concession by Eq. 2 necessarily *increases* the probability of its both-true world (Prop. 4). A fifth candidate — the probability that every asserted relation is satisfied — is maximized by asserting no relations whatever (Prop. 3). We select the mean marginal: it satisfies all five criteria, needs one query, and alone decomposes into per-claim contributions.

3.5 ESTIMATING RELATIONS FROM THE RESPONSE

Everything so far has taken the relation set $\mathcal{R}$ as given, which is what makes the model’s properties provable. In deployment it is not given.

In use $\mathcal{R}$ must be estimated from the text, and it is the weakest link in the pipeline (§5). A relation’s weight factors into a type term and a strength term, $p_r = \mathbb{P}(\tau \mid a_s, a_t, y) \cdot \mathbb{P}(a_t \mid a_s, \tau, y)$, estimated by two separate prompts (verbatim in App. K); conflating them is a mistake, since a confidently-identified weak link and a doubtfully-identified strong one are different epistemic situations one number cannot distinguish. *Prompt A* elicits the discourse sense with chain-of-thought *preceding* the label, so the token logprobs measure a decision already made, and the type confidence is the per-token geometric-mean probability of the label span. *Prompt B* elicits the conditional strength as a surrogate token, the renormalized $\mathbb{P}(\text{Yes}) / (\mathbb{P}(\text{Yes}) + \mathbb{P}(\text{No}))$ (Kadavath et al., 2022), asked in graded form so a weak but real link is not forced to answer No; a verbalized variant is kept only for comparison, verbalized confidence being weakly calibrated (Xiong et al., 2024).

Grounding is mandatory, not optional. Prompt A always receives the full response and asserts a coupling only where the response *itself* draws the connection — the single most consequential decision in the estimator. Judging a pair in isolation invites any abstractly plausible relation, and the failure is not mild: relation density reaches six to nine per claim, including spurious contradictions on paragraphs containing none. Since Eq. 1 is a product over relations, many soft mutually-constraining edges drive every marginal toward the prior, so an over-connected graph destroys the measure’s ability to distinguish coherent from incoherent rather than merely adding noise. Two mechanisms add precision: the *admissibility filter* restricts output to the nine senses marked $\star$ above, excluding Condition and Instantiation as broad enough that a model reaching for “related somehow” lands on them readily (13% of one relation source’s edges, zero in gold); and the *evidence-span requirement* obliges the model to quote the linking words verbatim, with the parser verifying the quotation really is a substring — the only precision mechanism that is *checked* rather than asked for.

Candidate pairs. Querying all $n(n-1)$ ordered pairs is quadratic and actively harmful, so a policy restricts which are scored: ALL_PAIRS; WINDOWED, forward pairs with $0 < j - i \leq w$ ($w = 4$); GATED, the window plus similar long-range forward pairs; and BIDIRECTIONAL, both arc directions within $|j - i| \leq w$. The last exists because of a structural limitation, not a tuning preference: a forward-only policy *cannot emit* a backward arc, so a gold relation that is both directed and backward is unreachable however the model answers — and 105 of our corpus’s 603 edge-producing gold relations are of that kind, capping forward-only recall at 82.6%. Each policy is then refined against the response, promoting discourse-adjacent out-of-window pairs and demoting in-window pairs that are not (App. E).

3.6 COMPOSING COHERENCE WITH FACTUALITY

The priors π are a free slot, and the natural thing to put in them is what a factuality pipeline already computed: $\pi_i \leftarrow q_i = \mathbb{P}(a_i = 1 \mid \mathcal{C})$ for retrieved evidence $\mathcal{C}$. A claim is then credited when it is both externally supported and internally upheld, and an unsupported claim enters the coherence graph already doubted, so an argument resting on it is penalized twice. Setting $q_i \equiv \frac{1}{2}$ recovers the pure coherence score of Rem. 3, so the composition is an option rather than a commitment. We keep two stages rather than one fused network for reasons of cost, ablation and caching that App. E sets out.

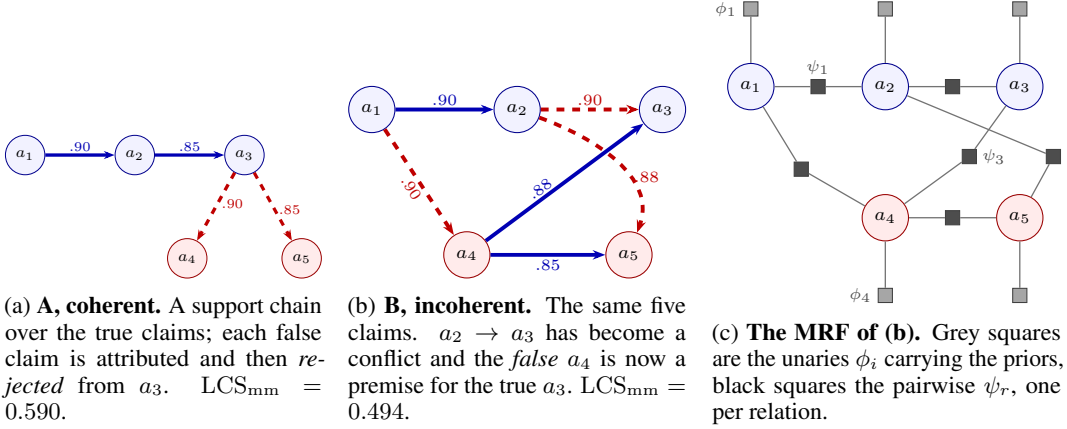


Figure 2: The five-claim example of §3.7. Blue nodes are claims true of the world ($\pi_i = 0.9$), red nodes false ($\pi_i = 0.1$); *the same five claims appear in both graphs*, so every per-claim factuality check returns the same verdict on both responses. Solid blue edges are ENTAILMENT, dashed red CONTRADICTION; line width grows with the relation weight p . The unary factors ϕ_i of panel (c) are identical for the two responses, so the entire $0.590 \rightarrow 0.494$ drop is produced by the pairwise ψ_r — by how the response arranges its claims, which is the whole of what coherence measures.

3.7 EXAMPLE

The pieces are easiest to see at work on a case small enough to enumerate exactly. This is the pair of Figure 1, revisited now that the model is defined: the same five claims and the same two arrangements, with the relation graphs read against §3.2’s vocabulary and the MRF they induce drawn explicitly. It also makes the orthogonality claim of §1 concrete, since it holds the claim set — and therefore every factuality verdict — fixed while varying only the arrangement.

Five claims about Voyager 1 are fixed once and for all, three true of the world (priors 0.9) and two false (0.1): a_1 , it crossed the heliopause in August 2012; a_2 , after the crossing its instruments measured a sharp rise in plasma density; a_3 , it is therefore in interstellar space; and, false, a_4 , its plasma wave instrument failed permanently before the crossing, and a_5 , it has left the Solar System entirely.

Two responses assert *these same five claims* and differ only in how they arrange them, so every per-claim factuality check returns the same verdict on both. **Response A** reports the density rise as what established the crossing, concludes the probe is in interstellar space, then names the two false claims and refuses each on a stated ground. **Response B** instead denies that the probe counts as being in interstellar space, asserts the instrument failure as fact, offers *that failure* as the reason the probe reached interstellar space, chains it into having left the Solar System, and finally denies both falsehoods in its closing sentence.

Figure 2 shows the resulting graphs and the MRF that B induces. Two relation changes carry the difference: the support edge $a_2 \rightarrow a_3$ in A becomes a CONTRADICTION in B, which denies its own conclusion, and the edge between a_3 and a_4 *reverses* — A rejects the false a_4 from a_3 , while B makes a_4 a premise for a_3 . Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. Exactly, by 2^5 enumeration, A scores $LCS_{mm} = 0.590$ against B’s 0.494: in A the true claims sit at 0.939, 0.989 and 0.992, each *above* its 0.9 prior because the chain supplies mutual support, and the false ones at 0.013 and 0.020, whereas in B the *true* a_3 is dragged to 0.513 because its only support arrives from a_4 , which the response itself denies — a premise the text refuses cannot transmit support. The priors enter only through the unary ϕ_i , identical in both responses, so the entire drop is attributable to arrangement alone. App. G gives both responses in full and two further examples.

Table 1: The four family types. Each holds the claim set, the topic and every unperturbed relation fixed, varying one structural dimension. ORDER and CONTROL apply only meaning-preserving edits and so assert *invariance*: CONTROL is the false-positive check, since a measure showing any monotone trend across it is tracking *having been edited* rather than coherence. The final column is the count in the released corpus (§4.5).

Type	Dimension varied	Rungs, $\rho_0 \rightarrow \rho_4$	Required pattern	n
CONFLICT	conflict structure	worse, base, concession_resolved, fix_one_conflict, coherent	strictly increasing	8
CHAIN	entailment-chain integrity	break_3, break_2, break_1, base, coherent	strictly increasing	2
ORDER	assertion order	shuffle_full, shuffle_block, shuffle_adjacent, base, ordering_only	invariant	3
CONTROL	nothing (meaning-preserving)	ordering_only, direction_reversal, base, direction_reversal, ordering_only	flat throughout	1

4 THE LOCOBENCH BENCHMARK

Validating a coherence measure requires responses that differ in coherence *while holding content fixed*, each accompanied by the relation graph a correct extractor would recover. Natural corpora supply neither. They offer no controlled variation in coherence at constant claim content, so an observed score difference cannot be attributed to coherence rather than to topic or length; and they carry no gold relation graph, without which *scoring* error cannot be separated from *estimation* error — the one distinction our evaluation is built to make. We therefore evaluate on a synthesized corpus, LoCoBench, and because the corpus is generated the burden is on us to state its construction completely. This section does so: the object generated (§4.1), the pipeline that generates it (§4.2), how each rung’s gold graph is derived (§4.3), the validation that gates admission (§4.4), and what the released corpus contains (§4.5).

4.1 COHERENCE LADDERS

Definition 3 (Coherence ladder). A ladder is a tuple $(\mathcal{P}, \langle \rho_0, \dots, \rho_4 \rangle, \Gamma)$ in which $\mathcal{P}$ is a base relation plan — a selected claim set together with a declared relation graph over it — each rung ρ_k is a sequence of perturbation operator calls applied to $\mathcal{P}$, realized as a response text and carrying its own gold relation graph, and Γ is a set of constraints on the relative ordering of the rungs’ scores. A ladder is admitted only if all five rungs pass every validation gate of §4.4.

Two properties of Def. 3 carry the experimental design. First, all five rungs share one claim set and one topic, so a measure that orders them is responding to relational structure and not to content. Second, each rung carries its own gold graph rather than inheriting the base plan’s, which is what makes a rung a genuine test of a readout rather than a restatement of the corpus; §4.3 gives the derivation.

Whole-family admission. A ladder is admitted entire or rejected entire. Four of the five rungs carry no standalone claim — only a claim relative to their neighbours — so a family missing a rung is not a partial success but a broken comparison. A family whose fourth rung fails validation is therefore rejected together with the four rungs that passed.

Table 1 gives the four types, and the mix is load-bearing rather than incidental. CONFLICT is the reference ladder: only the conflict structure varies, which is what makes an ordering attributable to coherence. CHAIN tests a different failure mode — breaking N links of an entailment chain and repairing them, so the defect is a *missing* relation rather than a present conflict. The distinction matters because a measure built on contradiction detection alone will do well on the first ladder and fail the second. ORDER and CONTROL are the negative controls, and a corpus containing only increase-type ladders cannot distinguish a coherence measure from an edit-count proxy: every rung of these two applies only meaning-preserving edits, so they assert that the score does *not* move. §5.3 reports that this is precisely where mined relation graphs fail, a result no increasing-order ladder could have exposed.

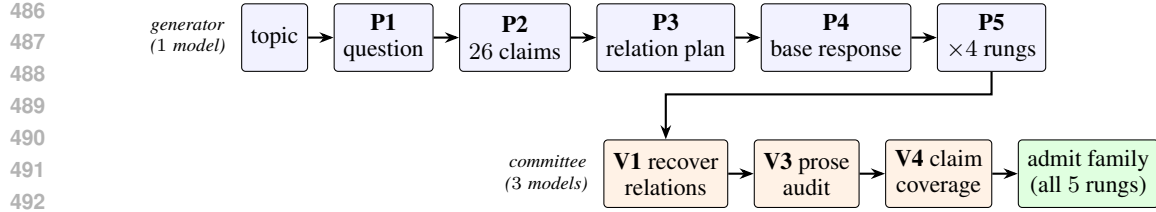


Figure 3: The generation pipeline. Five stages produce a family of five rungs from one topic; three validators, run by a committee that *excludes* the generating model, gate admission. A family is admitted whole or rejected whole (Def. 3).

Ordering constraints. A family declares its constraints in three classes, asserted per readout: **C1** a strict increase between consecutive rungs; **C2** a *predicted* decrease or invariance, which is where the concession dip of Prop. 4 becomes falsifiable and where an ordering-only edit must be exactly flat by Rem. 2; and **C3** endpoint separation between ρ_0 and ρ_4 . A missing score fails rather than passes. C2 is the only class that predicts the internal behaviour of a particular readout, which is why the baseline comparison of §5.5 excludes it: a single-score baseline makes no claim a C2 constraint could test.

4.2 THE GENERATION PIPELINE

Generation is a five-stage prompted pipeline followed by validation. The stages are separated so that a defect can be attributed to one of them and repaired there, and each stage’s prompt is fixed and versioned, with its numbered-instruction count and placeholder contract declared in Table 3 and asserted by a test against the prompt strings themselves. The mechanism matters because the alternative failure is silent: a corpus generated by prompts that no document describes.

The five stages, in order (Fig. 3), are as follows.

P1: topic to question. Given a topic from an inventory of 36 topics across four domains, emit one open-ended question. The prompt enumerates the target relations, so the question is chosen with the downstream plan in mind rather than for topical interest, and it prefers adjudicative domains — incident investigations, litigation, diagnostic reasoning, attribution disputes, competing hypotheses — because those are the settings in which exhaustive alternatives and resolving holdings arise naturally.

P2: question to claims. Emit 26 atomic claims answering the question, each a single proposition, tagged with a truth status. A surplus is generated deliberately: P3 selects a subset, and selection from a surplus is what lets the plan choose claims that support the required relation structure instead of forcing relations onto whatever claims exist.

P3: claims to relation plan. Select 14–16 claims (aiming for 15) and plan exactly 10 relations over them using the two-level scheme of §3.2, with at least one relation of each required sense, together with 4–6 pairs *explicitly declared unrelated*. Relations are kept local, preferring pairs within four positions, which is what makes a windowed candidate policy a reasonable baseline rather than a straw man. The declared non-relations are the corpus’s labelled negative set, and they are what makes a precision figure meaningful: precision against “every pair not in gold” is dominated by unlabelled pairs of unknown status, whereas precision against pairs the item asserts are unrelated is a real number. Each relation also carries an intended strength band and an intended *validity* — valid when the relation genuinely holds, invalid when the prose draws it unsoundly — targeting a valid fraction near 0.55, so a scorer cannot succeed by assuming every relation it finds is sound. The shipped items carry 7–11 relations each and 2–3 false claims, the spread arising because a rung’s perturbations add and remove edges relative to the plan.

P4: plan to response. Realize the plan as continuous prose of 550–650 words, asserting every selected claim and drawing every planned relation, with no headings or lists. The floor exceeds the default completion-token grant of some gateways; a model that hits the cap returns prose stopped mid-sentence, which every parser rejects but for the wrong stated reason, so the ceiling is raised explicitly rather than left to the backend default.

P5: response to rung. Apply exactly one perturbation to the base response, changing the minimum number of words and preserving everything else verbatim. Table 2 gives the six operators and the ten

Table 2: The six perturbation operators and the ten calls realizing them. Each call has a declared effect on the rung’s *gold relation graph* as well as on its text, which is what makes §4.3’s derivation well defined. The two O6 calls are factor-invariant by design: they are what the ORDER and CONTROL ladders exist to check.

Op	Calls	Effect on the text
O1 relabel	wrong_sense, direction_reversal, exhaustiveness_flip	rewrite the connective so the text draws a different sense, reverses the relation, or flips Contrast↔Alternative
O2 add conflict	inject_contradiction, spurious_relation	add one sentence directly contradicting a named claim, at least two sentences later; or assert a link between two claims the response kept separate
O3 break chain	break_chain	remove or negate the link realizing an edge so the chain no longer connects, keeping both endpoints
O4 drop relation	drop_relation	remove the connective realizing an edge, leaving the two claims as adjacent assertions
O5 resolve	add_resolution, remove_resolution	add or delete the holding sentence that settles a conceded tension
O6 reorder	shuffle_order, ordering_only	permute sentences at full, block or adjacent granularity; or exchange Precedence↔Succession preserving chronology — <i>no coupling changes</i>

Table 3: Prompt specification. Each prompt’s numbered-instruction count and required placeholder set are declared here and asserted by a test against the prompt strings, so a prompt cannot drift from the document describing it without the test failing. Instructions are counted by matching lines beginning at column one with a digit, period and space, so indented sub-items — P3’s per-sense glosses, P5’s per-call definitions — do not inflate the count. The validators carry no numbered instructions because their prompts specify an output schema rather than a procedure.

Stage	Role	Instructions	Required placeholders
P1	topic → question	9	TOPIC
P2	question → claims	7	QUESTION
P3	claims → relation plan	10	CLAIMS, QUESTION
P4	plan → base response	8	PLAN, QUESTION
P5	response → rung	5	OPERATOR, PLAN, RESPONSE
V1	recover relations	0	ATOMS, RESPONSE
V3	prose audit	0	RESPONSE
V4	claim coverage	0	ATOMS, RESPONSE

parameterized calls realizing them. The stage also emits a structured diff naming the operator, its target, the sentences changed and the relations added, removed or relabelled; length drift is capped at 15% except for the three calls that add or remove a sentence by definition.

The failure policy. What happens when a stage’s output is defective is a specification decision, and the policy is per stage with four outcomes. A **retry** handles a transient parse failure where the inputs are sound; because a byte-identical prompt re-sent to a temperature-zero backend cannot produce a different parse, retries are issued at successively higher sampling temperatures, which is what makes a retry budget worth more than one attempt. A **regenerate** handles an artifact that is wrong while its inputs are fine. A **drop edge** applies to V1 alone: an unrecovered relation must never be retained as gold, since gold the prose does not realize is exactly the defect that gate exists to catch. A **reject** applies when a family cannot carry its ranking claim, and stores the family with its reason rather than discarding it, so rejection rates remain auditable.

4.3 DERIVING EACH RUNG’S GOLD RELATION GRAPH

A rung’s gold graph is the base plan’s graph with that rung’s operator calls applied. Three mechanisms make this well defined, and each closes a way the derivation could otherwise produce a rung that does not match its own prose.

Table 4: Every acceptance threshold. V1 rates are computed over the *valid* planned relations only, since an invalid planned relation is one the prose is not required to draw soundly. The rightmost column is the committee decision rule for that gate.

Gate	Quantity	Threshold	Rule
V1	planned relations recovered with correct coupling	≥ 0.80	majority
V1	... and with correct sense	≥ 0.70	majority
V3	fluency, formality, organization (each, of 5)	≥ 4	majority
V3	leakage and hedging span lists	must be empty	majority
V4	claim coverage over gating statuses	$= 1.00$	majority
P3	claims selected	14–16	—
P3	relations planned (accepted band; 10 instructed)	8–12	—
P3	declared non-relations	4–6	—
P3	fraction of relations valid	0.55 ± 0.15	—
P3	false claims selected	≥ 2	—
P5	length drift	$\leq 15\%$	—

Target assignment. Each call in a rung is assigned a distinct eligible edge, so a rung composing two `drop_relation` calls drops two *different* edges. Eligibility is call-dependent: a `drop_relation` in a conflict ladder targets an unresolved conflict edge, an `add_resolution` a resolvable one. A `CONFLICT` plan therefore needs three unresolved conflict edges to support its full ladder — one for the resolution and two for the drops — and a plan with fewer is rejected at generation time rather than silently emitting a rung that duplicates its neighbour.

Derived-field re-derivation. When a call changes an edge’s sense, the coupling, directedness, ordering-only status and exhaustiveness are re-derived from the new sense through the compile map $\mathcal{C}$ rather than carried over, so a transformed edge cannot contradict $\mathcal{C}$.

The adjacency gate. A family is rejected when two *adjacent* rungs build the same edge set while the higher rung’s calls claim a change. Adjacency rather than parentage is the right comparison: every `CONFLICT` rung declares the base as its parent, so a parent-only check passes a rung that duplicates the rung immediately below it. The two factor-invariant calls of O6 are exempt, since they are *intended* to leave the edge set unchanged — which is exactly what makes them usable as invariance tests.

4.4 VALIDATION

An item is admitted only if an independent committee certifies it, and three properties of the arrangement matter more than the individual gates.

The committee excludes the generator. Validation is run by a panel of models, and the model that generated a family is removed from its own panel before voting. Without this, the dominant failure mode of an LLM-generated benchmark — the generator certifying its own idiosyncrasies as ground truth — is structurally unaddressed. Each gate is decided by *majority* of the remaining panel, and a facet counts as failed only when a majority objects, so one unusually strict auditor cannot reject a sound family.

Every threshold lives in one place. All acceptance thresholds are declared in Table 4 and nowhere else. A benchmark whose thresholds are scattered across the code applying them cannot be replicated from its description, nor re-run at a different operating point without a search for the constants.

Generation is rejected, never repaired. A family failing a gate is stored with its reason and discarded. No stage patches a defective artifact into an acceptable one, because a repaired item’s gold labels no longer describe a process we can state.

The three gates test different things. **V1**, the load-bearing one, hands an independent model the response and its claims and asks it to recover the relations the response *itself* draws, emitting for each ordered pair a sense, a coupling, a strength band and a resolved flag; admission requires ≥ 0.80 of valid planned relations returned with the correct coupling and ≥ 0.70 with the correct sense. This is what certifies that the gold graph describes the prose rather than the plan, and it is the gate that makes the GOLD relation source of §5.1 meaningful. **V3** audits the prose for fluency, formality and

Table 5: Composition of the released corpus. Every figure is computed from the shipped items rather than from the generation targets. *Edge-producing* excludes the 67 ordering-only relations, which compile to NONE and contribute no factor; *backward* counts edge-producing relations whose target precedes its source in assertion order, the relations a forward-only candidate policy cannot reach (§5.2).

Quantity	Value	Quantity	Value
Families	14	Gold relations, total	670
Items (rungs)	70	edge-producing	603
Claims	1,115	valid	408 (0.61)
Claims per item	15–16	backward	105
Words per item	609–698	Declared non-relations	342
Distinct topics	14	Ordering constraints per source	202
CONFLICT / CHAIN	8 / 2	C1 / C2 / C3 checks	112 / 30 / 45 + 15
ORDER / CONTROL	3 / 1	Generator / committee	1 / 3 models

organization, and requires the leakage and hedging span lists to be empty — leakage being any text that names the annotation scheme or the perturbation, which would let a scorer detect the rung without reading for coherence. V4 checks that every selected claim is actually asserted, gating on the missing and merged statuses, since a claim the response never makes cannot stand in a relation.

4.5 THE RELEASED CORPUS

Table 5 states what the corpus contains. It was produced by `claude-opus-5` and admitted by a three-model cross-vendor committee (`gpt-5.6-terra`, `gemini-3.1-pro`, `claude-sonnet-5`), none of which is the generator. The 14 families cover 14 distinct topics spanning all four domains — four Natural Sciences, four Engineering & Technology, three Humanities, three Social Sciences — so no single topic or domain carries a result. All four family types are represented, which is what allows §5.3 to report the increase and invariance axes separately; the 202 ordering constraints decompose as 112 over CONFLICT, 30 over CHAIN, 45 over ORDER and 15 over CONTROL.

Two properties of the corpus bound what can be concluded from it, and we state them here rather than in the limitations so that the results below are read correctly. It is LLM-generated throughout, so an idiosyncrasy shared by the generator and all three validators would be invisible to this construction. And at 70 items it is small in absolute terms: the per-source comparisons of §5.3 rest on 202 assertions, which is enough to separate sources differing by tens of points but not enough to resolve differences of a few.

5 EXPERIMENTS

The measure of §3 has two components that can fail independently: the probabilistic model, and the estimator that supplies its relation graph. An evaluation conflating them cannot say which to fix, so our experiments are built around one device — holding the readouts byte-identical and swapping only the relation graph (§5.1) — which attributes any shortfall to one component or the other by construction. We then ask three questions in turn: how accurate is the estimator (§5.2), how well does the model order responses given a correct graph (§5.3), and how does the measure compare against evidence-free alternatives (§5.5). §5.10 draws the answers together and §5.11 bounds them.

5.1 PROTOCOL

A *relation source* fixes where the relation graph comes from while holding claims, priors, readouts and scorer settings byte-identical — the central experimental device of the paper. GOLD takes each item’s own labelled relations, GOLD_VALID only those labelled valid, and a MINED source runs the estimator of §3.5 on the response text; because the readouts are identical across sources, any shortfall below gold is attributable to relation estimation and not to the probabilistic model. Mined relations are compared to gold at three nested levels, defined with the measurement in §5.2.

Table 6: Mined-versus-gold agreement, micro-averaged over the seventy items. The denominator is the 603 *edge-producing* gold relations (670 total, less the 67 ordering-only edges, which compile to NONE and contribute no factor). Agreement is scored at three increasingly strict levels: recovering the **pair**, then its **coupling** (which selects the factor table), then its **sense**. **By direction** decomposes coupling recall according to whether the gold relation runs forward or backward in assertion order — 105 of the 603 run backward, and no forward-only candidate set can reach them. **Errors** counts violations of the 342 declared-unrelated pairs (**non-rel**) and unordered pairs receiving two factors (**dup**). All figures are percentages except the three counts (edges, non-rel, dup). **Bold marks the best F1, not the best recall**: BIDIRECTIONAL wins every recall column under either miner and is nonetheless the weaker policy, which is the trade-off this table exists to show.

Policy	Edges	Pair			Coupling			Sense	Recall by dir.		Errors		
		P	R	F1	P	R	F1	F1	fwd	bwd	non-rel	dup	
<i>mined by llama-3.3-70b</i>													
WINDOWED	1,106	44.8	82.3	58.0	34.1	62.5	44.1	37.6	71.1	21.9	34	0	
BIDIRECTIONAL	3,172	28.5	91.7	43.5	17.9	79.3	29.3	24.7	79.9	76.2	119	1,231	
<i>mined by gpt-oss-120b</i>													
WINDOWED	1,094	46.0	83.4	59.3	40.2	73.0	51.9	47.0	82.1	29.5	38	0	
BIDIRECTIONAL	2,973	29.8	96.5	45.5	22.4	88.4	35.8	32.3	89.0	85.7	147	1,020	

The corpus is the one specified in §4: 14 families, 70 items, 1,115 claims and 670 gold relations, declaring 202 ordering constraints per source (Table 5). Mining is sampled once per cell rather than averaged, so we measure the resulting variability instead of estimating it: two *independent* runs of the same configurations moved every pair, coupling and sense F1 by at most 0.014, while the gold sources reproduced bit-for-bit. Per-source ladder agreement is the noisier statistic, moving by four to six of 202 assertions between those runs, so a single source’s percentage should not be read to the point and differences below roughly six assertions are not resolvable here. Exact values are by 2^{16} enumeration, corpus-scale ones by weighted mini-bucket (Liu & Ihler, 2011) at i-bound 6.

Five baseline families are chosen to fail in informative ways; four are implemented and share the LCS’s own decomposition, so each sees the same atoms and no retrieved evidence: ladder-blind controls (claim count, relation count, length) that catch a measure secretly tracking edit count or graph density (Steen & Markert, 2022); pairwise NLI with no global model (Manakul et al., 2023); ROSCOE’s Self-Consistency (Golovneva et al., 2023) in max, mean and symmetric variants; LLM judges — G-Eval (Liu et al., 2023) with the SummEval rubric (Fabbri et al., 2021) and a direct rating prompt, each a mean over five runs; and the three single-document DiscoScore metrics (Zhao et al., 2023) with an entity grid (Barzilay & Lapata, 2008).

5.2 RELATION ESTIMATION IS THE BOTTLENECK

We begin with the estimator, since every mined relation source inherits its errors. Measuring it against gold separates a model that scores a graph badly from an extractor that supplies the wrong graph.

How the numbers were obtained. Each source of Table 6 is one miner paired with one candidate-pair policy, run over the same seventy items. The policy fixes which claim pairs the miner is ever asked about, and the two differ structurally rather than by degree: WINDOWED proposes only *forward* arcs within an order radius of $w = 4$ (source before target in assertion order), yielding 1,693 candidate pairs over the corpus, while BIDIRECTIONAL proposes *both* arcs of every pair and applies no window, yielding 7,520 — a $4.4\times$ larger question set from identical inputs. Each surviving candidate then costs two LLM calls, Prompt A for the discourse sense and its coupling and Prompt B for the conditional strength (App. E); the measured rate is 1.65 calls per pair under WINDOWED and 1.41 under BIDIRECTIONAL, below two because Prompt B is skipped whenever Prompt A returns *none* — which it does for 593 and 4,447 pairs respectively. We then compare the surviving relations to the item’s gold graph at three *nested* levels, because they answer different questions: **pair** asks only whether the miner saw the two claims as related at all, isolating candidate selection from labelling; **coupling** additionally requires the correct Level-1 coupling, which is the headline because the coupling is what selects the factor table; and **sense** additionally requires the Level-2 discourse sense, which diagnoses taxonomy confusion rather than anything the MRF does. A mined edge

Table 7: **Left:** declared ordering constraints satisfied, by ladder type — 202 assertions per source. ORDER and CONTROL assert *invariance*, so a readout that trends there is tracking surface form rather than coherence; GOLD_VALID is near-flat on CONFLICT by construction, so its failures carry no information about the readouts and the GOLD source is what the constraints test. **Right:** each readout’s mean over the seventy items, with the standard deviation in grey. Averaging across a ladder’s rungs *deliberately collapses the ordering the ladder exists to test*, so the right-hand block is a **level comparison only** and carries no evidence about ordering — that is entirely the left block’s job; the spreads are given because a mean alone cannot say whether a source is stable or merely centred. The reified readout is omitted here; App. D compares all four.

Source / policy	Constraints satisfied					Mean readout value (sd)		
	Cfl	Chn	Ord	Ctl	%	LCS _{mm}	LCS _{cn}	LCS _{lp}
GOLD	89/112	18/30	45/45	15/15	82.7	0.795 $\pm$ 0.022	0.515 $\pm$ 0.135	0.410 $\pm$ 0.090
GOLD_VALID	37/112	12/30	45/45	15/15	54.0	0.794 $\pm$ 0.020	0.576 $\pm$ 0.151	0.485 $\pm$ 0.047
<i>mined by llama-3.3-70b</i>								
WINDOWED	53/112	17/30	0/45	0/15	34.7	0.796 $\pm$ 0.089	0.676 $\pm$ 0.239	0.224 $\pm$ 0.158
BIDIRECTIONAL	57/112	18/30	0/45	0/15	37.1	0.723 $\pm$ 0.171	0.542 $\pm$ 0.196	0.037 $\pm$ 0.026
<i>mined by gpt-oss-120b</i>								
WINDOWED	59/112	12/30	0/45	0/15	35.1	0.825 $\pm$ 0.074	0.500 $\pm$ 0.138	0.129 $\pm$ 0.096
BIDIRECTIONAL	53/112	17/30	0/45	0/15	34.7	0.808 $\pm$ 0.128	0.462 $\pm$ 0.095	0.033 $\pm$ 0.059

matches a gold one on the *ordered* pair when the coupling is asymmetric, where direction is part of the claim, and on the *unordered* pair when it is symmetric, where scoring direction would charge a policy’s arc ordering as a labelling error. Counts are micro-averaged over the seventy items against the 603 edge-producing gold relations.

What they mean. The decomposition matters more than any single F1, because it says *which* step fails. Take llama-3.3-70b under WINDOWED: of the 603 gold relations it finds 496 as pairs, types 377 of those with the right coupling, and gets the sense right on 321. So the estimator is not mostly failing to notice that two claims are related — it recovers four fifths of the pairs — it is failing to *type* what it found, losing 119 relations to the wrong coupling and 56 more to the wrong sense. That is what makes “extraction is the bottleneck” a claim about labelling rather than about recall, and it is why coupling F1, not pair F1, is the column to read. The second half of the story is precision. The same source emits 729 spurious couplings against its 377 correct ones, and under BIDIRECTIONAL that becomes 2,187 against 478. Because Eq. 1 is a *product* over relations, a spurious factor is not a diluted signal but an active one pulling marginals toward the prior, so precision is not tradeable against recall here in the way an F1 suggests; the 1,231 duplicate unordered pairs are the same defect seen structurally — two factors on one variable pair, which the model has no principled way to combine — and declared-non-relation violations rise from 34/342 to 119/342 in step. The two effects together explain the table’s central inversion. Forward-only mining is *structurally* capped, reaching 21.9–29.5% of backward-running gold relations against 76.2–85.7% for BIDIRECTIONAL, because 105 of the 603 relations run backward and no forward-only candidate set can emit them; but buying that reach costs more in precision than it returns in recall, so coupling F1 *falls* from 44.1 to 29.3. Reach and accuracy are separate axes, and on this corpus the wider policy is the weaker one. Finally, gpt-oss-120b is the better miner at every level — by 7.7 points of coupling F1, 9.5 of sense F1 and 7.6 of backward recall under WINDOWED, while emitting slightly *fewer* edges — but it is also the less informative one: 99% of its strength estimates sit at a clamp boundary against 59% interior for llama, and it returns just nine distinct strength values across 1,094 relations against llama’s 600. It decides confidently and supplies almost no graded uncertainty for the model to use.

5.3 THE MODEL ORDERS CORRECTLY WHEN THE GRAPH IS CORRECT

Holding the readouts fixed and varying only the graph now isolates the model’s own behaviour.

How a constraint is scored. A *relation source* fixes only where the relation graph comes from; the claims, the priors, the readouts and every scorer setting are byte-identical across sources (§5.1), so any difference between two rows of Table 7 is caused by the graph and nothing else. A single

constraint is one assertion about one readout on one pair of rungs: we score both rungs, take the difference, and ask whether it has the declared sign, counting a difference below 10^{-6} as flat. The assertions come in three classes — **C1** a strict increase between consecutive rungs (96 of them), **C2** a *predicted* decrease or exact flatness (64), and **C3** separation between the first and last rung (42) — for 202 per source, distributed over the four ladder types as the table’s columns show. The percentage is simply the fraction satisfied, and a missing score counts as a failure rather than being skipped. Because a constraint is a claim about a *difference*, the left-hand block of Table 7 is the only part that tests ordering; the mean readout values on the right average that difference away by construction and are reported for level comparison alone.

On gold relations the readouts satisfy 82.7% of these constraints — 167 of 202 — against 34.7–37.1% for every mined source. Since sources differ only in the graph, that 46–48 point gap is attributable to relation estimation alone, and its size is what §5.2 predicts: at 29–52% coupling F1 roughly half the factors in a mined network are wrong or missing, and a ladder whose consecutive rungs differ by one or two relations cannot survive that. This is the paper’s central empirical claim, and it is a claim about *where* the remaining problem is — the probabilistic model orders responses correctly when handed a correct graph, and extracting one is the open problem.

The per-ladder breakdown localizes the failure. On gold, ORDER and CONTROL are perfect (45/45, 15/15) since those rungs move no factor, CONFLICT is strong at 89/112, and CHAIN the weak ladder at 18/30. But *every mined source scores 0/45 on ORDER and 0/15 on CONTROL* — the most consequential number in the table, and not a scoring artefact: those ladders hold the claim set fixed and permute the prose, yet on one ORDER family the `gpt-oss` WINDOWED miner emitted 9, 16, 20, 20 and 17 edges across five rungs differing *only* in sentence order. LLM relation extraction is therefore not reorder-invariant, and any measure built on it inherits that sensitivity however sound its probabilistic layer. Mined sources are nonetheless *not* uniformly worse (CONFLICT 53–59 of 112, CHAIN 12–18 of 30), so a measure evaluated only on increase-type ladders would have called mined extraction far closer to gold than it is — precisely the false positive the CONTROL ladder was added to catch. Mined ladder agreement also does *not* track coupling F1 — `llama`’s BIDIRECTIONAL source has the worst coupling F1 (29.3) and the best agreement (37.1%) — so these sources are not separated by ladder agreement at all, and candidate policies should not be ranked on it.

5.4 WHAT THE BASELINES ARE

Table 8 sets the LCS against ten baseline columns, and reading it needs to know what each one computes. All are evidence-free and all consume the LCS’s own atom decomposition, so every row sees the same claims and no retrieved passages; the only thing varying across the table is how a measure turns those claims into a number. App. F gives the design rationale — why each family is chosen to fail informatively rather than to be a weak competitor — and what follows is the operational content.

Discourse representations (RC, LC, ENTITY GRAPH). Three single-document metrics from DiscoScore (Zhao et al., 2023). DiscoScore is presented as reference-based and most of its variants consume a reference a standalone response does not have; these three ignore the reference argument entirely, and the last *is* an entity grid (Barzilay & Lapata, 2008), so the classical baseline comes from the same implementation. RC is repeated noun mentions over distinct nouns; LC is nouns occurring in more than one sentence over distinct nouns; ENTITY GRAPH is the mean sentence-adjacency weight, an edge of weight $1/(j - i)$ wherever sentences i and j share a noun lemma. All three therefore reduce to noun-lemma repetition, with no representation of entailment, relation type or contradiction. A response can contradict itself in every sentence and still score at ceiling provided its nouns recur, which is what makes them a test of whether coherence is separable from *cohesion* rather than a weak competitor — and RC is nonetheless the strongest baseline in the table at 30/50.

Local contradiction detection (CONTRADICTION RATE). One minus the fraction of claim pairs the NLI extractor labels contradictory, in the spirit of SelfCheckGPT’s NLI variant (Manakul et al., 2023). It runs the *same* extractor over the *same* atoms as the LCS and is deliberately left ungrounded, so the only differences from our measure are relation *typing* and *propagation*: it isolates what joint inference adds over local detection. The prediction, which the table bears out at 12/50, is that it

does creditably where the planted defect is a *present* contradiction and poorly where it is a *missing* entailment, since a broken support chain leaves no contradictory pair to find.

Structured reasoning consistency (ROSCOE-SC, max and mean). ROSCOE’s Self-Consistency (Golovneva et al., 2023), $1 - \max_i \max_{j < i} p_{\text{contr}}(h_i, h_j)$. The two rows are one *ablation*, not two baselines, because the published measure differs from the LCS in three separable ways: it is a maximum, so a second conflict changes nothing; it is forward-only ($j < i$), so a relation running backward in claim order is unreachable, which is not hypothetical here — §3.5 measures 105 of the 603 scorable gold relations as both directed and backward; and it is untyped, so entailment, equivalence, exclusivity and co-necessity are invisible. Reporting a mean-aggregated variant alongside the faithful one says *which* difference is doing the work, and the answer is the aggregation. Under llama-3.3-70b the published max variant is exactly constant across all seventy items; under gpt-oss-120b it does vary, but every value it returns lies below 5×10^{-7} , so the maximum is saturated and the score is pinned at the floor either way. The mean variant, on the same items and the same pairwise probabilities, takes 57 and 63 distinct values respectively. ROSCOE targets step-by-step reasoning chains and a long-form response is not a chain, so we label it adapted.

LLM judges (JUDGE, G-Eval and direct rating). G-Eval’s coherence dimension (Liu et al., 2023) with the SummEval coherence rubric (Fabbri et al., 2021), and a direct “rate the logical coherence from 1 to 5” prompt as the no-rubric control. These are the practical incumbent and the comparison a reader will want. Each is reported as a mean over five runs, because the determinism of Rem. 2 is only an advantage next to a measurement of the judge’s instability, and because a spread wider than the gap under discussion bounds what the comparison can conclude. The ladder is a demanding test for them: consecutive rungs differ by a few words, so a judge must be sensitive at very small edit distances while staying flat where meaning is preserved.

Ladder-blind controls (CLAIM COUNT, LENGTH). Two quantities that ought *not* to track coherence: the atom count and the response length. They exist because Steen & Markert (2022) argue coherence measures must be checked against system-level confounders, and they catch a measure secretly tracking edit count or graph density. A control that *succeeds* is bad news for the corpus, not good news for the control.

Reading the vacuous rows. Three rows reach 20/20 on invariance without measuring anything, and the mechanism differs slightly between them. ENTITY GRAPH returns a single value on all seventy items — one distinct value, exactly constant — as does ROSCOE-SC (max) under llama. CLAIM COUNT is not constant: it takes two distinct values across the corpus (65 items at one, 5 at the other), but it never varies *within* a family, in 0 of 14. Since invariance is a within-family test, that is enough to pass it perfectly while scoring 0/50 on ordering. In all three cases the 20/20 is an artefact of a measure that cannot move where the test looks, which is why the final column of Table 8 takes the weaker of the two axes rather than their mean.

5.5 COMPARISON WITH EVIDENCE-FREE BASELINES, ON GOLD AND MINED GRAPHS

The gold relation source establishes what the model does with a correct graph; it does not establish that the exercise needs a probabilistic model at all. Answering that requires holding rival measures to the same ladders — measures that see the same atoms, consult no retrieved evidence, and return a single score.

The two axes, and why they are these sizes. A baseline returns one number per response, so only the *readout-independent* content of a constraint transfers to it. That leaves two axes. **Increase** takes the C1 consecutive-increase pairs together with the C3 endpoint separations over the ten increase-type families — eight CONFLICT and two CHAIN — deduplicated across readouts, since a rung pair is one assertion rather than one per readout. That is 50 assertions. The C2 class is excluded by construction: it predicts the internal behaviour of a *particular* readout, and a single-score baseline makes no claim a C2 constraint could test. **Invariance** takes every rung pair of the four invariance-type families — three ORDER and one CONTROL — and counts a pair satisfied when the score does not move, to 10^{-6} . That is 20 pairs. These 70 assertions are therefore a strict subset of the 202 of Table 7, and the percentages are *not* comparable between the two tables. The design is deliberately two-sided:

Table 8: **Everything on the same two axes.** Three conditions in one table: the LCS on *gold* relation graphs, the LCS on graphs *mined* from the response text, and the evidence-free *baselines*. Every cell is the assertion *count* with the same figure as a *percentage* in parentheses. **Increase** is the 50 readout-independent assertions (C1 + C3 over the ten increase-type families); **Invariance** the 20 rung pairs of the four invariance-type families, satisfied when the score is unchanged to 10^{-6} ; min takes the weaker of the two, as a lower bound. In block **b** the two Increase columns are the two *miners* and the rows are the two candidate policies, so a mined graph is identified by reading a row label together with a column head. In block **c** the five measures that call an LLM were likewise run once per backend and are reported per backend, while the model-free baselines are bit-identical across the two runs and span both columns. These percentages are *not* comparable to Table 7’s 34.7–37.1%, which runs over the fuller 202-assertion set including the C2 class. **The single most consequential column is Invariance:** every one of the twelve mined rows is 0/20, while gold is 20/20.

Measure	Increase (of 50)		Invariance	min
	llama-3.3	gpt-oss	(of 20)	
<i>a. LCS on GOLD relation graphs</i>				
LCS _{mm}	37 (74%)		20 (100%)	74
LCS _{lp}	36 (72%)		20 (100%)	72
LCS _{cn}	34 (68%)		20 (100%)	68
<i>b. LCS on MINED relation graphs — rows are the candidate policy, columns the miner</i>				
WIN / LCS _{mm}	20 (40%)	25 (50%)	0 (0%)	0
WIN / LCS _{lp}	35 (70%)	29 (58%)	0 (0%)	0
WIN / LCS _{cn}	23 (46%)	25 (50%)	0 (0%)	0
BID / LCS _{mm}	28 (56%)	21 (42%)	0 (0%)	0
BID / LCS _{lp}	25 (50%)	28 (56%)	0 (0%)	0
BID / LCS _{cn}	34 (68%)	24 (48%)	0 (0%)	0
<i>c. evidence-free baselines: model-free</i>				
RC (cohesion)	30 (60%)		16 (80%)	60
LC (cohesion)	28 (56%)		16 (80%)	56
LENGTH (control)	20 (40%)		12 (60%)	40
ENTITY GRAPH	0 (0%)		20 (100%) [†]	0
CLAIM COUNT (control)	0 (0%)		20 (100%) [†]	0
<i>evidence-free baselines: call an LLM</i>				
ROSCOE-SC (mean)	16 (32%)	17 (34%)	5–9 (25–45%)	25
CONTRADICTION RATE	12 (24%)	12 (24%)	8–9 (40–45%)	24
JUDGE, G-Eval	0 (0%)	21 (42%)	5–13 (25–65%)	0
JUDGE, direct rating	4 (8%)	18 (36%)	5–14 (25–70%)	8
ROSCOE-SC (max)	0 (0%)	0 (0%)	20 (100%) [†]	0

WIN/BID abbreviate the WINDOWED and BIDIRECTIONAL candidate policies. [†] vacuous: the measure never varies *within* a family, so it cannot fail an invariance pair (§5.4).

the first axis asks whether a measure can tell a more coherent response from a less coherent one, the second whether it stays put when nothing about coherence has changed, and a measure is useful only if it does both.

Gold graphs against the baselines (blocks a and c). Given a correct relation graph, the LCS is the only measure tested that passes both axes non-vacuously — 37/50 on ordering and 20/20 on invariance — and all three graded readouts sit above every evidence-free baseline in the weaker-axis column. The baselines fail in three distinguishable ways, and the differences between the failures are more informative than the ranking. The *cohesion* metrics are the strongest and the most instructive: RC reaches 30/50 and LC 28/50 while being defined over noun repetition alone, so they cannot separate a response that contradicts itself from one that does not when both repeat the same nouns — a neighbouring construct measured well, not this one measured badly. The *contradiction-based* measures detect conflict and then lose it in aggregation: ROSCOE-SC in its published maximum form scores 0/50, while the same pairwise probabilities aggregated by mean reach 16–17/50, so the loss is attributable to the aggregation rather than to the signal. The *judges* are the least stable: G-Eval moves from 0/50 to 21/50 and direct rating from 4/50 to 18/50 on a change of backend alone, on identical

prose and identical rubric, against a background in which every model-free baseline is bit-identical across those same two runs — the internal check that the harness is deterministic where it should be, and what makes the swing attributable to the judge rather than to the pipeline. Three baselines, finally, reach 20/20 on invariance without measuring anything, being unable to vary within a family at all, which is why the min column takes the weaker of the two axes rather than their mean.

Mined graphs (block b). On the increase axis the mined sources are more competitive than Table 7’s aggregate suggests, spanning 40–70%: two of the twelve rows clear the best evidence-free baseline’s 60% — llama/WINDOWED LCS_{lp} at 35/50 and llama/BIDIRECTIONAL LCS_{cn} at 34/50 — three more reach 56–58%, and the remaining seven fall below it. The invariance column is where the pipeline breaks: *every one of the twelve mined rows scores 0/20*, against 20/20 for all three gold rows. That single column is the substantive result of the table, and it is not a scoring artefact — those ladders hold the claim set fixed and permute the prose, so a measure reading a correct graph cannot move at all, and the mined graphs move because the miner is not reorder-invariant (§5.3). The gold rows’ advantage over the baselines is therefore concentrated entirely on the second axis, and the honest one-line summary of the pipeline as it stands is that mined relations buy competitive ordering and forfeit invariance completely.

One row invites a misreading worth forestalling. llama/WINDOWED LCS_{lp} at 35/50 sits just below the gold LCS_{lp} of 36/50, which suggests mined relations nearly match gold on this readout. They do not: that source’s per-rung sequences are not monotone — on one CONFLICT family the five rungs read 0.162, 0.169, 0.138, 0.125, 0.130 — so it satisfies the consecutive-increase and endpoint assertions the axis counts while wandering in between, and its mean value (0.224) is roughly half gold’s (0.410). The readout-independent subset is a coarse instrument by construction, which is exactly why it is paired with the invariance axis rather than reported alone.

Take-away. The gap between the LCS and the baselines is one of *kind* rather than of degree. No evidence-free baseline tested here both orders responses by coherence and stays invariant under meaning-preserving edits, and the measures that come closest on one axis are not the ones that come closest on the other: the cohesion metrics order tolerably and drift, the vacuous three never drift and never order. Given a correct graph the LCS does both, and the invariance half is structural rather than fitted — permuting sentences leaves the Markov network identical, so it is a property of the model rather than a result that happened to come out well. Given a *mined* graph it does neither reliably, and the table is arranged so that the two statements cannot be confused: what the model buys and what the current pipeline delivers are different claims, and only the first is settled.

From gold graphs to mined ones. The rows above hold the relation graph correct and ask whether a probabilistic model is needed at all. They do not say whether the end-to-end pipeline delivers, and that is a separate question with a different answer, which is why blocks **a** and **b** of Table 8 sit in the same units: only the provenance of the relation graph changes down the table, so a mined source can be read against gold and against a cohesion metric in a single glance.

5.6 REAL RESPONSES: THE FACTUALITY CORPORA

LoCoBench establishes what the model does when the ordering is known, at the cost of being synthetic. The complementary question — what the measure does on prose nobody constructed for it — needs a natural corpus, and answering it also exercises the two-stage composition of §3.6 end to end, which the ladder corpus never does because it supplies no evidence.

Corpus and cost. We draw a stratified sample of 250 responses, 50 from each of five long-form factuality datasets (Table 10), and run the full pipeline over each: retrieval, NLI, stage-one inference, relation mining under both candidate policies, all four readouts, and every baseline of §5.4. Sampling is not a convenience. The complete corpus is 1,137 items and 28,646 atoms, which prices at roughly 131K NLI calls plus 619K mining calls *per model*; the 250-item sample still took 25.6 and 61.4 hours of wall time for the two models reported here. The sample is a deterministic function of its seed and every item records its source file and line, so it can be regrown or extended without redoing completed work.

Stage one behaves, and one dataset can prove it. Table 11 reports the stage-one marginals. The BIO subset carries human supported / not-supported labels at the atom level, which makes it the only place in either corpus where an external ground truth exists, and the separation is clean under both backends: supported atoms average 0.93 and 0.89 against 0.58 and 0.52 for unsupported, a gap of +0.36 in both cases. That the two backends disagree on the absolute level while agreeing on the separation is the expected pattern — retrieval and NLI quality shift the scale, not the ordering — and it is what licenses using these marginals as coherence priors.

How Table 12 is computed. Each row is one measure scored on all 250 responses under one backend, and each cell reports three things about that set of 250 numbers: their mean, their standard deviation, and d , the count of *distinct* values the measure returned, computed at four decimal places. The last is the one that matters here, and it needs no ground truth to interpret: a measure that returns the same score for every response cannot rank any response above another, whatever that score is and however well it correlates with anything else. So d is a *precondition* for usefulness rather than a measure of accuracy, and it is the only question this corpus can settle, since it carries human labels at the atom level but no coherence gold standard. The LCS rows give all four readouts under both candidate policies, so the effect of the candidate set is read off directly rather than inferred.

The judges collapse where the measure does not. Both LLM judges saturate on natural prose. Under `llama-3.3-70b` the direct rating returns three distinct values across 250 responses, with a standard deviation of 0.000, and G-Eval five: they rate essentially everything at ceiling. ROSCOE-SC in its published maximum form collapses further still, to two distinct values under `gpt-oss-120b` — the max-aggregation failure of §5.4 appearing on real text rather than in an ablation. Against that, every LCS readout discriminates: across all sixteen readout-policy-backend cells the smallest count is 24, so even the least informative configuration separates more responses than the *best* judge configuration does at 16. The candidate policy shifts these counts but never collapses them — BIDIRECTIONAL returns fewer distinct values than WINDOWED in all eight comparisons, consistent with the extra spurious factors of §5.2 pulling marginals toward a common region, yet its worst cell still clears every judge. That the ordering is uniform across readouts, policies and backends is what makes it a property of the measure rather than of one configuration.

This corroborates §5.5’s judge result from the opposite direction: on constructed ladders the judges moved 42 points on a change of backend alone, and on natural prose they barely move at all. One measure, two corpora, two incompatible failure modes — and neither corpus alone would have shown both.

The two axes order the corpora differently. Factuality rises monotonically across the five datasets — 0.67, 0.90, 0.89, 0.96, 0.97 under `llama` — and LCS_{cn} does not follow it, running 0.92, 0.59, 0.82, 0.76, 0.93 over the same datasets in the same order (Table 13). ASKHIST is the sharpest case: its responses are well supported by retrieved evidence and among the least internally consistent in the sample. Both backends reproduce this ordering.

Separating agreement from inheritance: the uniform-prior ablation. Levels alone cannot settle whether coherence *agrees* with factuality or merely *inherited* it. Spearman correlation against the stage-one factuality score is +0.67 for LCS_{mm} and +0.35 for LCS_{cn} , and the first figure is partly expected by construction: §3.6 makes the factuality posteriors the coherence MRF’s priors, so a mean of posterior marginals must inherit some of them. The clean test is to re-run the same items with the prior slot filled by $\pi_i \equiv \frac{1}{2}$ instead, which removes stage one entirely while leaving the mined graph, the factor tables and the readouts untouched. Table 9 reports that condition on *both* complete runs — the same 250 responses, paired item by item, under both candidate policies — and it resolves the question in both directions.

First, inheritance is real but partial: LCS_{mm} ’s correlation with factuality falls from +0.665 to +0.398 once the priors are uniform under `llama-3.3-70b`, so roughly 40% of it came through the prior slot and the remainder is agreement between two independently computed quantities. Under `gpt-oss-120b` the same fall is far steeper, +0.674 to +0.169, so on that miner the association was inheritance almost in full; LCS_{re} behaves the same way on both (+0.393 $\rightarrow$ +0.020 and +0.407 $\rightarrow$ -0.027). The lesson is that how much of a coherence-factuality association is real rather than inherited is *miner-dependent*, and cannot be read off one backend. LCS_{lp} is the lone

Table 9: The uniform-prior ablation, on both complete runs. The same 250 responses, the same mined graphs and the same readouts are scored twice: once with the stage-one factuality posteriors in the prior slot, once with uniform priors $\pi_i \equiv \frac{1}{2}$, which removes stage one entirely. Items are paired, so each row is a within-item comparison of the two conditions. **d** counts the distinct values a readout takes over the 250 items — on a corpus with no ordering ground truth it is what separates a measure that can rank responses from one that cannot — and ρ is Spearman correlation against the stage-one factuality score, which is defined in both conditions because it always comes from the priors-on run. Two patterns hold in every block. Removing the priors *raises* **d**, in 14 of the 16 readout-policy cells (bold marks the larger of each pair; the exceptions are LCS_{re} under `gpt-oss-120b` on both policies). And it *lowers* ρ for the three readouts that inherit priors directly, sharply so for LCS_{mm} — which is the inheritance the ablation was built to quantify. LCS_{lp} is the exception in the other direction and is discussed in the text.

Miner	Policy	Readout	factuality priors			uniform priors		
			mean	d	ρ	mean	d	ρ
llama-3.3-70b	WIN	LCS_{mm}	0.976	90	+0.665	0.955	196	+0.398
		LCS_{cn}	0.803	74	+0.347	0.815	163	+0.212
		LCS_{lp}	0.685	45	+0.265	0.772	86	+0.339
		LCS_{re}	0.436	69	+0.393	0.451	86	+0.020
	BID	LCS_{mm}	0.980	79	+0.626	0.948	171	+0.369
		LCS_{cn}	0.804	59	+0.296	0.805	147	+0.184
		LCS_{lp}	0.661	35	+0.287	0.735	58	+0.297
		LCS_{re}	0.427	51	+0.381	0.441	63	+0.204
gpt-oss-120b	WIN	LCS_{mm}	0.948	148	+0.674	0.908	224	+0.169
		LCS_{cn}	0.711	134	+0.361	0.704	212	+0.080
		LCS_{lp}	0.671	85	+0.165	0.764	120	+0.188
		LCS_{re}	0.478	37	+0.407	0.491	32	-0.027
	BID	LCS_{mm}	0.960	111	+0.560	0.909	195	+0.103
		LCS_{cn}	0.721	87	+0.266	0.708	169	+0.046
		LCS_{lp}	0.620	58	+0.131	0.726	85	+0.173
		LCS_{re}	0.464	24	+0.380	0.481	15	+0.124

WIN is the forward-only WINDOWED candidate policy, BID the BIDIRECTIONAL one. All ρ are against the same quantity, the stage-one `factuality_score` of the priors-on run. One item of the 250 mines no relations under uniform priors and therefore scores at the prior throughout; it is retained rather than dropped.

readout whose correlation does *not* fall (+0.265 $\rightarrow$ +0.339 for llama); it is the one readout that is a normalized partition function *rather* than a function of the marginals, so it has no prior mass to inherit in the first place. Second, and less expected, removing the priors *sharpens* the measure rather than blunting it: distinct values rise from 90 to 196 of 250 for LCS_{mm} and from 74 to 163 for LCS_{cn} , because factuality posteriors on this corpus are themselves saturated near 0 and 1 (Table 11), and a saturated prior pins the posterior marginal regardless of what the relations say. The evidence-free configuration is therefore the more discriminative one here, which is worth stating plainly: the two-stage composition buys interpretability — a claim credited only when it is both supported and upheld — and costs dynamic range. Third, the per-dataset divergence of §5.6 survives the ablation: LCS_{cn} still takes its minimum on ASKHIST under uniform priors (0.620 against 0.589 with priors), so the finding that the two axes order the corpora differently does not depend on the prior handoff at all.

5.7 ABLATION: WHAT THE FACTUALITY STAGE CONTRIBUTES

§5.6 introduced the uniform-prior condition on llama-3.3-70b (Table 9) to separate agreement from inheritance. We now complete that ablation on the second miner, so every claim it supports is a two-model claim rather than a single-backend one, and report the parts that only a cross-model view exposes. The condition is the same: the prior slot is filled by $\pi_i = 0.5$, which removes stage one entirely — no retrieval, no claim-passage NLI, no factuality assessment of any kind — while leaving the mined graph, the factor tables and the readouts untouched. Everything else is held fixed: the same 250 responses, both candidate-pair policies, all four readouts, all twelve baselines, and 500

Table 10: The factuality sample: 50 responses from each of five long-form datasets, drawn with a fixed seed and stratified so that no dataset dominates. Atom counts are per response. Both models score the *same* 250 responses, so the corpus block is model-independent; what differs by model is the pipeline run over it (Tables 11 and 12).

Dataset	Domain	Items	Atoms/item	Relations/item*
BIO	biographies (human-labelled)	50	31.1	85.3
ASKHIST	history questions	50	23.2	87.5
BOOKS	book questions	50	24.0	89.0
ELI5	explanatory answers	50	21.1	76.0
LFOBJ	long-form objective	50	25.9	111.2
Total		250	25.1	89.8

* mined relations per response, llama-3.3-70b under the WINDOWED policy.

Table 11: Stage-one factuality marginals, per backend, over the 6,266 atoms of the sample. The right-hand block is the external check: on the BIO subset every atom carries a human supported / not-supported label, and **sep.** is the gap between the two means. Both backends separate the classes by the same margin while disagreeing on the absolute level.

Backend	All atoms		BIO: vs. human labels		
	mean	sd	supported	not supported	sep.
llama-3.3-70b	0.912	0.228	0.934	0.576	+0.357
gpt-oss-120b	0.853	0.289	0.887	0.523	+0.364
992 supported and 563 not-supported atoms.					

Table 12: Discriminative power on 250 real responses, by backend. **d** is the number of *distinct* values a measure returns over the 250 items (at four decimal places): a measure with *d* near 1 assigns nearly every response the same score and cannot rank them, whatever its mean. Values $d \leq 10$ are bold, marking collapse. The two judges and ROSCOE-SC (max) collapse; every LCS readout under either candidate policy and both cohesion metrics do not. All four readouts are given under both policies, so the effect of the candidate set can be read off directly: it shifts *d* but never collapses it. The four model-free baselines are identical across backends by construction and are given once.

Measure	llama-3.3-70b			gpt-oss-120b		
	mean	sd	d	mean	sd	d
<i>this work, WINDOWED candidate policy</i>						
LCS _{mm}	0.976	0.069	90	0.948	0.090	148
LCS _{lp}	0.685	0.450	45	0.671	0.433	85
LCS _{cn}	0.803	0.246	74	0.711	0.253	134
LCS _{re}	0.436	0.154	69	0.478	0.090	37
<i>this work, BIDIRECTIONAL candidate policy</i>						
LCS _{mm}	0.980	0.059	79	0.960	0.078	111
LCS _{lp}	0.661	0.465	35	0.620	0.460	58
LCS _{cn}	0.804	0.241	59	0.721	0.251	87
LCS _{re}	0.427	0.166	51	0.464	0.123	24
<i>baselines that call an LLM</i>						
CONTRADICTION RATE	0.997	0.008	48	0.996	0.009	54
ROSCOE-SC (mean)	0.996	0.011	73	0.994	0.012	69
ROSCOE-SC (max)	0.625	0.467	51	0.564	0.496	2
JUDGE, G-Eval	1.000	0.003	5	0.917	0.101	10
JUDGE, direct	1.000	0.000	3	0.829	0.162	16
<i>model-free baselines: identical across backends</i>						
RC (cohesion)		0.502	sd 0.180		$d = 224$	
LC (cohesion)		0.184	sd 0.064		$d = 209$	
ENTITY GRAPH		0.972	sd 0.119		$d = 17$	
LENGTH (control)		0.207	sd 0.046		$d = 136$	

Table 13: The two axes order the datasets differently. Every cell is a mean over the 50 items of a dataset, with the standard deviation in grey after it, WINDOWED policy. Factuality rises monotonically down each block under both backends. LCS_{cn} and LCS_{lp} do not follow it and instead agree with each other, both bottoming out on ASKHIST (bold): responses well supported by retrieved evidence and among the least internally consistent in the sample. LCS_{mm} tracks factuality closely, as its construction requires — its priors *are* the factuality posteriors — which is why the divergence is read off the other readouts. LCS_{re} is the least informative here, sitting near 0.5 throughout and, under gpt-oss-120b, varying by less than the standard deviation of any single dataset.

Backend	Dataset	factuality	LCS_{mm}	LCS_{cn}	LCS_{re}	LCS_{lp}
llama-3.3-70b	BIO	0.665 $\pm$ 0.211	0.945 $\pm$ 0.103	0.922 $\pm$ 0.171	0.498 $\pm$ 0.008	0.929 $\pm$ 0.232
	ASKHIST	0.897 $\pm$ 0.112	0.967 $\pm$ 0.078	0.589 $\pm$ 0.198	0.364 $\pm$ 0.205	0.287 $\pm$ 0.431
	BOOKS	0.891 $\pm$ 0.153	0.982 $\pm$ 0.069	0.820 $\pm$ 0.250	0.442 $\pm$ 0.145	0.721 $\pm$ 0.431
	ELI5	0.963 $\pm$ 0.065	0.995 $\pm$ 0.013	0.757 $\pm$ 0.252	0.405 $\pm$ 0.180	0.603 $\pm$ 0.472
	LFOBJ	0.974 $\pm$ 0.033	0.990 $\pm$ 0.020	0.927 $\pm$ 0.174	0.469 $\pm$ 0.107	0.885 $\pm$ 0.312
gpt-oss-120b	BIO	0.602 $\pm$ 0.194	0.877 $\pm$ 0.101	0.776 $\pm$ 0.238	0.472 $\pm$ 0.100	0.902 $\pm$ 0.234
	ASKHIST	0.769 $\pm$ 0.151	0.942 $\pm$ 0.091	0.522 $\pm$ 0.136	0.488 $\pm$ 0.046	0.377 $\pm$ 0.432
	BOOKS	0.789 $\pm$ 0.196	0.959 $\pm$ 0.110	0.729 $\pm$ 0.276	0.482 $\pm$ 0.075	0.658 $\pm$ 0.432
	ELI5	0.878 $\pm$ 0.138	0.982 $\pm$ 0.042	0.668 $\pm$ 0.238	0.481 $\pm$ 0.084	0.544 $\pm$ 0.471
	LFOBJ	0.935 $\pm$ 0.067	0.977 $\pm$ 0.033	0.859 $\pm$ 0.219	0.464 $\pm$ 0.123	0.875 $\pm$ 0.301

Table 14: The four arms: two miners $\times$ two prior sources, over the same 250 responses with the WINDOWED policy. The llama rows repeat Table 9’s WINDOWED block so the two miners can be read against each other; the gpt-oss rows are new. **d** counts the *distinct* values a readout takes across the corpus; on a corpus with no ordering ground truth it is the column that separates a measure which can rank responses from one which cannot. Bold marks the higher **d** within each miner-readout pair.

Miner	Priors	LCS_{mm}		LCS_{cn}		LCS_{lp}		LCS_{re}	
		mean	d	mean	d	mean	d	mean	d
llama-3.3-70b	factuality	0.9759	90	0.8032	74	0.6850	45	0.4358	69
	uniform	0.9547	196	0.8150	163	0.7719	86	0.4507	86
gpt-oss-120b	factuality	0.9475	148	0.7107	134	0.6713	85	0.4775	37
	uniform	0.9084	224	0.7037	212	0.7641	120	0.4914	32

of 500 runs completed without a failure. All four arms cover the *same items*, so every comparison below is paired.

The ablation is also substantially cheaper, which is itself informative about where the cost of the two-stage pipeline lies: $0.35\times$ the two-stage arm for llama-3.3-70b and $0.52\times$ for gpt-oss-120b. Stage one is most of the bill for the former and about half for the latter, whose reasoning-channel output makes relation mining the dominant expense instead.

Removing the priors *increases* the measure’s resolution. This is the ablation’s central and least expected result. In 7 of the 8 miner-readout cells of Table 14 the uniform-prior arm takes *more* distinct values than the two-stage arm, and the margins are large rather than marginal: 196 against 90 for LCS_{mm} under llama, 212 against 134 for LCS_{cn} under gpt-oss. The single exception is LCS_{re} under gpt-oss (32 against 37). Extending the count to the BIDIRECTIONAL policy as well (Table 9) gives 14 of 16 cells, the two exceptions being that same readout under that same miner on both policies — so the effect is general across readouts, policies and miners, with one localized counterexample rather than a scattering of them.

The mechanism is prior saturation, and it is measurable. The factuality posteriors are strongly ceiling-weighted — 82.7% of claims above 0.99 for llama and 72.1% for gpt-oss — so as unary factors they push the coherence marginals toward 1 and *compress* the range the readout can occupy. Replacing them with 0.5 returns that range to the relations. Consistent with this, the more saturated miner shows the larger relative gain. The implication for practice is not that priors are harmful but

that *saturated* priors cost resolution, and a factuality stage intended to prime a coherence model should be calibrated with that in mind rather than merely accurate.

The two arms are not interchangeable: the priors change *which* responses score low. Rank correlation between the arms over the 250 paired responses is +0.69 for LCS_{mm} , +0.78 for LCS_{cn} and +0.91 for LCS_{lp} — substantial but well short of the agreement that would make one arm a proxy for the other. The paired difference is systematic rather than noisy: on LCS_{mm} the uniform arm is lower on 87% of items, by -0.021 on average ($t = -5.3$). So the factuality stage does contribute, but what it contributes is a reordering of the low-scoring tail, not a shift in level.

The conflict separation owes nothing to the priors. §5.10 reports that the consistency readout separates responses whose mined graph contains a conflict edge from those whose does not. That result survives the ablation intact. With uniform priors, conflict-bearing responses score 0.519 against 0.976 for conflict-free ones under `llama` (a gap of +0.457, Welch $t = 32.9$) and 0.488 against 0.926 under `gpt-oss` (+0.438, $t = 32.2$) — statistically indistinguishable from the two-stage arm’s +0.489 and +0.438. The readout is therefore responding to relational structure it mined, not to factuality signal it inherited, and the ablation establishes this directly rather than by the correlational argument available before.

One association we cannot explain. The uniform arm consults no factuality signal whatsoever, yet its LCS_{mm} still correlates +0.379 with the two-stage arm’s mean prior ($t = 6.45$, $n = 250$). Mean-centring within dataset reduces this to +0.213 but does not remove it ($t = 3.43$), so a between-corpus offset is part of the story and not all of it. Per domain the association is significant on two of five (BIO +0.34, LFOBJ +0.34) and not on the other three (+0.06 to +0.14), and nothing we measured distinguishes those groups — notably *not* the spread of factuality within a domain, which is uncorrelated with the effect (+0.12 across the five). Controlling for contradiction count, relation count, claim count or mean relation strength does not account for it either. The plausible reading is a shared upstream cause: a response whose claims are well supported may also be one whose relations mine cleanly. We report the association and decline to assert the mechanism.

The baselines confirm the harness and sharpen the judge result. Under uniform priors every model-free baseline is bit-identical to its two-stage value — RC 0.5024, LC 0.1840, LENGTH 0.2066, claim count 0.3916 — as it must be, since none of them consults a prior; this is the internal check that the two runs differ only where intended. The LLM judges behave exactly as §5.10 describes and the ablation adds a cross-model contrast: `llama`’s direct rating takes 4 distinct values over 250 responses and its G-Eval 5, while `gpt-oss`’s take 17 and 12. As §5.10 notes, that apparent resolution is four rungs of a five-point scale plus seed-averaging noise, not measurement. Against either judge the LCS readouts take 120–224 values on the same responses.

Relation types and senses, and a structural limit made visible. Recording the mined relations themselves (not merely their count) lets the two candidate policies be compared by what they can *express* rather than only by the scores they produce. Two observations follow. First, the miners disagree markedly about what they find: over the same 250 responses `llama` mines 89.7 relations per response of which 3.9% are contradictions, while `gpt-oss` mines 58.8 of which 10.8% are — a third fewer relations at nearly three times the conflict rate, which is why its consistency mean is the lower of the two (0.704 against 0.815). Any claim about the absolute level of an LCS must therefore name its miner.

Second, and structurally: under the WINDOWED policy both miners emit **exactly zero** backward-running relations — 0 of 22,417 for `llama` and 0 of 14,695 for `gpt-oss` — against 46.7% and 45.3% backward under BIDIRECTIONAL. The consequence is visible at the sense level: EFFECT-CAUSE, the reversed form of CAUSE-EFFECT, appears 106 times forward and 0 backward under `llama`’s windowed policy but 73 forward and 303 backward under its bidirectional one (589/0 and 436/760 for `gpt-oss`). A forward-only window cannot express “B, because A” when B is asserted first. This is the same defect the ladder corpus measures as a backward-recall gap (§5.2), here confirmed on naturally occurring prose and localized to a particular discourse sense.

Table 15: The mined relation inventory over the same 250 responses, as a percentage of all relations the miner emitted under each policy. Level-1 couplings select the factor table; Level-2 senses compile to them through $\mathcal{C}$. WIN is the forward-only WINDOWED policy, BID the BIDIRECTIONAL one. Senses below 1% under both miners (ALTERNATIVE, PRECEDENCE, SUCCESSION) are omitted from the lower block; the last two are ordering-only and compile to no factor, which is why they should be rare by design.

	llama-3.3-70b		gpt-oss-120b	
	WIN	BID	WIN	BID
Relations mined	22,417	24,015	14,695	15,656
per response	89.7	96.1	58.8	62.6
contributing a factor	84.8	91.8	58.8	62.6
<i>Level-1 coupling (% of relations mined)</i>				
ENTAILMENT	90.1	88.0	84.9	84.2
CONTRADICTION	3.9	3.9	10.8	11.6
EQUIVALENCE	4.5	6.0	4.1	3.9
CO_NECESSITY	1.4	2.0	0.1	0.3
EXCLUSIVE	0.1	0.1	0.0	0.0
<i>Level-2 sense (% of relations mined)</i>				
Instantiation	48.0	49.3	32.9	34.6
Evidence	23.0	22.7	18.5	20.1
Cause-Effect	12.3	9.0	25.0	18.1
Restatement	4.3	5.7	4.1	3.9
Contrast	3.5	3.6	6.5	7.5
Condition	1.2	1.3	4.5	3.8
Disjunction	1.3	1.9	0.1	0.3
Effect-Cause	0.5	1.6	4.0	7.6
Concession	0.4	0.4	4.4	4.1
None	5.3	4.4	0.0	0.0

5.8 WHAT THE MINERS ACTUALLY EXTRACT: RELATION TYPES AND SENSES

The results so far compare miners by the scores they produce and, in §5.2, by their agreement with gold labels. Neither says what a miner *extracts*. Because the uniform-prior runs record every mined relation with its Level-1 coupling and Level-2 sense, not merely a count, the two miners can be compared by the content of their output over the same 250 responses. This turns out to separate them more sharply than any score does.

The two miners disagree about what coherence structure is, not merely how much of it there is. The volume difference is the visible one — llama-3.3-70b emits 89.7 relations per response against gpt-oss-120b’s 58.8 — but the composition difference matters more. llama places nearly half of everything it finds in a single sense, **Instantiation** (48.0%), which is the most generic link in the menu: one claim exemplifies another. gpt-oss distributes its output across the causal and adversative senses instead, with **Cause-Effect** at 25.0% against 12.3%, **Concession** at 4.4% against 0.4% and **Condition** at 4.5% against 1.2%. So the miner that finds 53% more relations finds a *narrower* variety of them, and a reader comparing two LCS values computed under different miners is not comparing two measurements of one quantity.

The contradiction gap is systematic, and it explains the score gap. Under gpt-oss 10.8% of mined relations are contradictions against llama’s 3.9%, and this is not driven by a few responses: computing the rate *per response* and pairing, gpt-oss is higher on 133 of 249 responses with a paired $t = +10.5$. This is the mechanism behind a result reported earlier without explanation. gpt-oss’s consistency mean is the lower of the two (0.704 against 0.815, Table 9) not because it scores the same graphs more harshly but because it *finds different graphs* — roughly two and a half times the conflict density on identical prose. Which miner is closer to correct is not settled by this corpus, which has no relation ground truth; §5.2 settles it on the corpus that does, where gpt-oss is the more accurate of the two at every level.

How the paired comparison is computed, and what it licenses. The statistic behind the preceding claim is a paired-samples t -test on the per-response contradiction rate, and the pairing is what makes the conclusion available. For each response i scored by both miners we form $r_i^{(m)}$, the fraction of that miner’s mined relations on that response labelled CONTRADICTION, and take the within-response difference $d_i = r_i^{(\text{GPT-oss})} - r_i^{(\text{LLAMA})}$. The statistic is $t = \bar{d}/(s_d/\sqrt{n})$ on $n - 1$ degrees of freedom, s_d being the sample standard deviation of the differences. On the factuality corpora $\bar{d} = +0.0593$, $s_d = 0.0892$ and $n = 249$ — one of the 250 responses mines no relations at all, leaving its rate undefined — giving $t = 0.0593/0.00565 = +10.50$ on 248 degrees of freedom.

Differencing within a response is the point. Length, topic, claim count and intrinsic difficulty are properties of the response, so they enter both terms and cancel; what survives is the miner effect alone. An unpaired comparison of the two rate distributions would instead have to overcome the between-response variance, which on this corpus is several times the effect being measured.

What the test licenses is a causal reading of a result reported earlier without one. `gpt-oss-120b`’s consistency mean is the lower of the two (0.704 against 0.815, Table 9), and two explanations were available: that it scores the same relational structure more harshly, or that it recovers different structure. The scoring model is byte-identical across the two arms — same factor tables, same readouts, same inference, same priors — so a difference in output must originate in the input graph, and the paired test shows the input graphs differ in precisely the way that moves this readout. CONTRADICTION is what the consistency readout penalizes, and `gpt-oss` finds roughly two and a half times the density of it.

The sign decomposition is the more robust form of the same evidence, and we prefer it. Of the 249 paired responses `gpt-oss` has the higher contradiction rate on 133 and `llama` on 4, with 112 ties — overwhelmingly responses where both miners find no contradiction at all. The two miners therefore agree exactly on 45% of the corpus and, on the 137 responses where they disagree, `gpt-oss` is higher 133 times. That asymmetry needs no distributional assumption, which matters because d is heavily zero-inflated and the normality premise behind t is correspondingly weak; at $n = 249$ the central limit theorem protects the mean difference, but the 133-to-4 split is the claim we would defend.

Two things the test does *not* establish. It says nothing about which miner is correct: on a corpus without relation ground truth a higher contradiction rate is equally consistent with better sensitivity and with over-firing, and §5.9 settles that on the corpus which does carry gold labels. And it does not license comparing absolute LCS values across miners — if two systems report different coherence scores for one response, the difference may lie entirely in what their extractors found, so a reported score must always name its extractor.

Not every mined relation becomes a factor, and the two miners differ there too. The senses that compile to Level-1 NONE — None, Precedence, Succession — record that two claims are related while coupling no truth values, so they contribute nothing to Eq. 1. Under `llama` these account for 5.4% of mined relations (1,208 of 22,417); under `gpt-oss`, one relation in total. The comparison that matters for the network is therefore 84.8 against 58.8 factors per response, narrowing `llama`’s apparent advantage from 53% to 44%. We note this because a relation count is the natural thing to report and is the wrong thing to compare: a miner can inflate it with links the model then ignores.

The sense inventory exposes the forward-only policy’s structural limit. §5.2 measures that limit against gold as a backward-recall gap. The sense distribution shows the same defect intrinsically, with no labels required. Effect-Cause is the reversed form of Cause-Effect: it asserts that the target causes the source, so realizing it requires an arc that runs backward in assertion order. Under WINDOWED both miners emit **exactly zero** backward-running relations — 0 of 22,417 and 0 of 14,695 — and Effect-Cause accordingly collapses to 0.5% and 4.0%, against 1.6% and 7.6% under BIDIRECTIONAL, where 46.7% and 45.3% of relations run backward. A forward-only candidate window cannot express “B, because A” when B is asserted first, whatever the prose says, and the resulting deficit is visible in the inventory of a single run rather than only in comparison with a gold graph. That is the more useful diagnostic, because it is available on any corpus.

A caveat on scope. This inventory is available for the uniform-prior runs only. The two-stage runs predate the change that records each relation’s labels rather than their count, so a priors-on

Table 16: The mined relation inventory on the ladder corpus (70 items), as a percentage of all relations emitted. *Gold* is the corpus’s own distribution over its 603 edge-producing relations — the column the factuality corpora have no counterpart for. The final block is gold-referenced recall of CONTRADICTION specifically: of the 164 gold contradictions, the fraction each source recovered with the correct coupling. WIN is the forward-only WINDOWED policy, BID the BIDIRECTIONAL one.

	llama-3.3-70b		gpt-oss-120b		
	WIN	BID	WIN	BID	Gold
Relations mined per item	15.8	45.3	15.6	42.5	8.6
<i>Level-1 coupling (% of relations mined)</i>					
ENTAILMENT	47.2	49.9	43.0	46.9	32.8
CONTRADICTION	22.5	22.4	27.1	28.3	27.2
EXCLUSIVE	15.1	12.6	12.4	10.2	12.9
CO_NECESSITY	8.0	10.2	6.2	6.6	11.6
EQUIVALENCE	7.1	5.0	11.2	8.0	15.4
<i>Gold-referenced CONTRADICTION recall (%)</i>					
of the 164 gold contradictions	48.2	56.7	70.1	76.2	—

comparison of the same kind would require re-mining. Nothing in the analysis above depends on the prior setting — mining precedes scoring and is unaffected by what fills the prior slot — so we expect it to transfer, but we have not measured that and do not claim it.

5.9 THE SAME INVENTORY ON LOCoBENCH, WHERE THE LABELS ADJUDICATE IT

The analysis of §5.8 can be repeated on the ladder corpus, and doing so adds the one thing the factuality corpora cannot supply: gold relations, against which a miner’s inventory can be scored rather than merely described. The four mined sources of §5.2 already recorded each relation’s coupling and sense, so no new mining was required.

The contradiction gap replicates, at almost exactly the same magnitude. Repeating the paired test of §5.8 on these 70 items gives $\bar{d} = +0.0577$ ($t = +3.98$, WIN) and $\bar{d} = +0.0574$ ($t = +5.26$, BID), against $+0.0593$ on the factuality corpora. Two corpora sharing no responses, no topics and no generation procedure yield the same 5.8-point difference between these two miners, which is better evidence that the effect is a property of the extractors than either corpus could furnish alone. The sign split is 46 to 21 (WIN) and 53 to 17 (BID), so the asymmetry is present but less extreme than on natural prose, where near-half the responses contained no conflict for either miner to disagree about.

Here the gold labels say which miner is right, and it is the one finding more. The corpus declares 27.2% of its edge-producing relations to be contradictions. gpt-oss-120b mines 27.1% under WINDOWED; llama-3.3-70b mines 22.5%. The higher rate is thus the better-calibrated one, and recall settles what a rate alone cannot: of the 164 gold contradictions gpt-oss recovers 70.1% against llama’s 48.2%, and 76.2% against 56.7% under BIDIRECTIONAL. The gap is *sensitivity* rather than *over-firing* — llama misses roughly half the conflicts its corpus contains. This resolves in gpt-oss’s favour the question §5.8 had to leave open, and it agrees with Table 6, where gpt-oss leads on every accuracy column at matched policy.

A caution against reading one corpus’s inventory through the other. The two inventories differ sharply, and the cause is the corpus rather than the miners. The ladder corpus is built to contain conflict — eight of its fourteen families vary conflict structure by construction — so both miners report contradiction rates above 22% here against below 11% on naturally occurring prose. Conversely Instantiation, which dominates the factuality inventory at 33–48%, accounts for under 10% of either miner’s output here. A miner’s inventory is therefore a joint property of the extractor and the text it reads, and a calibration established on one corpus does not transfer to the other. What does transfer is the *difference* between two miners on shared text, which is what both paired tests measure and why both are reported.

5.10 SUMMARY OF FINDINGS

We close by stating what the experiments establish, each finding paired with the measurement that supports it and with the boundary beyond which it does not hold.

(1) The probabilistic model orders coherence correctly when its input is correct. On gold relation graphs the readouts satisfy 167 of 202 declared constraints (82.7%), including 45/45 and 15/15 on the two ladder types where exact flatness is required (Table 7). Because sources differ *only* in where the relation graph comes from, this figure is a property of the scoring model in isolation.

(2) Relation extraction, not scoring, is the bottleneck — and the gap is large. The same readouts on mined graphs reach 34.7–37.1%, a 46–48 point shortfall attributable to estimation alone (Table 7). The magnitude is consistent with the estimator’s measured accuracy: at 29–52% coupling F1 (Table 6), roughly half the factors in a mined network are wrong or missing, and a ladder whose consecutive rungs differ by one or two relations cannot survive that. This is the paper’s central empirical claim, and it is a claim about *where* the remaining work lies.

(3) The invariance failure is the substantive one, and only a negative control could reveal it. Every mined source scores 0/45 on ORDER and 0/15 on CONTROL (Table 7), and 0/20 on the invariance axis (Table 8). This is real movement, not absent scores: on one ORDER family the gpt-oss WINDOWED miner emitted 9, 16, 20, 20 and 17 edges across five rungs differing *only* in sentence order. Invariance is structural in the model — permuting sentences changes neither which claims stand in which relation nor the Precedence $\leftrightarrow$ Succession compilation to NONE, so the Markov network is *literally identical* — so mining breaks the premise rather than the theorem. Notably, mined sources are *not* uniformly weak on the increase-type ladders (CONFLICT 53–59 of 112), so a corpus without CONTROL would have judged mined extraction far closer to gold than it is. That is the false positive the negative controls exist to catch.

(4) Candidate-pair policy trades recall against precision, and ladder agreement does not adjudicate it. Forward-only WINDOWED mining is structurally capped, recovering 21.9–29.5% of the 105 backward-running gold relations against 76.2–85.7% for BIDIRECTIONAL; admitting both directions triples the edges and halves coupling precision, inverting the F1 ordering (Table 6). Yet llama’s BIDIRECTIONAL source has the *worst* coupling F1 (29.3) and the *best* ladder agreement (37.1%), so these sources are not separated by ladder agreement and candidate policies should not be ranked on it.

(5) Given a correct graph the measure dominates every evidence-free baseline; the pipeline does not yet. All three graded readouts on gold relations sit above the best baseline’s 60%, and the LCS is the only measure tested that passes both the increase and invariance axes non-vacuously (Table 8). On mined graphs that dominance does not hold: two of twelve readout–policy combinations clear 60% on increase and none clears zero on invariance (Table 8). The distinction between what the model buys and what the current pipeline delivers is the one we most want not to be blurred.

(6) The baselines fail in three distinct ways, which is why the gap is one of kind. *Cohesion metrics* are the strongest and most instructive, reaching 60% and 56% while being unable to separate a response that contradicts itself from one that does not when both repeat the same nouns — a neighbouring construct, not this one measured badly. *Contradiction-based metrics* detect conflict and then lose it in aggregation, tracking response size. *LLM judges* fail in two ways at once, and it takes both corpora to see it: on the constructed ladders G-Eval moves 0/50 $\rightarrow$ 21/50 on a change of judge alone, on identical prose and rubric, while every model-free baseline is bit-identical across the same two runs; on real responses the same judges saturate instead, returning three and five distinct scores across 250 items (Table 12). And three baselines reach 20/20 invariance *vacuously*, being constant across all seventy items — trivially invariant and trivially unable to order anything, which is why the summary column takes the weaker axis.

(7) On real responses the measure discriminates where the incumbents do not. Over 250 responses from five long-form factuality datasets, all four readouts under both candidate policies return between 37 and 148 distinct values, while the two LLM judges return between 3 and 16 and ROSCOE-SC in its published maximum form returns as few as two (Table 12). A measure that assigns nearly every response the same score cannot rank responses whatever its mean or its reported correlation, so this is a precondition for usefulness rather than a margin of victory — and it is the one comparison the ladder corpus could not make, since there the judges were unstable rather than saturated.

(8) Factuality and coherence order real corpora differently, and the ablation says the separation is real. Across the five datasets, stage-one factuality rises monotonically while LCS_{cn} does not follow it; ASKHIST is well supported by retrieved evidence and among the least internally consistent in the sample, and both backends reproduce the ordering (Table 13). This is the orthogonality claim of §1 appearing in data rather than in construction. The obvious objection — that LCS_{mm} correlates with factuality at $\rho = +0.67$ simply because its priors *are* the factuality posteriors — is answerable rather than merely acknowledged: rescoring the same 250 items at $\pi_i \equiv \frac{1}{2}$ drops that correlation to +0.398 under llama-3.3-70b and to +0.169 under gpt-oss-120b, and leaves the per-dataset ordering intact under both (Table 9). Inheritance therefore accounts for roughly 40% of the association on one miner and nearly all of it on the other, and none of the ordering result. The same ablation carries an unwelcome finding in the other direction, reported in (9).

(9) The two-stage composition costs dynamic range, on both miners, and we would not have found that without the ablation. Removing the priors *raises* discriminative power sharply and in 7 of the 8 miner-readout cells: distinct values go from 90 to 196 of 250 for LCS_{mm} and 74 to 163 for LCS_{cn} under llama-3.3-70b, and from 148 to 224 and 134 to 212 under gpt-oss-120b (Tables 9 and 14); the lone exception is LCS_{re} under gpt-oss. The cause is measurable: the factuality posteriors are ceiling-weighted — 82.7% of claims above 0.99 for llama, 72.1% for gpt-oss — and a saturated prior pins the posterior marginal whatever the relations say, so the more saturated miner shows the larger gain. The composition of §3.6 therefore buys interpretability, in that a claim is credited only when it is both externally supported and internally upheld, and pays for it in resolution. On this corpus the evidence-free configuration is the more discriminative one, and a practitioner choosing between them should be told so. The finding also refines the design advice: the cost comes from prior *saturation* rather than from composition as such, so a better-calibrated stage one would pay less of it.

(10) The conflict separation owes nothing to the priors. With the prior slot uniform, the consistency readout still separates responses whose mined graph contains a conflict edge from those whose does not, by +0.457 under llama (Welch $t = 32.9$) and +0.438 under gpt-oss ($t = 32.2$) — indistinguishable from the two-stage arms’ +0.489 and +0.438 (§5.7). The readout is responding to relational structure it mined rather than to factuality it inherited, and the ablation shows this directly rather than by correlation.

(11) A forward-only candidate policy cannot express a backward discourse relation, and the sense labels show where. Under WINDOWED both miners emit *exactly zero* backward-running relations — 0 of 22,417 and 0 of 14,695 — against 46.7% and 45.3% under BIDIRECTIONAL. The effect localizes to a sense: EFFECT-CAUSE appears 106 times forward and 0 backward under llama’s windowed policy, 73 forward and 303 backward under its bidirectional one (§5.7). This is the ladder corpus’s backward-recall gap (finding 4) confirmed on naturally occurring prose.

(12) A property no scalar baseline offers: the score decomposes. As a mean of per-claim marginals, the count of claims dragged below their own prior localizes an incoherence to the claims responsible. A rubric scalar, however well correlated, cannot do this.

Lessons learned. Three of these findings only became visible because the evaluation used two corpora with opposite weaknesses, and that is the methodological lesson we would carry to the next measure of this kind. A constructed corpus can fix the ordering and therefore localize a failure — it is what tells us that scoring is sound and extraction is not (findings 1–2), and only its negative controls exposed the invariance collapse that the increase-type ladders alone would have hidden (finding 3). What it cannot do is say whether a measure is useful on text nobody built for it. A natural corpus answers that and nothing else: it has no coherence ground truth, so it cannot rank measures by correctness, but it does reveal that the practical incumbent assigns almost every real response the same score (finding 7) — a failure invisible on ladders, where the judges were unstable rather than flat. Neither corpus alone would have supported both conclusions, and a paper reporting only one of them would have been comfortably wrong in a different direction each time.

Three further lessons are narrower but cost us the most time. First, a readout’s dynamic range is a property of its factor tables, not of its definition: the no-priors tables collapse the measure’s range by more than half before any relation is mined (§3.3), and no amount of tuning downstream recovers it. Second, an aggregation choice can dominate a modelling choice — ROSCOE-SC differs from our measure in three separable ways, and the maximum is the one that matters, saturating to two distinct

values on real prose (§5.4). Third, a component added for interpretability can quietly cost accuracy elsewhere, and only an ablation shows it: filling the prior slot with factuality posteriors halves the measure’s distinct-value count on real responses (Table 9), a cost we would have reported as a feature had we not rescored the corpus without it. All three are the kind of thing an ablation finds and a leaderboard hides.

Taken together: the coherence model is sound and its guarantees are structural rather than fitted; the benchmark localizes the open problem in extraction rather than in scoring, with the negative-control ladders making that localization trustworthy; and on real responses the measure discriminates where the incumbent judges do not. The bounds are equally plain. The ladder corpus is LLM-generated and small (70 items, 202 assertions per source), so differences of a few assertions are not resolvable; the factuality sample has no coherence gold standard, so it bounds usefulness rather than correctness. §5.11 bounds the rest.

5.11 LIMITATIONS

Four of the results above are strong enough to be worth bounding precisely. The invariance guarantee holds only given a correct graph, the pairwise factorization excludes a class of real incoherence, two constants in the design are set by hand rather than derived, and the natural-corpus results bound usefulness rather than correctness. On that last point: the factuality sample carries human labels at the *atom* level only, so it supports the discriminative-power comparison of §5.6 and the stage-one check, but it cannot rank measures by agreement with human coherence judgements, which no corpus we know of supplies at this response length. The uniform-prior ($\pi_i \equiv \frac{1}{2}$) condition *has* now been run on that sample (Table 9), which settles the inheritance question, but on one backend only: the `gpt-oss-120b` counterpart is incomplete, so the ablation’s two findings — that inheritance accounts for about half of LCS_{mm} ’s correlation with factuality, and that uniform priors are the more discriminative configuration — rest on a single miner and should be read as such until the second run finishes. Both remaining gaps are additions to the corpus rather than to the model.

Proposition 2 (Jointly inconsistent, pairwise consistent). *Let $\mathbb{P}(\mathbf{a})$ factorize as in Eq. 1 over a_1, a_2, a_3 with non-negative factors. If $\mathbb{P}(a_1=1, a_2=1, a_3=1) = 0$ then $\mathbb{P}(a_i=1, a_j=1) = 0$ for some pair $i \neq j$, so no member of the family represents “the three claims cannot all hold, but any two can” (proof in App. H).*

Genuinely k -ary conflicts are therefore unrepresentable rather than merely approximated badly, and this is not hypothetical: a planted arithmetic inconsistency — 356 teams forming 32 conferences of exactly 10 members, where no two claims conflict but $32 \times 10 \neq 356$ requires all three — yields *zero* mined conflict edges and is reported as coherent, unchanged under `ALL_PAIRS`. Transitivity is likewise not enforced. Both defects are expressible as first-order rules in the Markov logic view of App. I, so Rem. 1’s closure guarantee is exactly what buys tractability and what costs us this defect. Two constants remain hand-set ($\lambda = 0.45$ and the resolver heuristic), and production inference is weighted mini-bucket at i-bound 6 against 2^n enumeration ($n \leq 20$), an approximation we have not characterized. At 52% coupling F1 at best roughly half a mined network’s factors are wrong or missing, so mined-source numbers measure current extraction quality rather than the measure’s ceiling; human scoring of the invariance rungs, human correlation generally, and judge dependence (42 points between judges) remain open, and the corpus is LLM-generated throughout.

6 CONCLUSION

Coherence is a property of a joint distribution over what a response asserts, readable off a Markov random field whose factors are the relations the response itself draws. Three things follow. First, the factor tables matter: those right for grounding a claim against evidence charge a premise for being a premise, so a coherence measure built on them spends its dynamic range in the wrong place (Prop. 1). Second, the relation vocabulary can be rich at the annotation layer and must be narrow at the inferential one: twelve senses compiling onto five couplings keeps the model a closed pairwise family while leaving the labels interpretable. Third, not every reading of “how coherent is this?” is admissible: two of four candidates are non-monotone in conflict resolution because they measure whether a conflict is active rather than what should be believed, and a fifth is maximized by asserting no relations at all. Empirically the probabilistic layer is adequate and grounded extraction is not, which is where the open problem lies.

AI USE STATEMENT

Generative AI is both an object of study and a component of the artifacts released here, and we disclose both roles. The relation estimator of §3.5 is LLM-based by design: its prompts, its logprob-derived confidences and its failure modes are contributions of the paper. The evaluation corpus of §5.1 is LLM-generated throughout, including its gold relation labels, and validated by an independent LLM committee; §5.11 states the consequences for treating it as ground truth. We additionally used AI assistants for code implementation and for editorial assistance on this manuscript. All LLM-generated code was reviewed by the authors and is covered by a test suite that pins the exact numerical values reported here; every number in §5 was traced to a stored experiment record. The research questions, the taxonomy, the readout definitions, the propositions and their proofs are the authors’ own. We take responsibility for the final content of this work.

ETHICS STATEMENT

This work involves no human subjects and no personally identifying data. The fixtures include a summary of a published appellate judgment, used as a worked example of ordering-dependent coherence; it is a matter of public record and no private individual is the subject of analysis. The principal ethical risk is misreading: a coherence score is not a truth score, and a response can be perfectly coherent and entirely false. That is precisely why §3.6 keeps the two axes separable, and we caution against deploying LCS alone as a factuality filter — a fluent, internally consistent fabrication is exactly the artifact it scores highest. The released fixtures and corpus deliberately contain false claims, labelled as such, and should not be redistributed without those labels.

REPRODUCIBILITY STATEMENT

The model is fully specified in the paper: Def. 1 gives the joint, §3.3 the potentials, Def. 2 the penalized sets, §3.2 the compile map, and Eq. 4 the selected readout, with the three alternatives and the auxiliary-variable constructions two of them require in App. D. Algorithms 2 and 1 give candidate selection and the end-to-end procedure. App. K reproduces all four estimation prompts verbatim, including the trailing token boundary the surrogate readout depends on. The exact values in §3.7 and App. G are computed by explicit 2^n enumeration — 2^5 for the five-claim pair, 2^{16} for the recall report — and are pinned by regression tests in the released code, so they are reproducible without LLM access. The k -ary representability figures of App. H are likewise exact by 2^3 enumeration, with the seeded random search and the published table both reproduced by a released script. §5 names every model used and states the sampling regime and its measured variance. Code, fixtures and corpus are released.

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A LOCoBENCH AT A GLANCE

This section is a self-contained summary of the benchmark, for a reader who wants the whole of it in one page; §4 gives the full treatment, with the tables and thresholds this section only names.

The problem it solves. Evaluating a coherence measure requires knowing which of two responses is more coherent — exactly what no natural corpus supplies, since human judgements are expensive, disagree, and do not say *why* one text is worse. Perturbing a document and assuming the perturbation hurt is weaker than it looks: the edit lands on raw text and its effect on the thing being measured goes unverified. LoCoBench inverts that. Perturbations apply to a *declared relation graph* rather than to prose, so each edit’s intended effect is known per item before any text exists, and the ordering is a property of the construction rather than an assumption about it.

The unit of evaluation. The unit is not a response but a *family*: five responses (*rungs*) over one fixed claim set and one fixed topic, differing along exactly one structural dimension (Def. 3). Holding the claim set fixed is what makes an ordering attributable to coherence rather than to content: every rung asserts the same claims, so every per-claim factuality verdict is identical across the family and cannot explain any score difference. Four family types divide the work (Table 1). CONFLICT varies conflict structure and CHAIN breaks and repairs an entailment chain; both assert a strictly *increasing* score, and the pair matters because a measure built on contradiction detection alone passes the first and fails the second, where the defect is a *missing* relation rather than a present conflict. ORDER and CONTROL apply only meaning-preserving edits and so assert *invariance* — the negative controls, without which no corpus can tell a coherence measure from an edit-count proxy. Constraints come in three classes: a strict increase between consecutive rungs, a *predicted* decrease or flatness, and endpoint separation. A missing score fails rather than passes.

How a family is built. Generation is a five-stage prompted pipeline (Fig. 3): a topic becomes a question (P1), the question becomes 26 atomic claims (P2), a subset of them a relation plan (P3), the plan a base response (P4), and six perturbation operators realized by ten calls (Table 2) turn that base into the four remaining rungs (P5). The surplus at P2 is deliberate — selecting from more claims than are needed lets the plan choose claims supporting the required structure rather than forcing relations onto whatever claims exist. Every prompt is fixed, versioned, and has its instruction count and placeholder contract asserted by a test against the prompt string (Table 3), so a defect is attributable to one stage and repairable there.

How a family earns admission. Three validators gate it, run by a committee that *excludes* the generating model (Table 4). **V1** is the load-bearing one: an independent model is shown the prose and asked to recover the relation graph, and the family is admitted only if enough planned relations come back with the right coupling and sense. This is what makes the gold graph a claim about the text rather than about the generator’s intentions: a plan the prose does not realize fails here. **V3** audits fluency, formality and organization and rejects any leakage of the construction into the text; **V4** checks that every planned claim is present. Families are admitted or rejected whole, so no rung is scored against a ladder whose siblings were discarded.

What shipped. Fourteen families, seventy items, 1,115 claims at 15–16 and 609–698 words per item, over fourteen topics spanning four domains (four Natural Sciences, four Engineering & Technology, three Humanities, three Social Sciences), so no topic or domain carries a result. The families declare 670 gold relations, 603 of them edge-producing — the other 67 are ordering-only senses compiling to NONE — and 342 declared *non*-relations, which matter because a measure must be tested on pairs it should leave alone as well as on pairs it should link. Of the 670, 408 (0.61) are marked valid; and 105 of the 603 run *backward*, their target preceding their source in assertion order, which is what makes candidate-pair policy a substantive choice rather than a tuning detail. Generation was by `claude-opus-5`, admission by a cross-vendor committee of `gpt-5.6-terra`, `gemini-3.1-pro` and `claude-sonnet-5`. Together the families yield 202 ordering constraints per scored source (Table 5).

What it can and cannot show. Fixing the claim set isolates coherence from factuality, and acting on a declared graph lets the benchmark say *which* defect a measure misses rather than only that it

scored poorly — which is how §5.3 separates a scoring failure from an extraction failure. Two limits bound it: the corpus is LLM-generated throughout, so an idiosyncrasy shared by the generator and all three validators would be invisible here, and at seventy items its 202 assertions separate sources differing by tens of points but not by a few. Both are reasons to read LoCoBench as a diagnostic instrument rather than a leaderboard.

B STAGE ONE: PROBABILITIES FOR INDIVIDUAL CLAIMS

The coherence model consumes claim-level probabilities and reuses the inference machinery that produces them. That stage is FactReasoner (Marinescu et al., 2025); the account here is a summary sufficient to make the composition of §3.6 precise, and it also fixes the label-span estimator whose construction the relation miner of §3.5 reuses.

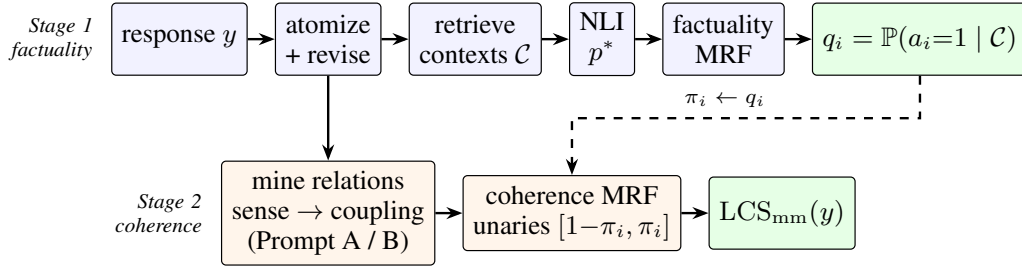


Figure 4: The two-stage pipeline. Stage 1 (§B) is FactReasoner (Marinescu et al., 2025): it grounds each atom against retrieved evidence and returns a posterior marginal q_i . Stage 2 (§3) builds a second MRF over the atoms *alone* and adds one pairwise factor per mined atom–atom relation; the mined relation already carries its Level-1 coupling, so no separate compile step intervenes. The dashed arrow is the only coupling between the stages, and carries the composition $\pi_i \leftarrow q_i$ into the $\phi_i = [1 - \pi_i, \pi_i]$ of Def. 1. Atomization is shared, so the response is decomposed once and both stages score the same claim set; setting $\pi_i \equiv \frac{1}{2}$ instead recovers a pure coherence score that consults no external evidence (Rem. 3).

B.1 DECOMPOSITION, RETRIEVAL, AND RELATION EXTRACTION

A response y is decomposed by an LLM *atomizer* into atomic claims $\mathcal{A} = \{a_1, \dots, a_n\}$, each a single independently verifiable proposition. A *reviser* then decontextualizes each one, resolving pronouns and anaphora so that the claim is checkable outside its original sentence; near-duplicates are collapsed. For each claim a retriever returns supporting passages, which are deduplicated into an evidence set $\mathcal{C} = \{c_1, \dots, c_m\}$.

An LLM-based NLI extractor then labels admitted pairs. For a claim–evidence pair the label is drawn from $\{\text{entailment}, \text{contradiction}, \text{neutral}\}$; for an evidence–evidence pair both directions are scored and reconciled, with entailment in both directions relabelled *equivalence*. Neutral verdicts are dropped rather than encoded: a claim about which the evidence is silent contributes no factor and is left at its prior, which is what makes the factor set of §3 contain only the relations someone actually asserted.

B.2 READING A PROBABILITY OFF THE GENERATION

Each retained relation carries a probability p^* . Rather than asking the model to verbalize a confidence, we read it from the distribution over the tokens it actually emitted. The extractor is prompted to end with a bracketed label; let T be the set of generated tokens overlapping the span of that label in the decoded text. Then

$$p^* = \exp\left(\frac{1}{|T|} \sum_{t \in T} \log p_t\right), \quad (3)$$

the per-token geometric-mean probability of the label the model produced. Three details of this construction are load-bearing, and we record them because each was a source of silent error.

Span alignment. The probability is measured over exactly the span the label *parser* reports, reconstructing the decoded string from the token list and tracking each token’s character offsets. Measuring the same span the label came from makes the label and its probability structurally unable to disagree. Requiring standalone bracket-delimiter tokens instead does not work: tokenizers routinely fuse [,] and " into subwords with adjacent characters, so a delimiter-based rule silently fails on some backends and not others.

Token-array conventions. An earlier version of this code stripped the final entry of the logprob array as an end-of-sequence token. OpenAI-compatible and vLLM logprob arrays contain only emitted content tokens — the stop is signalled out of band — so dropping the last entry deleted a real content token, and for a bracketed label that token is the one closing the label whose confidence is being measured.

Degenerate cases. When the logprob array is empty, no label span is found, or no token aligns to it, the estimator returns 0.5 rather than 0.0. Returning zero would be read downstream as a *degenerate*, *impossible* relation for a label the model in fact generated, which is a stronger claim than “we could not measure the confidence”.

Backends without logprobs. Some serving stacks expose no token logprobs at all. There p^* instead comes from SIMBA-UQ (Bhattacharjya et al., 2025), which samples the extractor repeatedly across a temperature schedule, builds a pairwise similarity matrix over the samples, and aggregates each sample’s similarity row into a self-consistency confidence. The winning sample supplies the label and its aggregated confidence supplies p^* , so the downstream model is unchanged.

B.3 THE FACTUALITY NETWORK AND ITS VARIANTS

The labelled graph becomes a pairwise MRF with one binary variable per claim and per evidence passage. Unary factors are $\phi_X(x) = [1 - \pi_X, \pi_X]$ with π_X the prior on X ; each retained relation contributes one pairwise factor from Table 20 below, oriented so the evidence passage is the premise. Marginals $\mathbb{P}(A_i=1)$ are computed by weighted mini-bucket elimination (Liu & Ihler, 2011) with i-bound 6, and a claim is reported supported when $\mathbb{P}(A_i=1) > \frac{1}{2}$.

Three graph variants differ in which factors are admitted: FR1 connects each claim only to the passages retrieved for it; FR2 deduplicates passages and scores every claim against all of them; FR3 adds evidence–evidence factors. We use FR2. Because scoring every claim against every passage is $O(nm)$ LLM calls, a candidate-pair policy can restrict the pairs actually scored; the calibration of that policy is instructive for the analogous choice in §3.5. Replaying policies against recorded verdicts on a 20-claim narrative with 60 passages, a provenance-restricted policy with an embedding similarity gate holds recall at 1.000 through a threshold of 0.20, slipping to 0.985 at 0.30 and 0.833 at 0.40. Substituting token-Jaccard similarity for sentence embeddings, however, loses 22 of 72 non-neutral relations and moves the factuality score by 0.05 — and *no* threshold recovers full recall, even 0.05. Cheap surrogates for semantic similarity are not safe here, and the saving is workload-dependent rather than constant: roughly $5\times$ on unrelated subtopics but $1.2\times$ on a narrative where every claim shares characters.

Conventions. We use the with-priors factor tables (Table 20) throughout, with $\pi_C = 0.9$ for evidence and $\pi_A = 0.5$ for claims. The published factuality model uses the no-priors tables and a higher evidence prior. The difference is immaterial to the coherence results below, which set claim priors explicitly, but the convention must be stated because the two table families are not interchangeable — that is the subject of §3.3.

C THE RELATION TAXONOMY IN FULL

§3.2 states the two-level taxonomy compactly. This appendix grounds every entry in a realistic response excerpt, gives the compile map with its glosses and flags, and works one coupling through by hand.

Table 17: One realistic response excerpt per Level-1 coupling of Def. 2. As in Table 19, each excerpt is prose containing both claims — A is the source, underlined, B **the target, bold**, the *italic* words drawing the link — and $V(\tau)$ is repeated from Def. 2 so the excerpt can be read against the world it rules out. The *Why* column names the world the response denies, which is what distinguishes each coupling from the one above it; the three conflict-adjacent couplings CONTRADICTION, EXCLUSIVE and CO_NECESSITY are illustrated on one two-part weld failure so that only exhaustiveness varies. NONE produces no factor, so its excerpt is one where the text relates the claims without constraining their truth.

Coupling	Reading	$V(\tau)$	Response excerpt, and the world it denies
ENTAILMENT	$a_s \Rightarrow a_t$	$\{(1, 0)\}$	“The alloy is chemically identical to the certified <u>reference</u>, <i>and therefore it meets the airframe specification</i>.” <i>Why</i>: the response rules out the world where the source holds and the target fails — $(1, 0)$ — and says nothing about the case where the alloy differs, which is why the $a_s=0$ row returns the target to its prior.
CONTRADICTION	$a_s \Rightarrow \neg a_t$	$\{(1, 1)\}$	“The weld <u>passed the tensile test</u>, <i>yet the same joint was porous along its full length</i>.” <i>Why</i>: only the both-true world $(1, 1)$ is denied. Both can still fail together — if the wrong joint was tested, neither claim holds — so $(0, 0)$ must stay available.
EXCLUSIVE	exactly one holds	$\{(0, 0), (1, 1)\}$	“The joint was destructively examined, so <i>exactly one of the weld met the tensile minimum and the weld fell short of it must be the finding</i> — <i>there is no third outcome</i>.” <i>Why</i>: both same-value worlds are denied. The added $(0, 0)$ is the whole difference from CONTRADICTION: once the joint has actually been examined the pair cannot both fail, so leaving that world open — as CONTRADICTION does — would under-penalize the conflict.
CO_NECESSITY	at least one holds	$\{(0, 0)\}$	“The fracture began at the joint, so <i>at least one of the weld and the fastener must have given way</i> — conceivably both.” <i>Why</i>: the dual of EXCLUSIVE. Only the both-false world is denied, so asserting the disjunction is evidence <i>for</i> both endpoints (Example 1).
EQUIVALENCE	$a_s \Leftrightarrow a_t$	$\{(0, 1), (1, 0)\}$	“The shipment was certified to specification — <i>that is to say, every contractual requirement had been met</i>.” <i>Why</i>: the two disagreeing worlds are denied, in both directions; unlike ENTAILMENT there is no vacuous row, since a false source now also forces a false target.
NONE	no coupling	$\emptyset$	“The replacement units were installed in March, <i>some three months before the first in-flight failures were reported</i>.” <i>Why</i>: no world is denied. The response genuinely relates the claims, but ordering them in time constrains neither one’s truth, so the pair contributes no factor and both claims stay at their priors.

C.1 ONE EXCERPT PER INFERENTIAL COUPLING

Table 17 gives a response excerpt for each coupling, set against the world that coupling denies; since a coupling *is* its penalized set, reading the two together is the quickest way to see what each one commits the response to. The first three couplings are exactly the NLI edge types of §B, which is what lets a mined relation reuse the existing graph machinery unchanged. The last two are additions

Table 18: The compile map $\mathcal{C}$ of §3.2, with glosses. *Ordering-only* senses record assertion order but induce **no factor**, which is what makes an order-only edit provably score-invariant (Rem. 2). **Concession** is a conflict the text may itself resolve, handled by Eq. 2. The nine senses marked $\star$ form the admissible vocabulary of §3.5.

Level-2 sense	$\mathcal{C}$ coupling	prior	dir.	flag	gloss
Cause-Effect $\star$	ENTAILMENT	$\perp$	yes		source brings target about
Effect-Cause $\star$	ENTAILMENT	$\perp$	yes		source is the effect, target the cause
Evidence $\star$	ENTAILMENT	$\perp$	yes		source is evidence for target
Condition	ENTAILMENT	$\perp$	yes		source conditions target
Instantiation	ENTAILMENT	$\perp$	yes		target instantiates source
Restatement $\star$	EQUIVALENCE	0.90	no		definitional identity
Contrast $\star$	CONTRADICTION	$\perp$	no		opposed, <i>not</i> exhaustive
Concession $\star$	CONTRADICTION	$\perp$	yes	conc.	a tension the text resolves
Alternative $\star$	EXCLUSIVE	$\perp$	no		exhaustive: exactly one holds
Disjunction $\star$	CO_NECESSITY	$\perp$	no		at least one holds
Precedence $\star$	NONE	—	yes	ord.	temporal order only
Succession	NONE	—	yes	ord.	temporal order only

required by argumentative prose, and the distinction between them and CONTRADICTION is the one the taxonomy exists to draw.

Why exhaustiveness needs its own coupling. CONTRADICTION forbids only the both-true world: two claims that cannot both hold, but might both fail. EXCLUSIVE additionally forbids the both-false world, which is what *exhaustive* alternatives assert. “No one was harmed” and “three people died” can neither both hold nor both fail — either someone was harmed or no one was — so a model that treats them as mere opposition has mis-described the situation in a way that matters for scoring: it leaves available a world in which *neither* claim is true, and so under-penalizes. Conversely CO_NECESSITY is the dual, penalizing only the both-false world, and is what a disjunction asserts (“at least one of these two findings must hold”). Adjudicated disputes — incident investigations, litigation, competing hypotheses — are precisely the domains where exhaustive pairs occur, because the function of a finding is to close one. The CONTRADICTION, EXCLUSIVE and CO_NECESSITY rows of Table 17 state one two-part weld failure three times over, so that the only thing varying across them is which same-value worlds the response leaves standing.

Example 1 (A CO_NECESSITY factor in isolation). *Take a single CO_NECESSITY relation with $p = 0.9$ over two claims with priors $\pi = 0.5$, and no other factors. From Table 20 the pairwise table is $[1 - p, \pi_s, \pi_s, p] = [0.1, 0.5, 0.5, 0.9]$, and multiplying in the unaries and normalizing gives world masses*

$$\mathbb{P}(0, 0) = 0.05, \quad \mathbb{P}(0, 1) = 0.25, \quad \mathbb{P}(1, 0) = 0.25, \quad \mathbb{P}(1, 1) = 0.45.$$

The both-false world is suppressed from the 0.25 it would receive under independent fair priors to 0.05, while the three worlds satisfying “at least one holds” share the recovered mass in proportion to how many endpoints they assert. Each marginal rises from 0.5 to 0.7: asserting that at least one of two claims holds is evidence for both, which is the intended reading.

C.2 THE COMPILE MAP, WITH GLOSSES

Couplings are the right vocabulary for a factor table and the wrong one for annotation. Asked whether two claims stand in ENTAILMENT, an annotator — human or model — has been handed a question about logic, when the fact actually observable in the text is a question about discourse: does this sentence give a cause, offer evidence, concede a tension, or lay out alternatives? We therefore mine an interpretable layer and compile it down.

C.3 ONE EXCERPT PER DISCOURSE SENSE

Table 19 grounds each sense in a response excerpt that contains both claims and the wording that licenses the link, since what an annotator must actually recognize is a form of words rather than a truth-functional pattern. Several entries encode judgements worth defending. Restatement carries

Table 19: One realistic response excerpt per Level-2 sense of Table 18, in the same row order and grouped by the coupling each compiles to. Each excerpt is prose of the kind the estimator actually sees and *contains both claims*: *A* is underlined, *B* is **bold**, and the *italic* words are the ones that draw the link — the span §3.5 obliges the model to quote, which is why it must be findable in the response and not in either claim. Senses are marked * when admissible; the three unmarked ones are included because their breadth is the argument for excluding them (§3.5). The Contrast, Alternative and Disjunction rows describe one two-part weld failure three ways, separated only by exhaustiveness (§3.2); they reuse the scenario of Table 17 deliberately, so that a discourse sense and the coupling it compiles to can be compared on the same prose.

Level-2 sense	<i>C</i>	Response excerpt, and why this sense rather than a neighbour
Cause-Effect*	ENTAILMENT	“<u>The company shipped a batch it knew to be out of tolerance</u>, and <i>as a direct result</i> its share price fell 15% that quarter.” Why: the response itself presents <i>A</i> as bringing <i>B</i> about; the marker names the direction.
Effect-Cause*	ENTAILMENT	“<u>The entire fleet was grounded on 14 March</u>. The order came <i>because inspectors had found a cracked wing spar on two airframes</i>.” Why: the same support runs from the stated effect to its cause. Direction of <i>explanation</i> is reversed; direction of inferential support is not.
Evidence*	ENTAILMENT	“<u>The maintenance log records no inspection between January and June</u>, <i>which shows that</i> the mandatory 500-hour check was skipped.” Why: <i>A</i> is offered in support of <i>B</i> without any claim that it caused it — a record is evidence, not a mechanism.
Condition	ENTAILMENT	“A part is cleared for airframe use <i>only if</i> <u>its alloy meets the reference specification</u>.” Why: <i>B</i> is asserted contingent on <i>A</i>. Note how weakly the cue constrains: nearly any dependence can be reread as a condition, which is why the sense is inadmissible.
Instantiation	ENTAILMENT	“<u>Several suppliers failed the 2023 audit</u> — <i>for instance</i>, the Tucson plant failed on two counts.” Why: <i>B</i> is one case falling under <i>A</i>’s generalization. Like Condition, broad enough to fit almost any pair that shares a topic.
Restatement*	EQUIVALENCE	“<u>The shipment was certified to specification</u> — <i>in other words</i>, every requirement in the contract had been met.” Why: a <i>marked</i> paraphrase asserts identity of content, which is why the strength prior is fixed at 0.90 rather than estimated.
Contrast*	CONTRADICTION	“<u>The weld passed the tensile test</u>, <i>but</i> the same joint was porous along its full length.” Why: opposed and <i>not</i> exhaustive. A third possibility — the wrong joint was tested — leaves both false, so the both-false world must stay available.
Concession*	CONTRADICTION	“<i>Although</i> <u>the airline argued throughout that pilot error was to blame</u>, the final report identified a metallurgical defect; the tribunal <i>ultimately held</i> for the latter.” Why: a tension the text itself settles. The resolving holding triggers the discount of Eq. 2 rather than counting as a live conflict.
Alternative*	EXCLUSIVE	“The two accounts cannot be reconciled: <i>either</i> <u>no one aboard was harmed</u>, or three passengers died, and <i>exactly one of them is true</i>.” Why: exhaustive. Unlike Contrast the pair can neither both hold nor both fail, so the both-false world is penalized too.
Disjunction*	CO_NECESSITY	“Metallurgy confirms the fracture began at the joint, so <i>at least one of</i> <u>the weld</u> and the fastener must have failed — possibly both.” Why: at least one holds and both may. The dual of Alternative: only the both-false world is penalized.
Precedence*	NONE	“The replacement units were installed in March, <i>some three months before</i> the first in-flight failures were reported in June.” Why: the text genuinely relates the two claims, and says nothing about either one’s truth — so the sense induces no factor.
Succession	NONE	“The first in-flight failures were reported in June, <i>fully three months after</i> the replacement units had been installed.” Why: the mirror of Precedence — the same two claims, the same absence of truth-functional content. Offering both senses invites an arbitrary choice on a relation that produces no factor either way.

Table 20: Pairwise potentials $\psi_{\tau,p}(a_s, a_t)$, listed row-major over the assignments $(0, 0), (0, 1), (1, 0), (1, 1)$; π_s is the source claim’s prior and bold marks the penalized set $V(\tau)$ of Def. 2. The *no-priors* family is correct for claim–evidence factuality, where the source passage’s own truth is at stake; the *with-priors* family is what coherence requires (Prop. 1) and what Def. 1 uses. The two symmetric couplings EQUIVALENCE and EXCLUSIVE coincide in both families, having no source-prior term to correct.

Coupling τ	No priors	With priors	Symmetry
ENTAILMENT	$[p, p, \mathbf{1-p}, p]$	$[1-\pi_s, \pi_s, \mathbf{1-p}, p]$	asymmetric
CONTRADICTION	$[p, p, p, \mathbf{1-p}]$	$[1-\pi_s, \pi_s, p, \mathbf{1-p}]$	asymmetric
EQUIVALENCE	$[p, \mathbf{1-p}, \mathbf{1-p}, p]$		symmetric
EXCLUSIVE	$[\mathbf{1-p}, p, p, \mathbf{1-p}]$		symmetric
CO_NECESSITY	$[\mathbf{1-p}, p, p, p]$	$[\mathbf{1-p}, \pi_s, \pi_s, p]$	symmetric

a fixed strength prior of 0.90 because a marked paraphrase (“in other words”, “equivalently”) is a claim about identity of content, and a model asked to estimate its strength has little to add; the estimator’s value is used when available, and the prior seeds it otherwise. Five senses collapse onto ENTAILMENT, which is a deliberate loss: the model does not distinguish causation from evidence, because a pairwise potential over truth values cannot. The grouped ENTAILMENT block of Table 19 shows exactly what compilation discards — five distinct discourse facts, one factor table. What survives compilation is the inferential force, and the sense is retained alongside it for interpretability and for error analysis.

The Contrast/Alternative boundary is exactly the exhaustiveness question of §3.2, and the Contrast, Alternative and Disjunction rows of Table 19 are that question in concrete form: one two-part weld failure written up three ways — as mere opposition, as an exhaustive choice, and as a disjunction — differing in nothing but which same-value worlds the response leaves available. The two ordering-only senses are the taxonomy’s way of representing a relation the text genuinely draws while asserting that it has no truth-functional content: a narrative that says one event preceded another has related the two claims, and has said nothing about whether either is true.

D POTENTIALS AND THE FOUR COHERENCE READOUTS

D.1 BOTH FAMILIES OF PAIRWISE POTENTIALS

§3.3 states Prop. 1 and the collapse it prevents. The table above gives both families side by side: the *no-priors* family is correct for claim–evidence factuality, where the source passage’s own truth is at stake, and the *with-priors* family is what coherence requires. The two symmetric couplings EQUIVALENCE and EXCLUSIVE coincide in both families, having no source-prior term to correct.

D.2 THE FOUR READOUTS

Definition 1 specifies a distribution; a coherence score is a functional of it, and several are defensible. We state five criteria, define four candidates with their exact constructions, and then decide between them by exact inference rather than by preference.

Definition 4 (Criteria). A coherence readout should be: **(R1)** a genuine functional of the joint $\mathbb{P}(\mathbf{a})$, not of the relation list alone; **(R2)** valued in $[0, 1]$ and interpretable; **(R3)** monotone — resolving or removing a conflict must not decrease it; **(R4)** sensitive to conflict without hand-tuned constants; **(R5)** computable from standard inference queries.

R1 deserves comment, since it is the criterion that does the most work. A readout that depends only on $\mathcal{R}$ — how many relations there are, what types, what weights — is not measuring the response; it is measuring its own annotation. Proposition 3 below shows this is not a hypothetical concern.

D.3 (A) MEAN POSTERIOR MARGINAL

$$\text{LCS}_{\text{mm}}(y) = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(a_i = 1). \quad (4)$$

The marginals come directly from a single MAR query. The interpretation is the expected fraction of the response’s claims that the argument, taken as a whole, upholds. A conflict suppresses the marginals of its endpoints; a satisfied entailment chain propagates support along itself; a claim no relation touches stays at its prior and contributes neutrally.

Alongside the scalar we report three diagnostics that come free with the same query: the per-claim marginals themselves, which localize an incoherence to specific claims; the count of claims dragged *below their own prior*, which is the sharpest single indicator of an active conflict; and the mean normalized binary entropy $\mathcal{E} = \frac{1}{n} \sum_i H_2(q_i)$, lower being more decisive.

D.4 (B) CONSISTENCY: A TWO-TERM PROBABILITY

The obvious alternative to averaging marginals is to ask for the probability that the response is consistent. Making that precise turns out to require two terms:

$$\text{LCS}_{\text{cn}} = \frac{1}{2}(\mathcal{K} + \mathcal{S}), \quad \mathcal{K} = \mathbb{P}\left(\bigwedge_{r: \tau_r = \text{CONTRADICTION}} \neg(a_{s_r} \wedge a_{t_r})\right), \quad \mathcal{S} = \frac{\sum_{r \in \mathcal{R}_S} p_r U_r}{\sum_{r \in \mathcal{R}_S} p_r}, \quad (5)$$

where $\mathcal{R}_S$ ranges over $\{\text{ENTAILMENT}, \text{EQUIVALENCE}, \text{EXCLUSIVE}\}$ and U_r is the probability that relation r is *actively upheld*:

$$U_r = \mathbb{P}(a_{s_r}=1 \wedge a_{t_r}=1) \text{ for ENTAILMENT, EQUIVALENCE, } \quad U_r = \mathbb{P}(a_{s_r} \neq a_{t_r}) \text{ for EXCLUSIVE.}$$

The two coupling sets are deliberately different. The conflict event $\mathcal{K}$ keys on CONTRADICTION alone, even though $\mathcal{T}_{\text{conf}} = \{\text{CONTRADICTION}, \text{EXCLUSIVE}\}$. The reason is that an EXCLUSIVE spreads its incoherence over *both* same-value worlds, so reading only its both-true half would describe half the coupling and silently report the other half as satisfactory. EXCLUSIVE therefore earns credit in $\mathcal{S}$ instead, where its exactly-one world is simultaneously where the exclusion is honoured and where it is informative. CO_NECESSITY enters neither term: its defect is the both-false world, which (a)’s marginals already see directly.

Computing $\mathcal{K}$ exactly. $\mathcal{K}$ is an event probability over a conjunction of $|\{r : \tau_r = \text{CONTRADICTION}\}|$ pairwise events, which a naive encoding would represent as a factor of exponential arity. Instead we augment the network with deterministic auxiliary variables: one u_r per conflict edge with a ternary factor enforcing $u_r = a_{s_r} \wedge a_{t_r}$, then a chained running disjunction $U_k = U_{k-1} \vee u_k$ with ternary factors, and finally $\mathcal{K} = \mathbb{P}(U_{|\cdot|}=0)$. Every auxiliary factor is deterministic and at most ternary, so no $2^{|\mathcal{R}|}$ table is ever built and the augmented network is read with one MAR.

Computing $\mathcal{S}$ exactly. $\mathcal{S}$ is not an event probability but a weighted sum of *pairwise joints* — a linear functional of the joint — so it cannot be read off the same accumulator chain. It needs one deterministic indicator variable per supported edge and its own MAR. Because the indicators are mutually independent given the base network, they may be batched across several queries without approximation, which matters only for inference backends that enumerate.

Why “actively upheld” and not “satisfied”. The definition of U_r credits a relation only in the informative world, not in every world where it is technically satisfied. An entailment $a_s \Rightarrow a_t$ is satisfied vacuously whenever $a_s = 0$, and crediting that is fatal:

Proposition 3 (The vacuous-truth trap). *Let $\mathcal{K}' = \mathbb{P}(\text{every relation in } \mathcal{R} \text{ is satisfied})$, the apparently natural extension of $\mathcal{K}$ to all couplings. Then a response with no mined relations satisfies $\mathcal{K}' = 1$, the maximum attainable value.*

Proof. $\mathcal{K}'$ is the probability of an intersection of $|\mathcal{R}|$ events; for $\mathcal{R} = \emptyset$ the intersection is the whole space. $\square$

The proposition is immediate, but its consequence is not, and it is the reason LCS_{cn} has the form it does. Measured on the family of §G.2, a disconnected bag of claims scores $\mathcal{K}' = 1.000$ while a response whose causal spine is fully satisfied scores 0.221 — an inversion, not a tie. Nor is the defect repaired by count normalization: the geometric mean over relations gives 0.883 for the base response against 0.882 for the coherent one, non-monotone *and* still 1.000 on the bag; the expected fraction of satisfied relations gives 0.857 against 0.852, likewise inverted; and normalizing against a prior ceiling leaves $[0, 1]$ altogether (8.82 against 5.46) while still running backwards. The common cause is that “are the asserted relations honoured?” is a property of $\mathcal{R}$ rather than of y — a violation of R1 — and it is maximized by asserting nothing. Crediting only the informative world removes the free lunch: a bag of claims upholds nothing and scores $S = 0$.

The two terms are averaged rather than multiplied so that neither can annihilate the other. A live contradiction should not erase a well-supported spine, and a relation-free response should not inherit a perfect score from an empty conflict set.

D.5 (C) A REIFIED COHERENCE NODE

The most direct reading of “is this response coherent?” as a single probability is to introduce a variable for it. Add a binary R with Bernoulli prior ρ and, for each relation, a ternary *vote* factor

$$h_r(R, a_s, a_t) = \begin{cases} 1 & R = 0 \quad (\text{the branch is flat}) \\ 1 & R = 1, (a_s, a_t) \notin V(\tau_r) \\ 1 - p_r & R = 1, (a_s, a_t) \in V(\tau_r), \end{cases} \quad \text{LCS}_{\text{re}} = \mathbb{P}(R = 1). \quad (6)$$

In the $R = 1$ branch each violated relation charges $1 - p_r$, so R behaves as a noisy-AND over relation satisfaction; in the $R = 0$ branch nothing is charged, so $R = 0$ carries no commitment. With no relations R is decoupled and $\text{LCS}_{\text{re}} = \rho$, correctly signalling that nothing was measured. Note that the violated set is $V(\tau)$ from Def. 2, so ENTAILMENT is treated as satisfied at $a_s = 0$ — vacuously — which is where (c) inherits a weaker form of Prop. 3’s problem. Appendix J works a three-claim subnet through by hand, giving $\mathbb{P}(R=1) = 0.453$.

D.6 (D) NORMALIZED LOG-PARTITION

The partition function measures how much probability mass the constraints leave standing, which is a global coherence signal requiring no marginals at all. It is unnormalized, so we grade it between two references built from the same edge skeleton:

$$\text{LCS}_{\text{lp}} = \frac{\log Z - \log Z_{\min}}{\log Z_{\max} - \log Z_{\min}}. \quad (7)$$

For the ceiling, $Z_{\max}$ is the partition function of the same network with every conflict edge removed: deleting constraint factors only adds mass, so $\log Z \leq \log Z_{\max}$ always. For the floor we take the *MAP world mass* of the base network, $Z_{\min} = \max_{\mathbf{a}} \prod_i \phi_i \prod_r \psi_r$.

Lemma 1 (The MAP world mass is a valid, coherence-graded floor). $\log Z_{\min} \leq \log Z$ for every network, and $Z_{\min}$ is non-increasing in conflict strength.

Proof. $Z = \sum_{\mathbf{a}} \text{mass}(\mathbf{a})$ is a sum of non-negative terms, so its single largest term is a lower bound. For the second claim, increasing p_r on a conflict edge multiplies the mass of every world in $V(\tau_r)$ by a factor below one and leaves other worlds unchanged, so the maximum over worlds cannot increase. $\square$

Lemma 1 is not a technicality; it replaces two floors that look more natural and are wrong. Retyping every edge to CONTRADICTION, or saturating conflicts to $p = 1$, both fail to bound $\log Z$ below for the row-stochastic tables of Prop. 1: an isolated conflict removes no mass at all when its endpoints sit at the uniform prior — the setting in which both candidates were checked — and little away from it, whereas a mass-concentrating entailment backbone concentrates a great deal, so a network combining a strong backbone with several soft conflicts can have a base $\log Z$ *below* either candidate floor. We observed exactly this on real graphs, which put Eq. 7 outside $[0, 1]$ until the MAP floor replaced them. In practice we clamp the ratio to $[0, 1]$ regardless, since approximate inference computes $\log Z$ and the MAP bound with finite i-bound and small tolerance violations are possible.

Table 21: All four readouts across a coherence-ordered family of five variants of one response, by exact inference over 2^{16} worlds. The variants run from least to most coherent left to right in the intended ordering. Bold marks a value that moves the *wrong* way relative to its predecessor. Only (a) and (d) are monotone throughout.

Variant	(a) LCS_{mm}	(b) LCS_{cn}	(c) LCS_{re}	$\log Z$	(d) LCS_{lp}
<i>worse</i>	0.586	0.778	0.116	-10.54	0.68
<i>base</i>	0.607	0.781	0.148	-9.64	0.78
<i>concession</i>	0.612	0.772	0.134	-9.64	0.80
<i>fixcasualty</i>	0.613	0.764	0.156	-8.95	0.89
<i>coherent</i>	0.620	0.752	0.181	-8.25	1.00
monotone?	yes	no	no	yes	yes

Table 22: The four readouts against the criteria of Def. 4, with the inference queries each requires.

Readout	Queries	R1	R2	R3	R4	R5	Note
(a) mean marginal	MAR	✓	✓	✓	✓	✓	selected
(b) consistency	$\text{MAR} \times 2$	✓	✓	×	✓	✓	concession dip
(c) reified	MAR	✓	✓	×	×	✓	free p ; dip
(d) norm. $\log Z$	PR, MAP	✓	✓	✓	✓	✓	needs 2 references

D.7 DECIDING BETWEEN THE FOUR

Table 21 settles the question empirically and Table 22 summarizes. Readouts (b) and (c) both fail R3, and they fail it at the *same* variant — the one where a conceded tension is resolved — for a shared reason that is worth stating precisely, because it looks like an implementation bug and is not.

Proposition 4 (Belief versus event activity). *Softening a resolved conflict via Eq. 2 increases the probability of its both-true world. For a single conflict edge with uniform priors, discounting p from 0.80 to 0.55 moves $\mathbb{P}(a_s=1, a_t=1)$ from 0.10 to 0.225.*

Proof. Under both conflict couplings the $(1, 1)$ cell of Table 20 is $1 - p$, which increases as p falls. Take a lone CONTRADICTION edge with $\pi_s = \pi_t = \frac{1}{2}$, so the with-priors table is $[\frac{1}{2}, \frac{1}{2}, p, 1 - p]$ and each world carries an equal unary factor of $\frac{1}{4}$. The unnormalized masses are then proportional to the table itself, giving $\mathbb{P}(1, 1) = (1 - p) / (\frac{1}{2} + \frac{1}{2} + p + (1 - p)) = (1 - p)/2$, which is monotone decreasing in p . Evaluating at $p = 0.80$ and $p = 0.55$ gives 0.100 and 0.225. $\square$

The consequence sorts the four readouts into two kinds. A readout that asks *is the conflict active* — (b) counts exactly that event, (c) charges $1 - p_r$ for it — must register a softened conflict as *more* often active, and therefore dips. A readout that asks *how strongly should each claim be believed* — (a) through marginals, (d) through mass — sees less suppression and rises. Since resolving a concession is by construction an increase in coherence, a coherence score must track belief rather than event activity. That is the substantive finding of this section, and it is not specific to our parameterization: any readout defined as the probability of a conflict-free configuration will invert under any discount-based treatment of resolved concessions.

A second failure mode of (b). Table 21 shows (b) failing worse than a single dip, and the reason is instructive about $\mathcal{K}$ and $\mathcal{S}$ jointly. In this family both conflicts are EXCLUSIVE rather than CONTRADICTION, so $\mathcal{K} \equiv 1.000$ throughout by the coupling-set decision above, and the score is carried entirely by $\mathcal{S}$. But removing an exclusion removes *support* credit as well as conflict, so the score runs backwards from 0.778 to 0.752 across the whole ladder. On three-coupling variants of the same response, where the conflicts are CONTRADICTION and $\mathcal{K}$ is live, (b) instead runs $0.576 \rightarrow 0.654 \rightarrow \mathbf{0.582} \rightarrow 0.672 \rightarrow 0.752$: correctly ordered apart from the concession dip. Either way it is non-monotone, and the lesson is that $\mathcal{K}$ and $\mathcal{S}$ should be read separately as diagnostics rather than averaged into a headline.

Algorithm 1 COMPUTELCS(y) — the end-to-end coherence score.

Require: response y ; relation estimator $\mathcal{M}$; discount λ ; priors π
Ensure: LCS $\in [0, 1]$, marginals $\{q_i\}$, diagnostics

- 1: $\mathcal{A} \leftarrow \text{ATOMIZEANDDECONTEXTUALIZE}(y)$
- 2: $P \leftarrow \text{SELECTCANDIDATEPAIRS}(\mathcal{A}, y, \dots)$ ▷ Algorithm 2
- 3: $\mathcal{R} \leftarrow \emptyset$
- 4: **for all** $(a_i, a_j) \in P$ **do** ▷ concurrent, rate-limited
- 5: $(\text{sense}, \tau, c) \leftarrow \text{PROMPTA}(a_i, a_j, y)$ ▷ $c = \mathbb{P}(\tau \mid \cdot)$ from Eq. 3
- 6: **if** $\tau = \text{NONE}$ **then continue**
- 7: $\sigma \leftarrow \text{PROMPTB}(a_i, a_j, \tau, y)$ ▷ surrogate-token strength
- 8: $\mathcal{R} \leftarrow \mathcal{R} \cup \{(i, j, \tau, \text{clip}_{[0,1]}(c \cdot \sigma))\}$
- 9: **for all** $r \in \mathcal{R}$ with $\tau_r \in \mathcal{T}_{\text{conf}}$ sensed as **Concession** **do**
- 10: **if** a resolving holding h exists **then** $p_r \leftarrow p_r(1 - \lambda\pi_h)$
- 11: **else** mark r unresolved
- 12: $\mathcal{N} \leftarrow$ network of Eq. 1 over $(\mathcal{A}, \mathcal{R}, \pi)$ ▷ with-priors tables
- 13: $\{q_i\} \leftarrow \text{INFER}(\mathcal{N}, \text{MAR})$
- 14: **return** $\frac{1}{n} \sum_i q_i, \{q_i\}, (\#\{q_i < \pi_i\}, \mathcal{E}, \log Z)$

Selection. We report (a) as the headline. It satisfies all five criteria, needs a single MAR query, and is the only one of the four whose value decomposes into per-claim contributions — which is what makes an incoherence *locatable* rather than merely detectable. We compute and report all four, since (d) is a genuine alternative satisfying the same criteria through a different route (mass rather than marginals), and $\mathcal{K}, \mathcal{S}$ carry diagnostic information the mean marginal does not expose.

D.8 ALGORITHM AND COST

Algorithm 1 is dominated by steps 4–9: $|P|$ Prompt-A calls plus one Prompt-B call per surviving edge, which the candidate policy keeps near-linear in n rather than quadratic. Inference is negligible by comparison.

Answering all four readouts requires seven inference calls: the base MAR and PR, the $\mathcal{K}$ accumulator MAR, the $\mathcal{S}$ indicator MAR, the reified MAR, the conflict-free ceiling PR, and the base MAP. Computed independently they would cost fourteen, since each readout re-runs the shared base pair. Production inference is weighted mini-bucket (Liu & Ihler, 2011) with i-bound 6; every *exact* value reported in this paper instead comes from explicit 2^n enumeration, which we use as a regression oracle for $n \leq 20$ and which is the source of Tables 21 and the values in §G.

E RELATION MINING IN DETAIL

E.1 CANDIDATE-PAIR SELECTION

Each claim is aligned to its originating sentence; a pair is *discourse-adjacent* if the two claims share content tokens above a Jaccard threshold, if their sentences are within a short span, or if the target’s sentence opens with a discourse connective. Out-of-window pairs that are discourse-adjacent are *promoted* into the candidate set; in-window pairs that are not are *demoted* out of it. The refinement is a correlated filter — it selects for the same surface adjacency that the sense prompt reads as evidence of relatedness — which is why we report it as a policy component and measure its effect rather than treating it as neutral preprocessing.

E.2 WHY THE EVIDENCE SPAN MUST QUOTE THE RESPONSE

The evidence-span requirement of §3.5 obliges the model to quote the words that draw the link from the *response*, not from either claim, and the parser then verifies the quotation is genuinely a substring of the response. Quoting the claims would not work: claims are decontextualized rewrites, and only about 27% of them appear in the response verbatim even after normalization, so a claim-quoting rule would reject most genuine relations.

Algorithm 2 SELECTCANDIDATEPAIRS: response-anchored candidate selection.

Require: claims $\mathcal{A} = \langle a_1, \dots, a_n \rangle$ in assertion order; response y ; policy; window w ; gate threshold θ_g ; discourse thresholds θ_d, δ

Ensure: ordered candidate pairs P , plus a coverage record

- 1: **if** policy = ALL_PAIRS **then return** every ordered pair $(a_i, a_j), i \neq j$
- 2: $W \leftarrow \{(a_i, a_j) : 0 < j - i \leq w\}$ $\triangleright$ forward order window
- 3: **if** policy = BIDIRECTIONAL **then** $W \leftarrow W \cup \{(a_j, a_i) : 0 < j - i \leq w\}$
- 4: $G \leftarrow \emptyset$
- 5: **if** policy = GATED **then**
- 6: $G \leftarrow \{(a_i, a_j) : j - i > w, \text{sim}(a_i, a_j) \geq \theta_g\}$
- 7: align each claim to its best-matching sentence of y
- 8: $Prom \leftarrow \{(a_i, a_j) \notin W \cup G : \text{DISCOURSEADJACENT}(i, j)\}$
- 9: $Dem \leftarrow \{(a_i, a_j) \in W : \neg \text{DISCOURSEADJACENT}(i, j)\}$
- 10: **return** $(\text{DEDUP}(W \cup G \cup Prom) \setminus Dem), (|W|, |G|, |Prom|, |Dem|)$

E.3 THE TWO PROMPT GENERATIONS

Two prompt generations. The first-generation Prompt A over-asserts badly: against a corpus in which 7.2% of candidate pairs are genuinely related, it answers “related” on 72% of the pairs a windowed policy hands it. Five properties push it that way, and the revision addresses each. (i) Its few-shot balance was three related to one unrelated; class balance in the exemplars is a strong prior on the answer distribution, and this one pointed the wrong way by an order of magnitude. The revision is two related to three unrelated. (ii) Its single negative exemplar had the response *explicitly disavow* a link (“the two processes were not related this year”), which teaches that NONE requires denial, whereas real prose simply stays silent; the revision’s negatives are all silence. (iii) It asked for grounding but gave the model nowhere to show it and nothing to check it — hence the evidence span. (iv) Its first reasoning step was a leading question listing eight affirmative relation types before mentioning NONE; the revision asks the discriminating question first, whether the text draws the link or the claims are merely near each other. (v) It gave NONE one line against nine for relation-bearing senses; the revision gives it first position and equal prose. The revision also names *transitivity* explicitly as a non-relation, since roughly 85% of the first generation’s false positives are the model closing a chain the text laid out pairwise, and 96.6% of them touch an endpoint of a real relation. Both generations are retained permanently so that any claim about the prompt can be measured rather than asserted; §5 reports both.

E.4 COUNTING FAILURES

Failures must be counted, not absorbed. Mining issues one LLM call per candidate pair for Prompt A and one more for each pair that yields an edge, run concurrently under a rate limit. A throttled or failed call returns no parse, and a parser that reads “no parse” as “no relation” converts an infrastructure failure into a sparse graph — which then scores as a *more* coherent response, since fewer conflicts are asserted. We therefore tally call errors separately from genuine NONE answers and report them, and distinguish an explicit NONE from an unparseable coupling for the same reason.

E.5 ALIGNING CLAIMS ACROSS THE TWO STAGES

Aligning claims across stages. The two stages must agree on which claim is which, and the obvious key is the wrong one. Matching proceeds by object identity first (the shared-atomization path, where the response is decomposed once and both stages see the same objects), then by normalized text (casefolded, whitespace-collapsed, terminal punctuation stripped), and only then by identifier. Text precedes identifier deliberately: near-duplicate removal keeps the first-seen claim and leaves the surviving identifier set *sparse* — $a_0, a_1, a_3, \dots$ — so two independent decompositions of the same response can disagree about which claim occupies which index. Matching on identifiers alone would then attach a prior to the *wrong* claim, which is strictly worse than attaching none, since it silently corrupts the score rather than degrading it. When coverage falls below one half the run is flagged

degraded, and the configured policy is to warn, to raise, or to discard every prior and fall back to a clean coherence-only score rather than report a half-primed mixture.

F EVALUATION PROTOCOL AND BASELINES IN FULL

F.1 THE FIVE BASELINE FAMILIES

A coherence measure should be compared against controls chosen to fail in informative ways. We describe five families and state what each is diagnostic of. Four are implemented and share the LCS’s own decomposition — each baseline is handed the same atoms the LCS was scored on and, like the coherence MRF, consults no retrieved evidence — so the comparison is an ablation rather than a separate experiment. Metrics requiring a source document or a reference are therefore excluded by construction, and we say so where the exclusion is not obvious.

1. *Ladder-blind controls*: claim count, relation count, response length, and number of perturbation operators applied. These cost nothing and they catch the most likely way a coherence measure is secretly wrong — by tracking edit count or graph density rather than coherence. A measure that cannot beat them on C3, or that fails to stay flat where they do, has not earned its complexity. Steen & Markert (2022) make the same argument for coherence measures generally, and introduce intra-system correlation and bias matrices to identify measures whose apparent performance comes from system-level confounders; we adopt the controls in that spirit. A control that *succeeds* is bad news for the item, not good news for the control, and §F reports one such case.
2. *Local contradiction detection*: pairwise NLI over claim pairs with no global model, scored as one minus the fraction of pairs labelled contradictory, in the spirit of SelfCheckGPT’s NLI variant (Manakul et al., 2023). This isolates precisely what joint inference adds over local detection. The prediction is that it does creditably where the defect is a *present* contradiction and poorly where the defect is a *missing* entailment, since a broken chain has no contradictory pair to find. We run it with the same NLI extractor as §B, over the same atoms, so the only differences from the LCS are relation *typing* and *propagation*.
3. *Structured reasoning consistency*: ROSCOE’s Self-Consistency (Golovneva et al., 2023), $1 - \max_i \max_{j < i} p_{\text{contr}}(h_i, h_j)$, which is evidence-free like ours but differs in three separable ways — it is a *maximum*, so a second conflict changes nothing; it is *forward-only*, so a relation running backward in claim order is unreachable (§3.5 measures 105 of 603 gold relations as both directed and backward); and it is *untyped*. We therefore also report a mean-aggregated and a symmetric variant, which turn one baseline into an ablation identifying *which* of the three differences accounts for any gap. ROSCOE targets step-by-step reasoning chains, and a long-form response is not a chain, so we label it adapted.
4. *LLM judges*: G-Eval’s coherence dimension (Liu et al., 2023), with the SummEval coherence rubric (Fabbri et al., 2021), and a direct “rate the logical coherence from 1 to 5” prompt as the no-rubric control. These are the practical incumbent and the comparison reviewers will want. The ladder is a demanding test for them, since consecutive rungs differ by a few words, so a judge must be sensitive at very small edit distances while remaining flat where meaning is preserved. We report each judge as a mean over five runs with its standard deviation, because the determinism of Remark 2 is only an advantage next to a measurement of the judge’s instability, and because a spread wider than the gap being discussed bounds what the comparison can conclude.
5. *Discourse representations*: DiscoScore (Zhao et al., 2023), with an entity-grid model (Barzilay & Lapata, 2008) as the classical reference. DiscoScore is presented as reference-based and most of its variants consume a reference, which a standalone response does not have; three of them (RC, LC and ENTITYGRAPH) ignore the reference argument entirely and score a single document, and those are the three we use — the last of which *is* an entity grid, so the classical reference comes from the same implementation. All three reduce to noun-lemma repetition across sentences, with no representation of entailment or contradiction, which makes them a sharp test of whether coherence is separable from *cohesion* rather than a weak competitor: §F reports a pair of responses with identical nouns and identical repetition structure, one self-consistent and one self-contradictory, on which all three are exactly flat.

6. *Internal ablations*: uniform versus factuality priors (§3.6); the four readouts of §D; the four candidate-pair policies; the two Prompt-A generations; and the gold-relation ceiling. These separate the model’s design choices from the estimator’s accuracy.

Beyond ladder ordering we recommend Spearman correlation against human coherence ratings on the same items, reported alongside per-family ordering agreement. Ordering agreement is the sharper instrument, being a controlled contrast with a known answer; human correlation is what establishes that the construct being ordered is the one readers care about. Neither substitutes for the other.

A baseline we cite but do not run. The closest existing work to our relation-mining stage is Liu & Strube (2025), who also build a graph over sentence pairs labelled with PDTB 3.0 senses — via a discourse parser rather than an LLM — and add entity links. We do not report a head-to-head number, and the reason is a difference in output type rather than a convenience: their model emits a *three-way* label (low, medium, high), and a three-way label cannot be checked against the ordering constraints of §5.1, which ask whether consecutive rungs move in a declared direction. The comparison would also require their trained classifier and a PDTB parser, neither of which is a pretrained scorer one can apply to a new response. Three differences are worth recording regardless, because they are the axes on which the two approaches diverge: their model performs no probabilistic inference, so a local conflict cannot propagate; it retains only forward-order pairs ($i < j$), so a backward relation is unreachable, which is the same structural cap we measure in §3.5; and it produces no per-claim attribution, so an incoherence is detected but not located.

F.2 WHAT THE BASELINES SHOW BEFORE ANY LADDER SWEEP

Two properties of these baselines are worth establishing before any ladder sweep, because both bear on what the comparison in §5.5 can support.

First, *cohesion is not coherence, and the discourse floor cannot tell the difference*. On a pair of two-sentence responses with identical nouns and identical repetition structure — “The weld passed the tensile test. The weld held under load.” against “... The weld *failed* under load.” — all three single-document DiscoScore metrics return *exactly* equal values, while the pairwise-NLI and ROSCOE baselines separate the pair. The metrics are defined over noun repetition, so this is a property of their definitions rather than an accuracy failure; it is also the cleanest available demonstration that the axis we measure is not the one entity-based coherence models measure.

Second, *a controlled contrast has to be checked, not assumed*. A measure can reproduce a declared ordering for reasons that have nothing to do with coherence, so a contrast is only informative once the model-free proxies have been shown *not* to reproduce it. The claim-identical ordering pair of §G.3 is the clean case: claim count is exactly flat there by construction, response length differs by four tokens (265 against 269) in the direction *opposite* to the declared ordering, and all three discourse metrics rank the adversarial summary *above* the faithful one, the self-serving-first ordering carrying more noun repetition. A cohesion metric prefers the less coherent response of the two. The ladder corpus of §5 generalizes exactly this property: every rung of a family shares one claim set, so a claim-count proxy is flat by construction across all seventy items.

F.3 MEAN READOUT VALUES PER SOURCE

Table 7 of §5 reports only the constraint counts, since averaging a readout across the rungs of a ladder collapses the ordering the ladder exists to test. The mean values are recorded here as a level comparison. The BIDIRECTIONAL density cost is visible quantitatively: LCS_{lp} collapses to 0.037 and 0.033 against gold’s 0.410 on the same seventy responses, from adding roughly two thousand spurious factors — the log-partition readout registers added factor mass most directly.

Table 23: Declared ordering assertions satisfied on the ladder corpus, under two different judge models. Only the readout-independent content of a constraint is scorable by a single-score baseline: the four C1 consecutive-increase pairs and the C3 endpoint separation, giving five per family. C2 is excluded because it predicts the internal behaviour of particular readouts (§D.7), which a baseline makes no claim about. The ORDER and CONTROL families assert invariance rather than increase and so contribute *no* scorable assertion, leaving ten families and 50 assertions; the LCS rows are scored on *exactly* those 50, so they are comparable to the rows below them, while the 82.7% of Table 7 is computed over the fuller 202-assertion set including invariance and is not. Judges are the mean of five runs. Model-free baselines are identical under both judges, as they must be; the two judge rows are the entire difference.

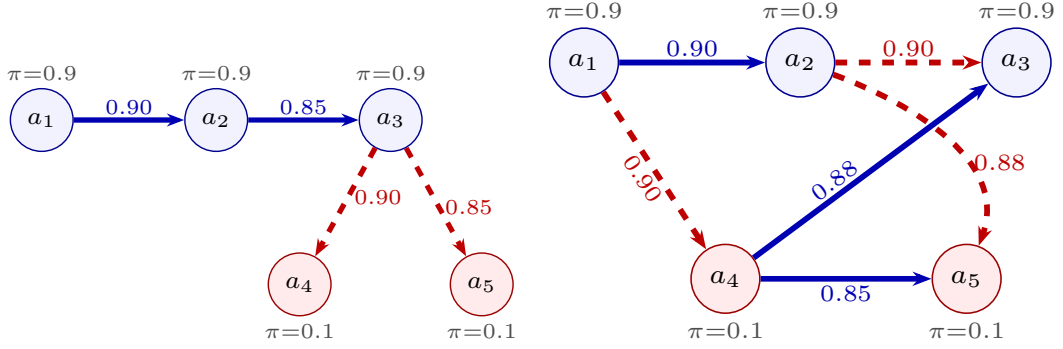
Measure	Assertions satisfied				Uses LLM
	llama-3.3-70b		gpt-oss-120b		
LCS _{mm} (gold relations)	37/50	74.0			no
LCS _{lp} (gold relations)	36/50	72.0			no
LCS _{cn} (gold relations)	34/50	68.0			no
RC (cohesion)	30/50	60.0	30/50	60.0	no
LC (cohesion)	28/50	56.0	28/50	56.0	no
LENGTH (control)	20/50	40.0	20/50	40.0	no
ROSCOE-SC (mean)	16/50	32.0	17/50	34.0	yes
CONTRADICTION RATE	12/50	24.0	12/50	24.0	yes
JUDGE, direct rating	4/50	8.0	18/50	36.0	yes
JUDGE, G-Eval	0/50	0.0	21/50	42.0	yes
ROSCOE-SC (max)	0/50	0.0	0/50	0.0	yes
ENTITY GRAPH	0/50	0.0	0/50	0.0	no
CLAIM COUNT (control)	0/50	0.0	0/50	0.0	no

Source	Policy	LCS _{mm}	LCS _{cn}	LCS _{re}	LCS _{lp}
GOLD	—	0.795	0.515	0.238	0.410
GOLD_VALID	—	0.794	0.576	0.316	0.485
<i>mined by llama-3.3-70b</i>					
	WINDOWED	0.796	0.676	0.384	0.224
	BIDIRECTIONAL	0.723	0.542	0.205	0.037
<i>mined by gpt-oss-120b</i>					
	WINDOWED	0.825	0.500	0.425	0.129
	BIDIRECTIONAL	0.808	0.462	0.251	0.033

Two further measurements belong with these. The cohesion baselines drift by at most 0.0096 across a permuted family, which is why they score 16/20 rather than 20/20 on invariance. And the 0.014 run-to-run F1 movement of §5.1 supersedes the ± 0.06 we previously estimated on a ten-item corpus: the larger corpus tightens reproducibility rather than merely enlarging the denominator.

F.4 JUDGE SATURATION, MEASURED

The two judges of §5.5 fail in opposite ways, and the mechanism is saturation rather than noise. llama-3.3-70b’s G-Eval is astonishingly stable and useless: mean per-item standard deviation 0.0002, only three distinct values across the seventy items, and a mean score of 0.996 — it rates almost everything at ceiling, so it cannot order anything. gpt-oss-120b’s G-Eval uses its range (fourteen distinct values, mean 0.749) and pays in stability (mean sd 0.0800, max 0.187). Reporting judge variance alone would have called the first judge the reliable one, which is exactly backwards for the task. Both judges also violate invariance freely: gpt-oss’s G-Eval moves by up to 0.55 across rungs that differ only in sentence order.



(a) **Response A, coherent.** The three true claims form a support chain ($a_1 \rightarrow a_2 \rightarrow a_3$) and each false claim is *attributed and rejected* by a CONTRADICTION edge from a_3 . Every claim ends on the correct side of its own prior: $\text{LCS}_{\text{mm}} = 0.590$, $\text{LCS}_{\text{cn}} = 0.963$, $\log Z = -0.95$.

(b) **Response B, incoherent.** The same five claims. The response contradicts its own chain ($a_2 \not\rightarrow a_3$) and makes the false a_4 a load-bearing premise for the true a_3 , while a_1 and a_2 deny a_4 and a_5 . The *true* claim a_3 is dragged to 0.513 — far below its own 0.9 prior: $\text{LCS}_{\text{mm}} = 0.494$, $\text{LCS}_{\text{cn}} = 0.414$, $\log Z = -3.78$.

Figure 5: Relation graphs for the five-claim worked pair (§B). Blue nodes are claims true of the world ($\pi_i = 0.9$), red nodes false ($\pi_i = 0.1$); the *same* five claims appear in both panels. Solid blue: ENTAILMENT; dashed red: CONTRADICTION. Line width grows with the relation weight p . The panels differ only in *how the response arranges* the claims, which is the whole of what coherence measures — no per-claim factuality check can separate them.

F.5 PER-MEASURE LADDER AGREEMENT

G WORKED EXAMPLES IN FULL

We work four examples. The first is the smallest complete instance of the two-stage model: one claim set, two responses, exact inference. The second is a richer sixteen-claim graph that exercises the conflict taxonomy. The third is a claim-identical pair that isolates ordering. The first three are hand-authored, which bounds what they can show: they demonstrate that the model reads a graph correctly, not that the estimator finds such graphs in text it has never seen. The fourth is therefore a corpus rather than an example — one hundred machine-generated responses whose incoherence is planted by construction, scored end to end with no hand-drawn relations anywhere.

G.1 A FIVE-CLAIM PAIR, END TO END

The point of this example is that factuality and coherence are separable, so we hold the first fixed and vary only the second. Five claims about the Voyager 1 mission are fixed once and for all, three of them true of the world and two false; stage one is taken as given, contributing the priors $\pi = 0.9$ for each true claim and $\pi = 0.1$ for each false one through the composition of §3.6:

	π_i	Claim
a_1	true 0.9	Voyager 1 crossed the heliopause in August 2012.
a_2	true 0.9	After the crossing, Voyager 1’s instruments measured a sharp rise in plasma density.
a_3	true 0.9	Voyager 1 is therefore in interstellar space.
a_4	false 0.1	Voyager 1’s plasma wave instrument failed permanently before the crossing.
a_5	false 0.1	Voyager 1 has left the Solar System entirely.

Two responses then assert *these same five claims* and differ only in how they arrange them. Because the claim set is identical, every per-claim factuality check returns the same verdict on both; the relation graphs (Figure 5) and the MRFs they induce (Figure 6) are what differ. Table 24 gives both responses in full, with the span asserting each claim marked $[a_i]$ at its first mention.

The italicized spans are the ones the estimator reads as relations, and they are the whole difference between the two. Table 25 lays the two relation sets side by side, one row per claim pair, with

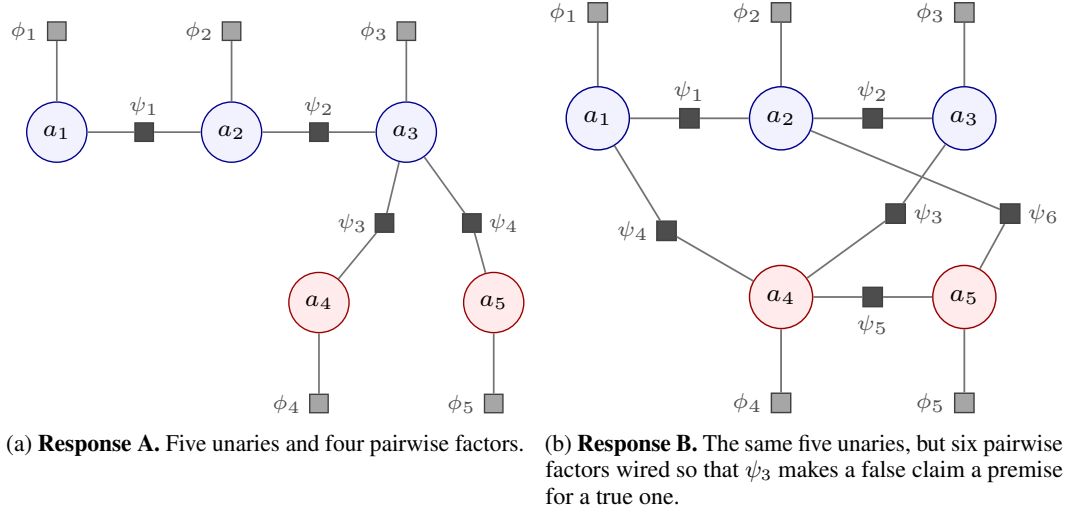


Figure 6: The coherence MRF of Eq. 1 for the two responses, as an explicit factor graph. Circles are the binary claim variables; grey squares the unary factors $\phi_i = [1 - \pi_i, \pi_i]$ that carry the priors; black squares the pairwise factors ψ_{τ_r, p_r} of Table 20, one per mined relation. **The priors enter only through the ϕ_i :** the claim-level potentials are identical across the two panels, and every difference in score comes from the ψ_r — their number, their endpoints, and their couplings.

Response A (coherent) — $\text{LCS}_{\text{mm}} = 0.590$, $\text{LCS}_{\text{cn}} = 0.963$, $\log Z = -0.95$

Voyager 1 crossed the heliopause in August 2012, $[a_1]$ and in the days that followed its instruments measured a sharp rise in plasma density $[a_2]$ — the signature of the denser medium beyond, and *the measurement that established the boundary had been passed*. Voyager 1 is therefore in interstellar space. $[a_3]$ Two persistent misconceptions are worth clearing up. The first is that the probe’s plasma wave instrument had failed permanently before the crossing; $[a_4]$ *it cannot have, since it is precisely that instrument which returned the density profile the conclusion rests on*. The second is that Voyager 1 has left the Solar System altogether. $[a_5]$ *Being in interstellar space is not the same thing: the Sun’s gravitational reach extends through the Oort cloud, far beyond the probe’s present distance*.

Response B (incoherent) — $\text{LCS}_{\text{mm}} = 0.494$, $\text{LCS}_{\text{cn}} = 0.414$, $\log Z = -3.78$

Voyager 1 crossed the heliopause in August 2012, $[a_1]$ and after the crossing its instruments measured a sharp rise in plasma density. $[a_2]$ *That said, the probe cannot really be counted as being in interstellar space*. $[a_3]$ The plasma wave instrument had in fact failed permanently before the crossing, $[a_4]$ *and it is precisely that failure which tells us the probe reached interstellar space*. The instrument failure also means Voyager 1 has left the Solar System entirely. $[a_5]$ *Of course, the August 2012 crossing rules out any such permanent failure, and the density measurement is flatly inconsistent with the probe having left the Solar System*.

Table 24: The two responses in full. They assert the *same* five claims of §G.1; each claim’s first mention is marked $[a_i]$. Italicized spans are the ones the estimator reads as relations. In A they discharge the two false claims; in B they reason *from* them.

each relation’s discourse sense and the coupling it compiles to. In A the false claims are named, attributed, and then refused on a stated ground, which compiles to a CONTRADICTION edge from a_3 . In B the same falsehoods are named and then *used* — a_4 is offered as the reason a_3 holds — while the response also denies a_3 outright and, in its last sentence, denies a_4 and a_5 too. Mentioning a falsehood is not what costs B its score; asserting it and reasoning from it is. All values below are exact, by $2^5 = 32$ -world enumeration.

Example 2 (Coherent). *Response A’s four relations (Table 25, column A) are a two-edge support chain over the true claims and one rejection of each false claim. The model returns $\text{LCS}_{\text{mm}} = 0.590$ and $\text{LCS}_{\text{cn}} = 0.963$ with $\log Z = -0.95$. Every claim finishes on the correct side of its own prior: the true claims sit at 0.939, 0.989 and 0.992 — each above its 0.9 prior, because the chain supplies*

Table 25: The mined relations of both responses, one row per claim pair. s is the Level-2 discourse sense, τ the Level-1 coupling it compiles to (Table 18), p the weight. E abbreviates ENTAILMENT, C CONTRADICTION. These are exactly the edges of Figure 5 and the factors ψ_r of Figure 6; a dash means the response draws no relation over that pair.

Pair	Response A (coherent)				Response B (incoherent)				What differs
	Edge	s	τ	p	Edge	s	τ	p	
$\{a_1, a_2\}$	$a_1 \rightarrow a_2$	Cau-Eff	E	.90	$a_1 \rightarrow a_2$	Cau-Eff	E	.90	The one relation both agree on.
$\{a_2, a_3\}$	$a_2 \rightarrow a_3$	Evidence	E	.85	$a_2 \rightarrow a_3$	Contrast	C	.90	A supports its conclusion; B denies it. Support becomes conflict.
$\{a_3, a_4\}$	$a_3 \rightarrow a_4$	Contrast	C	.90	$a_4 \rightarrow a_3$	Evidence	E	.88	The arrow reverses. A rejects a_4 ; B makes it a premise.
$\{a_3, a_5\}$	$a_3 \rightarrow a_5$	Contrast	C	.85	—	—	—	—	A refuses a_5 ; B never draws the distinction.
$\{a_4, a_5\}$	—	—	—	—	$a_4 \rightarrow a_5$	Cau-Eff	E	.85	B alone chains one falsehood into the other.
$\{a_1, a_4\}$	—	—	—	—	$a_1 \rightarrow a_4$	Contrast	C	.90	B denies a_4 , which it had just used as a premise — this strands a_3 .
$\{a_2, a_5\}$	—	—	—	—	$a_2 \rightarrow a_5$	Contrast	C	.88	Likewise for a_5 : asserted, then denied.
Totals		2 E, 2 C				3 E, 3 C			A’s conflicts point <i>outward</i> , at the false claims; half of B’s point <i>inward</i> .

mutual support — and the false ones at 0.013 and 0.020, pushed below 0.1 by the rejections. Nothing in the response fights anything else, so no claim is penalized.

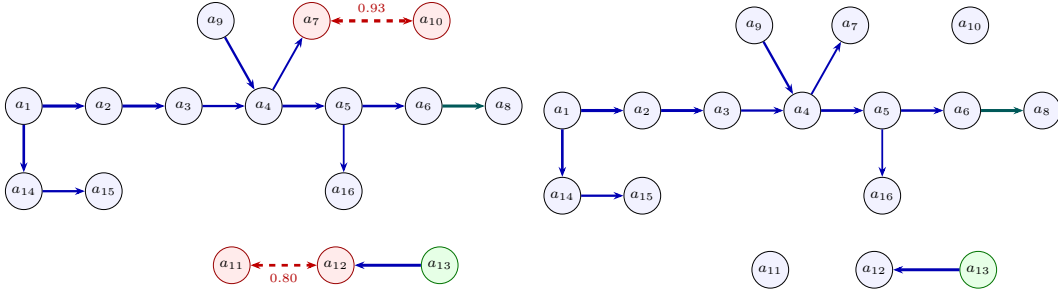
Example 3 (Incoherent). *Response B keeps only $a_1 \rightarrow a_2$ and replaces the rest with six relations that pull against each other (Table 25, column B): it denies its own conclusion, promotes the false a_4 to a premise for that same conclusion, chains a_4 into a_5 , and then denies both falsehoods in its closing sentence. Three of its six relations are conflicts, and two of those point at claims the response itself asserts. The score falls to $\text{LCS}_{\text{mm}} = 0.494$ and $\text{LCS}_{\text{cn}} = 0.414$, with $\log Z = -3.78$. The diagnostic that localizes the damage is *per-claim*: the true a_3 is dragged to 0.513, far below its own 0.9 prior, because its only support arrives from a_4 , which the response itself denies — and a premise the text refuses cannot transmit support. Three claims finish below their own priors, against two in Response A, and a_3 is the one that matters: a claim true of the world, demoted to a coin flip by the argument built around it.*

Two things are worth reading off this pair. First, the priors enter *only* through the unary factors ϕ_i of Figure 6, which are identical in both panels; the entire $0.590 \rightarrow 0.494$ drop is produced by the ψ_r , so it is attributable to arrangement alone. Second, a claim being true of the world does not protect it: coherence is a property of the joint distribution, and a_3 is suppressed by *how* it is argued for, not by what it says. This is the mechanism Remark 2 anticipates and the ordering pair of §G.3 exhibits at scale.

A caution on readouts. LCS_{lp} is 0.194 for Response A and 0.267 for Response B, which inverts the order — correctly, and for a reason Eq. 7 makes plain: its references $\log Z_{\text{max}}$ and $\log Z_{\text{min}}$ are recomputed *per response* (-0.49 and -1.06 for A, -1.65 and -4.55 for B), so it reports how close a response comes to the ceiling of *its own* edge skeleton, not how the two responses compare. It is a within-response diagnostic. Raw $\log Z$ is comparable across responses and moves the right way, by 2.83 nats.

G.2 A SIXTEEN-CLAIM RECALL REPORT

The second example is a sixteen-claim aviation recall report, which exercises the conflict taxonomy more fully than the five-claim pair: it carries both an unresolved conflict and a conceded one. Its relation graph is twelve support edges — an entailment spine $a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_6$ (shipment certified to spec, installation, entry into service, in-flight failures, regulatory probe, grounding order), branches



(a) **Incoherent** (base). Two EXCLUSIVE conflicts (dashed): $a_7 \leftrightarrow a_{10}$ is *unresolved*; $a_{11} \leftrightarrow a_{12}$ is a *concession* the holding a_{13} (green) resolves. $\text{mm} = 0.607$, $\log Z = -9.64$. (b) **Coherent**. The identical support structure with both conflict edges removed; a_{10} and a_7 are no longer suppressed. $\text{mm} = 0.620$, $\log Z = -8.25$, $\text{lp} = 1.00$.

Figure 7: Relation graphs for the AeroParts recall report (§B). Solid blue: ENTAILMENT; teal: EQUIVALENCE; dashed red, double-headed: EXCLUSIVE. Line width grows with the relation weight p . **The two panels differ only in the two conflict edges**, so the $0.607 \rightarrow 0.620$ lift and the 1.39 nats of recovered mass are attributable to conflict structure alone.

to the pulled-from-service claim, the maintenance-log evidence, the tribunal’s damages award and the regulator’s bulletin, and one EQUIVALENCE edge — plus two conflicts. Every value below is exact, by 2^{16} enumeration, and the graph is reproduced edge-by-edge in Figure 7 with each edge’s weight shown by line thickness.

Example 4 (Incoherent). *The report asserts both that no passengers or crew were harmed and that a wire report claimed three passengers died. These are exhaustive alternatives rather than mere opposition: they cannot both hold and cannot both fail, so the relation is EXCLUSIVE with $p = 0.93$ — and nothing in the response resolves it. A second tension, between the airline’s pilot-error argument and the final report’s metallurgical-defect finding ($p = 0.80$), is a **Concession** that the tribunal’s holding does resolve, and Eq. 2 discounts it accordingly. The model returns $\text{LCS}_{\text{mm}} = 0.607$ with $\log Z = -9.64$. The diagnostic that localizes the incoherence is per-claim: the marginal of the claim on the losing side of the unresolved conflict is dragged below its own prior, which no aggregate score would reveal. Applying the concession discount alone — resolving the second tension while leaving the first — raises the score to 0.612, as C2 requires.*

Example 5 (Coherent). *Removing the two conflict edges and changing nothing else leaves the identical support structure (Figure 7b). The score rises to $\text{LCS}_{\text{mm}} = 0.620$ with $\log Z = -8.25$ and $\text{LCS}_{\text{lp}} = 1.00$: the response is now as coherent as its own edge skeleton permits, which is what the ceiling of Eq. 7 means. The suppressed claim recovers to its prior.*

Because the two panels of Figure 7 differ *only* in the two conflict edges, the $0.607 \rightarrow 0.620$ lift and the 1.39 nats of recovered probability mass are attributable to conflict structure alone — which is the property the ladder construction of §5.1 generalizes from one pair to a family. The full ladder over this family is monotone throughout: $0.586 < 0.607 < 0.612 < 0.613 < 0.620$ over the rungs *worse, base, concession-resolved, one-conflict-fixed, coherent*.¹

G.3 A CLAIM-IDENTICAL ORDERING PAIR

This pair makes the order-sensitivity claim concretely. Two summaries of one appellate judgment are built from an *identical* set of eighteen claims — the same atom texts, one a permutation of the other — differing only in whether self-serving assertions precede or follow the findings that qualify them. The faithful ordering scores $\text{LCS}_{\text{mm}} = 0.95$ and the self-serving-first ordering 0.90. No per-claim

¹Treating both tensions as plain CONTRADICTION rather than EXCLUSIVE — the three-coupling model that omits the exhaustiveness distinction — gives $\text{LCS}_{\text{mm}} = 0.587$ and $\log Z = -9.75$ for Example 4, with Example 5 unchanged at 0.620 and -8.25 . The distinction of §3.2 is worth 0.020 of score here, and the reason it matters more than that figure suggests is Table 21: with EXCLUSIVE conflicts, readout (b)’s conflict term is pinned at 1.000 and the readout fails outright.

factuality check can separate them, because every claim is byte-identical; what differs is which claims the response leaves standing unqualified at the point where it asserts them.

It is worth being exact about *how* the two scores differ, because Remark 2 might seem to forbid it: nothing in Eq. 1 references position, so an edit preserving the relation set cannot move any readout. The pair does not preserve the relation set. Mining is response-grounded (§3.5), so the relations the estimator draws are a function of the prose, not of the claim list: the faithful summary states each tension beside the truth or holding that settles it, which the estimator reads as a **Concession** the text resolves and Eq. 2 discounts; the adversarial summary leaves the same claims standing unqualified until a later holding overrides them, which reads as a live conflict. The claim set is what factuality sees and it is identical; the relation set is what coherence sees and it is not.

H k -ARY REPRESENTABILITY

This appendix proves Prop. 2 of §5.11 and quantifies the gap it describes.

Proof of Prop. 2. By Eq. 1, $\mathbb{P}(1, 1, 1) = Z^{-1} \prod_i \phi_i(1) \prod_{i < j} \psi_{ij}(1, 1)$, a product of non-negative terms, which vanishes only if some term does. If $\phi_i(1) = 0$ then a_i is false almost surely and both pairs containing i have probability zero. If $\psi_{ij}(1, 1) = 0$ then every world with $a_i = a_j = 1$ carries that factor, so $\mathbb{P}(a_i=1, a_j=1) = 0$ directly. A ternary factor that vanishes only at $(1, 1, 1)$ has neither consequence. $\square$

The gap is quantitative as well as qualitative. With uniform priors, a ternary factor supported off $(1, 1, 1)$ gives $\mathbb{P}(1, 1, 1) = 0$ while leaving each pair jointly true with probability 0.143; the best pairwise attempt that drives $\mathbb{P}(1, 1, 1)$ below 10^{-4} leaves the pairs at 0.028, and a random search over 2×10^5 pairwise parameterizations found nothing better. The pairwise model can only suppress the bad world by also suppressing the good ones that share a $(1, 1)$ cell with it. Every figure here is exact by 2^3 enumeration and reproduced by a released script and regression test, so the proposition is checked and not merely asserted.

On the three-claim arithmetic instance of §5.11 — a league of 356 teams forming 32 conferences of exactly 10 members — the readouts report the passage as coherent at $\text{LCS}_{\text{mm}} = 0.68$ and $\text{LCS}_{\text{lp}} = 1.00$. Prop. 2 says no reweighting of the pairwise factors could have helped. Of the two milder consequences named in §5.11, transitivity is the more visible: the model will happily hold $a \Rightarrow b$ and $b \Rightarrow c$ at high weight while leaving a and c unrelated, since it has no rule tying them.

I CORRESPONDENCE WITH MARKOV LOGIC

The coherence MRF has a natural first-order encoding (Richardson & Domingos, 2006), with $\mathbb{P}(\mathbf{a}) = \frac{1}{Z} \exp(\sum_k w_k n_k(\mathbf{a}))$ for $n_k(\mathbf{a})$ the number of true groundings of formula F_k . Evidence predicates $\text{Entail}(i, j)$, $\text{Contradict}(i, j)$, $\text{Equiv}(i, j)$ and $\text{Resolves}(h, i, j)$ carry the mined graph; $\text{Holds}(i)$ is the query predicate. Working out the correspondence is instructive in both directions.

I.1 THE NAIVE ENCODING IS A DIFFERENT MODEL

The obvious encoding gives each relation one weighted clause, e.g. $\text{Entail}(i, j) \wedge a_i \Rightarrow a_j$ with $w = \logit(p)$. This is *not* equivalent to Definition 1. Its induced pairwise factor is $[e^w, e^w, 1, e^w]$, which is not row-stochastic in the sense of Prop. 1: it rewards the vacuous $a_s = 0$ rows with e^w on *both* target states, so a clause is best satisfied by making its antecedent false. Brute force over the 2^{16} worlds of the recall report of §G.2 gives mean marginal support 0.455 under the naive MLN against 0.587 under the corresponding MRF, with per-claim marginals differing by up to 0.31. These are two different models, not two encodings of one.

The failure is sharpest at the mode. The all-false world satisfies every implication vacuously, so under the naive encoding the most probable configuration of any purely-implicational graph is the one in which the response asserts nothing — the maximum-a-posteriori reading of a response is that none of it holds. This is Prop. 3 appearing in a second guise, and it is the same lesson: an encoding that scores rule satisfaction rather than belief can be maximized by having nothing to satisfy.

I.2 A THREE-CLAUSE EXPANSION IS EXACTLY EQUIVALENT

Writing the log-potential in the general pairwise form $\ln \psi(a_s, a_t) = \alpha + \beta a_s + \gamma a_t + \delta a_s a_t$ and matching it against the with-priors tables of Table 20 gives, for π_s the source prior and $L(x) = \text{logit}(x)$:

τ	α (unit)	β (source)	γ (target)	δ (conjunction)
ENTAILMENT	$\ln(1 - \pi_s)$	$\ln \frac{1-p}{1-\pi_s}$	$L(\pi_s)$	$+L(p) - L(\pi_s)$
CONTRADICTION	$\ln(1 - \pi_s)$	$\ln \frac{p}{1-\pi_s}$	$L(\pi_s)$	$-L(p) - L(\pi_s)$
EQUIVALENCE	$\ln p$	$\ln \frac{1-p}{p}$	$\ln \frac{1-p}{p}$	$+2L(p)$

At the atom prior $\pi_s = \frac{1}{2}$ the $L(\pi_s)$ terms vanish and the table reduces to $\gamma = 0$ with $\delta = \pm L(p)$: there the conjunction weight is exactly $\pm \text{logit}(p)$, the interaction the implication was meant to carry all along, and the unit and source clauses are the correction that makes each row a conditional. Away from $\frac{1}{2}$ the $L(\pi_s)$ terms are load-bearing, which matters for the two-stage composition of §3.6, whose priors are factuality posteriors rather than uniform. With this expansion the ground Markov logic network reproduces the MRF exactly: at $\pi_s = \frac{1}{2}$, mean support 0.5866 and $\log Z = -9.751$ for both, with zero per-claim difference to machine precision; the agreement is verified against the brute-force oracle at priors on and off $\frac{1}{2}$.

We establish the equivalence for the three original couplings only. EXCLUSIVE and CO_NECESSITY are not tabulated here, so the claim “the MLN encoding reproduces the MRF” should be read as scoped to the three-coupling fragment. That is a real limitation for the recall report of §G.2, whose conflicts are EXCLUSIVE; the five-claim pair of §G.1 uses CONTRADICTION throughout and so falls inside the tabulated fragment.

I.3 WHICH VIEW TO KEEP, AND WHY BOTH

The reason to prefer the MRF for the measure itself is Remark 1: its pairwise factor family is closed at three parameters, so the inferential vocabulary cannot quietly grow as senses are added, and every relation type is guaranteed to be a 2×2 potential. The reason to keep the Markov logic view is that the rules our model cannot express are naturally written there — a transitivity rule $\text{Entail}(i, j) \wedge \text{Entail}(j, k) \Rightarrow \text{Entail}(i, k)$, a concession-cancellation rule $\text{Resolves}(h, i, j) \wedge a_h \Rightarrow \neg \text{Penalize}(i, j)$ replacing the fixed discount of Eq. 2, and a double-conflict rule — and their weights could be *learned* from labelled ladders rather than set by hand. That is the route past the two limitations §5.11 names, and it requires an MLN solver (MC-SAT for marginals, MaxWalkSAT for the mode) rather than the bucket-elimination stack used here.

J THE REIFIED COHERENCE NODE, WORKED

To make Eq. 6 concrete, take the three-claim subnet consisting of the unresolved casualty conflict of Example 4 together with one entailment feeding it: claims a_7, a_{10} joined by a conflict at $p = 0.93$, and $a_4 \rightarrow a_7$ an entailment at $p = 0.65$, all priors $\frac{1}{2}$, with $\rho = \frac{1}{2}$. Writing out the eight-cell vote factor for each relation from Eq. 6 and enumerating the 2^4 worlds of (a_4, a_7, a_{10}, R) gives $\mathbb{P}(R=1) = 0.453$: below its prior, because both relations have violating worlds that carry appreciable mass, and the conflict’s violating world $(1, 1)$ carries the most.

Two properties of the construction are worth noting. The $R = 0$ branch is flat, so $R = 0$ asserts nothing and the node cannot be driven up merely by adding relations that happen to be satisfied. And the factor is a *noisy* AND rather than a hard one: a relation with $p_r < 1$ leaves some mass on its violating worlds even when $R = 1$, so a single strongly-violated relation reduces $\mathbb{P}(R=1)$ without pinning it to zero. The reason we do not adopt the readout despite this reasonable behaviour is Table 21: it dips at the concession variant for the reason of Prop. 4, and it requires the free parameter ρ , violating R4.

K THE ESTIMATION PROMPTS, VERBATIM

The four prompts below are reproduced exactly as they are issued, extracted programmatically from the released source so that this appendix cannot drift from the code. Placeholders in double braces are substituted at call time; `{{sense_menu}}` is substituted at build time from one of the two menus in §K.5.

K.1 PROMPT A, FIRST GENERATION — DISCOURSE SENSE AND COUPLING

6,148 characters. Placeholders: `{{sense_menu}}`, `{{response}}`, `{{atom_a}}`, `{{atom_b}}`.

```

1
2
3 Instructions:
4 You are given the full RESPONSE a model produced, and two atomic claims, A and B, both
  ↳ taken FROM THAT RESPONSE, in their order of appearance (A comes before B). Your task is
  ↳ to decide the discourse/logical relation FROM A TO B, following the steps below.
5
6 IMPORTANT -- ground your decision in the response. Assert a coupling ONLY if the response
  ↳ ITSELF draws that connection between A and B (as written, or as a clear step in the
  ↳ author's argument/narrative). Do NOT assert a relation that is merely plausible in
  ↳ general but that the response does not actually make. If A and B both appear in the
  ↳ response yet the response draws no logical or discourse dependence between them, the
  ↳ answer is None.
7
8 1. Reason step by step, referring to the response: does the response present A as causing,
  ↳ enabling, providing evidence for, restating, elaborating, temporally preceding,
  ↳ contrasting with, or contradicting B? Is B a claim the response later withdraws or that
  ↳ a holding resolves? Consider the direction (A to B). If the response links A and B only
  ↳ indirectly through other claims, or not at all, that is None.
9
10 2. Name the DISCOURSE SENSE, one of: {{sense_menu}}.
11   - Cause-Effect: A causes/leads to B. Effect-Cause: A is the effect, B its cause.
12   - Evidence: A provides evidence for B. Condition: A is a condition for B.
13   - Restatement: A and B assert the same thing. Instantiation: A is a general claim, B a
  ↳ specific instance (or vice versa).
14   - Contrast: A and B are in opposition but NOT exhaustive (they need not cover all
  ↳ possibilities; both could conceivably be false).
15   - Concession: A and B are in tension but the text concedes/resolves it ("although A,
  ↳ still B", or a holding settles it).
16   - Alternative: A and B are EXHAUSTIVE competing options -- EXACTLY ONE holds (they are
  ↳ mutually exclusive AND together cover the possibilities: not both, and not neither).
  ↳ E.g. "no one was harmed" vs "three people died"; "the cause was pilot error" vs "the
  ↳ cause was a metallurgical defect".
17   - Disjunction: AT LEAST ONE of A and B holds (they may both hold, but the response rules
  ↳ out neither being true) -- e.g. two supporting findings at least one of which must
  ↳ be present.
18   - Precedence/Succession: A and B are ordered in time with no truth dependence.
19   - None: the response draws no logical or discourse dependence between A and B.
20
21 3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive,
  ↳ co_necessity, none.
22   - Cause-Effect, Effect-Cause, Evidence, Condition, Instantiation -> entailment
23   - Restatement -> equivalence
24   - Contrast, Concession -> contradiction
25   - Alternative -> exclusive (exactly one of A, B is true)
26   - Disjunction -> co_necessity (at least one of A, B is true)
27   - Precedence, Succession, None -> none
28   Prefer "exclusive" over "contradiction" when the two claims are not just incompatible
  ↳ but EXHAUSTIVE (one of them must be true); prefer "contradiction" when they merely
  ↳ cannot both hold but could both be false.
29
30 4. Give your final answer as two bracketed tags on ONE line, sense first:
31 [sense=Cause-Effect] [coupling=entailment]
32 A JSON object {"sense":"Cause-Effect","coupling":"entailment"} is also acceptable.
33
34 Use the following examples to better understand your task.
35
36 Example 1 (the response makes the causal link):
37 RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns
  ↳ spiked, and the company's stock price fell 15 percent over the same period.
38 A: The company launched a flawed product last quarter.
39 B: The company's stock price fell 15 percent last quarter.
40 1. Reasoning: the response presents the flawed launch as the head of a chain (panning,
  ↳ returns) that ends in the stock decline, so the response itself draws a causal link
  ↳ from A to B.

```

2916 41 2. Discourse sense: Cause-Effect.
 2917 42 3. Coupling: A causing B is a positive inferential link, i.e. entailment.
 2918 43 4. Final answer:
 2919 44 [sense=Cause-Effect] [coupling=entailment]
 2920 45
 2920 46 Example 2 (both claims present, but the response draws NO connection -> None):
 2921 47 RESPONSE: The quarterly report was published in April. Separately, the annual audit was
 2922 48 ↳ scheduled for December. The two processes are run by different teams and were not
 2923 49 ↳ related this year.
 2924 50 A: The quarterly report was published in April.
 2925 51 B: The annual audit was scheduled for December.
 2926 52 1. Reasoning: both claims appear in the response, and one might imagine a
 2927 53 ↳ reporting-to-audit link in general, but this response explicitly treats them as
 2928 54 ↳ separate and unrelated. The response draws no dependence from A to B.
 2929 55 2. Discourse sense: None.
 2930 56 3. Coupling: no dependence the response asserts, i.e. none.
 2931 57 4. Final answer:
 2932 58 [sense=None] [coupling=none]
 2933 59
 2934 60 Example 3 (the response states an EXHAUSTIVE alternative -> exclusive):
 2935 61 RESPONSE: The official statement said no one was harmed in the incident. However, the
 2936 62 ↳ coroner's report confirmed that three people died in the incident.
 2937 63 A: No one was harmed in the incident.
 2938 64 B: Three people died in the incident.
 2939 65 1. Reasoning: the response sets A and B against each other ("However, ...") and they cannot
 2940 66 ↳ both be true; but they also cannot both be false -- either people were harmed or they
 2941 67 ↳ were not -- so exactly one holds. This is exhaustive, not a mere contrast. No holding
 2942 68 ↳ resolves it.
 2943 69 2. Discourse sense: Alternative.
 2944 70 3. Coupling: exactly one of A, B is true, i.e. exclusive.
 2945 71 4. Final answer:
 2946 72 [sense=Alternative] [coupling=exclusive]
 2947 73
 2948 74 Example 4 (at least one must hold -> co_necessity):
 2949 75 RESPONSE: The defect was caught in review: at least one of the two independent checks --
 2950 76 ↳ the vibration analysis or the metallurgical assay -- flagged it.
 2951 77 A: The vibration analysis flagged the defect.
 2952 78 B: The metallurgical assay flagged the defect.
 2953 79 1. Reasoning: the response asserts the defect WAS caught by at least one check, so A and B
 2954 80 ↳ cannot both be false; but both could hold (both checks may have flagged it). This is a
 2955 81 ↳ disjunction, not an exclusion.
 2956 82 2. Discourse sense: Disjunction.
 2957 83 3. Coupling: at least one of A, B is true, i.e. co_necessity.
 2958 84 4. Final answer:
 2959 85 [sense=Disjunction] [coupling=co_necessity]
 2960 86
 2961 87 Your task:
 2962 88 RESPONSE: {{response}}
 2963 89 A: {{atom_a}}
 2964 90 B: {{atom_b}}

2952 K.2 PROMPT A, SECOND GENERATION — REBALANCED FOR PRECISION

2954 6,361 characters. Same placeholders. The five changes relative to the first generation, and their
 2955 motivation, are given in §3.5: few-shot balance from 3:1 related to 2:3; negative exemplars showing
 2956 silence rather than explicit disavowal; a verifiable evidence span; the discriminating question asked
 2957 first; and NONE given first position and equal prose. Transitivity is named as a non-relation.

2958 1
 2959 2
 2960 3 Instructions:
 2961 4 You are given the full RESPONSE a model produced, and two atomic claims, A and B, both
 2962 5 ↳ taken FROM THAT RESPONSE. Decide the discourse/logical relation FROM A TO B.
 2963 6
 2964 7 Most pairs of claims in a text are NOT related. They are simply near each other. Your
 2965 8 ↳ default answer is None, and you should depart from it only when the response itself
 2966 9 ↳ draws a dependence between A and B.
 2967 10
 2968 11 Three things that are NOT relations, however plausible they look:
 2969 12 - Topical adjacency. A and B being about the same subject, or sitting in neighbouring
 ↳ sentences, is not a relation.
 - World knowledge. If the link is true in general but the response does not draw it, the
 ↳ answer is None.
 - Transitivity. If the response links A to some third claim C, and C to B, that does NOT
 ↳ make A related to B. Report only the links the text draws directly.

2970 13 1. First ask the discriminating question: does the response ITSELF assert a dependence from
2971 ↪ A to B -- as written, or as an explicit step in the author's argument? Identify the
2972 ↪ words in the response that do it. If you cannot point to such words, the answer is None.

2973 14 2. If and only if you can, name the DISCOURSE SENSE, one of: {{sense_menu}}.
2974 15 - None: the response draws no dependence between A and B. This is the common case:
2975 ↪ unrelated, merely adjacent, related only by world knowledge, or related only through
2976 ↪ a third claim.
2977 16 - Cause-Effect: the response says A causes/leads to B. Effect-Cause: A is the effect, B
2978 ↪ its cause.
2979 17 - Evidence: the response offers A as evidence/support for B.
2980 18 - Restatement: A and B say the same thing -- a paraphrase, a converse, or a restatement
2981 ↪ the text marks ("in other words", "equivalently", "that is").
2982 19 - Contrast: A and B are opposed but NOT exhaustive (both could be false).
2983 20 - Concession: A and B are in tension and the text concedes or resolves it ("although A,
2984 ↪ still B"; a holding settles it).
2985 21 - Alternative: A and B are EXHAUSTIVE competing options -- exactly one holds (not both,
2986 ↪ not neither).
2987 22 - Disjunction: AT LEAST ONE of A and B holds; both may.
2988 23 - Precedence: A and B are ordered in time with no truth dependence.

2989 24 3. Map the sense to a COUPLING, one of: entailment, contradiction, equivalence, exclusive,
2990 ↪ co_necessity, none.
2991 25 - Cause-Effect, Effect-Cause, Evidence -> entailment
2992 26 - Restatement -> equivalence
2993 27 - Contrast, Concession -> contradiction
2994 28 - Alternative -> exclusive (exactly one of A, B is true)
2995 29 - Disjunction -> co_necessity (at least one of A, B is true)
2996 30 - Precedence, None -> none
2997 31 Prefer "exclusive" when the two claims are incompatible AND exhaustive; "contradiction"
2998 ↪ when they merely cannot both hold but could both be false.

2999 32 4. Answer on ONE line, with three bracketed tags. The evidence must be a SHORT VERBATIM
3000 ↪ QUOTE COPIED FROM THE RESPONSE -- the words that draw the link. Do not quote claim A or
3001 ↪ claim B; quote the response. Use [evidence=none] when the sense is None.
3002 33 [sense=Evidence] [coupling=entailment] [evidence="which is why the panel held"]
3003 34

3004 35 Use the following examples to better understand your task.

3005 36 Example 1 (adjacent and on-topic, but the response draws no link -> None):
3006 37 RESPONSE: The survey was administered in March to a sample of 400 households. Response
3007 ↪ rates in rural districts have declined over the past decade. The analysis weighted
3008 ↪ results by district population.
3009 38 A: The survey was administered in March to a sample of 400 households.
3010 39 B: Response rates in rural districts have declined over the past decade.
3011 40 1. The two sentences are adjacent and both concern the survey, but the response never says
3012 ↪ the March administration bears on rural response rates. There are no words that draw a
3013 ↪ dependence.
3014 41 2. Discourse sense: None.
3015 42 3. Coupling: none.
3016 43 4. [sense=None] [coupling=none] [evidence=none]

3017 44 Example 2 (the response makes the causal link explicitly):
3018 45 RESPONSE: The company launched a flawed product last quarter. Reviewers panned it, returns
3019 ↪ spiked, and as a direct result the company's stock price fell 15 percent over the same
3020 ↪ period.
3021 46 A: The company launched a flawed product last quarter.
3022 47 B: The company's stock price fell 15 percent last quarter.
3023 48 1. The response presents the launch as the head of a chain ending in the stock decline, and
3024 ↪ marks it: "as a direct result".
3025 49 2. Discourse sense: Cause-Effect.
3026 50 3. Coupling: entailment.
3027 51 4. [sense=Cause-Effect] [coupling=entailment] [evidence="as a direct result"]

3028 52 Example 3 (true in general, but this response does not draw it -> None):
3029 53 RESPONSE: Higher interest rates raise the cost of corporate borrowing. The firm opened
3030 ↪ three distribution centres last year. Its logistics costs fell by eight percent.
3031 54 A: Higher interest rates raise the cost of corporate borrowing.
3032 55 B: The firm opened three distribution centres last year.
3033 56 1. One could argue borrowing costs affect expansion decisions, but that is world knowledge.
3034 ↪ This response states the two facts side by side and links neither to the other.
3035 57 2. Discourse sense: None.
3036 58 3. Coupling: none.
3037 59 4. [sense=None] [coupling=none] [evidence=none]

3038 60 Example 4 (the response states an exhaustive alternative -> exclusive):
3039 61 RESPONSE: The two readings are mutually exclusive and exactly one must be right: either no
3040 ↪ one was harmed in the incident, or three people died in it.
3041 62 A: No one was harmed in the incident.
3042 63 B: Three people died in the incident.

3024 71 1. The response sets A and B against each other and says exactly one must be right:
 3025 ↪ "mutually exclusive and exactly one must be right: either ... or".
 3026 72 2. Discourse sense: Alternative.
 3027 73 3. Coupling: exclusive.
 3028 74 4. [sense=Alternative] [coupling=exclusive] [evidence="mutually exclusive and exactly one
 ↪ must be right"]
 3029 75
 3030 76 Example 5 (linked only through a third claim -> None):
 3031 77 RESPONSE: The alloy is chemically identical to the certified reference. It therefore meets
 ↪ the reference specification. Meeting that specification is a precondition for airframe
 ↪ use.
 3032 78 A: The alloy is chemically identical to the certified reference.
 3033 79 B: Meeting that specification is a precondition for airframe use.
 3034 80 1. The response links A to the middle claim (it meets the specification), and the middle
 ↪ claim to B. It does not link A to B directly; that would be transitivity, which is not
 ↪ a relation to report.
 3035 81 2. Discourse sense: None.
 3036 82 3. Coupling: none.
 3037 83 4. [sense=None] [coupling=none] [evidence=none]
 3038 84
 3038 85 Your task:
 3039 86 RESPONSE: {{response}}
 3040 87 A: {{atom_a}}
 88 B: {{atom_b}}

K.3 PROMPT B, DEFAULT — SURROGATE-TOKEN CONDITIONAL STRENGTH

2,182 characters. Placeholders: {{response}}, {{atom_a}}, {{atom_b}}, {{coupling}} (twice). This is the one prompt that does not end in a newline: it terminates with Answer: including the trailing space, so that the model's next token is the surrogate whose logprobs are read. Reproducing it without that trailing space changes the tokenization of the answer position and therefore the estimate.

3050 1
 3051 2
 3052 3 Instructions:
 3053 4 You are given the full RESPONSE, two claims A and B drawn from it, and a {{coupling}}
 ↪ relation that holds from A to B. Assuming A is TRUE, judge -- IN THE CONTEXT OF THIS
 3054 ↪ RESPONSE -- whether the relation's implication about B is credible: at least plausible
 ↪ / more likely than not, NOT whether it is certain.
 3055 5
 3056 6 - entailment or equivalence: given A and how the response uses it, is B at least plausibly
 ↪ TRUE (more likely than not)?
 3057 7 - contradiction or exclusive: given A and how the response uses it, is B at least plausibly
 ↪ FALSE (more likely than not)? (For "exclusive", A and B are exhaustive alternatives, so
 3058 ↪ A being true makes B false.)
 3059 8 - co_necessity: A and B are a pair of which at least one holds. Given the response rules
 ↪ out "neither", is it at least plausible that B holds when A does NOT?
 3060 9
 3061 10 Answer with a SINGLE WORD, the very first word of your reply: Yes or No.
 3062 11 - Answer "Yes" if the implication is credible/plausible (even if not certain).
 3063 12 - Answer "No" only if the implication is implausible or the response does not actually
 ↪ support it.
 3064 13 Do not output anything before the word Yes or No.
 3065 14
 3065 15 Example 1 (response supports a near-certain entailment):
 3066 16 RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it
 ↪ meets the certified reference specification.
 3067 17 A: The new alloy is chemically identical to the certified reference alloy.
 3068 18 B: The new alloy meets the certified reference specification.
 3069 19 coupling: entailment
 20 Answer: Yes
 3070 21
 3071 22 Example 2 (weak but plausible, and the response draws the link -- still Yes):
 3072 23 RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15
 ↪ percent over the same period.
 24 A: The company launched a flawed product last quarter.
 3073 25 B: The company's stock price fell 15 percent last quarter.
 26 coupling: entailment
 3074 27 Answer: Yes
 3075 28
 3076 29 Example 3 (clear contradiction stated by the response):
 3076 30 RESPONSE: The official statement said no one was harmed, but three people died in the
 ↪ incident.
 3077 31 A: No one was harmed in the incident.
 32 B: Three people died in the incident.

```

3078 33 coupling: contradiction
3079 34 Answer: Yes
3080 35
3081 36 Your task:
3082 37 RESPONSE: {{response}}
3083 38 A: {{atom_a}}
3084 39 B: {{atom_b}}
3085 40 coupling: {{coupling}}
3086 41 Answer:

```

K.4 PROMPT B, BASELINE — VERBALIZED CONDITIONAL STRENGTH

2,112 characters. Retained only for the comparison reported in §5; verbalized confidence is weakly calibrated (Xiong et al., 2024).

```

3090 1
3091 2
3092 3 Instructions:
3093 4 You are given the full RESPONSE and two claims A and B drawn from it. Assume claim A is
3094 5 ↪ TRUE and, IN THE CONTEXT OF THIS RESPONSE, estimate how strongly A determines B under a
3095 6 ↪ {{coupling}} relation from A to B.
3096 7
3097 8 - For an entailment or equivalence coupling: how likely is B to be TRUE given A?
3098 9 ↪ (exclusive = A and B are exhaustive alternatives, so A true forces B false.)
3099 10 - For a co_necessity coupling (at least one of A, B holds): how likely is B to be TRUE when
3100 11 ↪ A is FALSE?
3101 12
3102 13 Answer with a single probability in [0, 1] to two decimals, after one short sentence of
3103 14 ↪ justification. A near-certain link is close to 1.00; a merely plausible link is around
3104 15 ↪ 0.60-0.70. End your answer with the probability wrapped in brackets on its own, exactly
3105 16 ↪ in the form: [p=0.NN]
3106 17
3107 18 Example 1:
3108 19 RESPONSE: The new alloy is chemically identical to the certified reference alloy, so it
3109 20 ↪ meets the certified reference specification.
3110 21 A: The new alloy is chemically identical to the certified reference alloy.
3111 22 B: The new alloy meets the certified reference specification.
3112 23 coupling: entailment
3113 24 Justification: chemical identity to a certified reference almost guarantees the
3114 25 ↪ specification is met, so B follows very strongly from A.
3115 26 [p=0.95]
3116 27
3117 28 Example 2:
3118 29 RESPONSE: The company launched a flawed product last quarter, and its stock price fell 15
3119 30 ↪ percent over the same period.
3120 31 A: The company launched a flawed product last quarter.
3121 32 B: The company's stock price fell 15 percent last quarter.
3122 33 coupling: entailment
3123 34 Justification: a flawed product can plausibly drive a stock decline, but many other factors
3124 35 ↪ affect price, so the link is only moderate.
3125 36 [p=0.65]
3126 37
3127 38 Example 3:
3128 39 RESPONSE: The official statement said no one was harmed, but three people died in the
3129 40 ↪ incident.
3130 41 A: No one was harmed in the incident.
3131 42 B: Three people died in the incident.
3132 43 coupling: contradiction
3133 44 Justification: if no one was harmed, it is almost certain that B (three deaths) is false.
3134 45 [p=0.93]
3135 46
3136 47 Your task:
3137 48 RESPONSE: {{response}}
3138 49 A: {{atom_a}}
3139 50 B: {{atom_b}}
3140 51 coupling: {{coupling}}

```

K.5 THE TWO SENSE MENUS

The full menu offers all twelve senses of Table 18 plus NONE:

```

3130 1 Cause-Effect, Effect-Cause, Evidence, Condition, Restatement, Instantiation, Contrast,
3131 2 ↪ Concession, Alternative, Disjunction, Precedence, Succession, None

```

The restricted menu offers only the senses the evaluation corpus labels, which is the admissibility filter of §3.5 applied at elicitation time rather than after it:

Cause-Effect, Effect-Cause, Evidence, Restatement, Contrast, Concession, Alternative, ↪ Disjunction, Precedence, None
--

Note that the restricted menu omits **Succession** as well as **Condition** and **Instantiation**, leaving nine relation-bearing senses. **Succession** is dropped because it is the mirror of **Precedence** and both compile to NONE, so offering both invites an arbitrary choice on a relation that produces no factor either way. The second-generation Prompt A’s body is written against this restricted menu; when it is run with the full menu, three senses are offered with no gloss and no coupling mapping in the prompt body, and we report which menu each source used.

L SERIALIZATION AND THE VARIABLE-ORDERING TRAP

One implementation detail is worth recording because it is silent when wrong. Networks are serialized in UAI format, whose variables are identified by integer index; the index order is the sort order of (cardinality, name), and with uniformly binary claim variables that reduces to *lexicographic order by name*. So a_{10} sorts before a_2 . Inference returns marginals by index, and mapping them back by any other convention — source order, numeric order — silently permutes marginals across claims. The score is unchanged, since Eq. 4 is a mean, but every per-claim diagnostic is then attributed to the wrong claim, and the “claim dragged below its prior” readout points at an innocent claim. The auxiliary variables of §D.4 are named with a prefix that sorts after all claim names for the same reason, so that adding them cannot shift the claim indices.